

Principal Bundles, Connections and Gauge Theory

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School of Mathematical Sciences

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Cameron R. D. Bunney

Supervisor: Dr. Alexander Schenkel

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Abstract

This study into principal bundles, gauge theory and connections introduces these topics in a rigorous setting whilst providing examples of the ideas and structures introduced throughout. A brief introduction to the relevant differential geometry is presented to provide the playground and tools used throughout the rest of the work. The Hopf fibration is introduced and shown to be a principal $U(1)$ -bundle. De Rham and Čech cohomologies are introduced to prove that magnetic charges are quantised and are found within $\check{H}^2(\mathcal{U}, \mathbb{Z})$. As a corollary of this result, the integral of the curvature form of principal $U(1)$ -bundles over the 2-sphere is shown to be quantised. Equivariant principal bundles are defined and exemplified through $\mathbb{Z}_2$ -equivariant principal bundles over $\mathbb{R}^n$. Moreover, $\mathbb{Z}_2$ -equivariant principal $U(1)$ -bundles over $\mathbb{R}^n$ are classified through equivariant cohomology. A $\mathbb{Z}_2$ -equivariant Hopf fibration is developed with equivariant sections of the associated line bundles being analysed.

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1 Introduction

Mathematics and physics have long enjoyed a symbiosis, this being no more highly apparent than with the advent of calculus by Newton and Leibniz: physics provided the question of the motion of free falling objects and mathematics provided the tools to pave the way to the answer. Since then, mathematics and physics developed and flourished into disciplines in their own right, with their study being in the pursuit of knowledge and deepening understanding in the respective field. That notwithstanding, the two subjects have remained close, with physics driving new mathematics, and new mathematics being discovered soon to be followed by applications within physics.

“I had finally learned the beauty of fiber-bundle theory. . . I found it amazing that gauge fields are exactly connections on fiber bundles, which the mathematicians developed without reference to the physical world.” - Chen Ning Yang [30]

Fibre bundle theory, connections and gauge theory are examples of the close relationship between mathematics and physics, studied because the concepts “were natural and real” [30, p. 339] and purely geometric in origin. They were later thrust to the forefront of theoretical physics due to Yang-Mills theory forming the basis for the standard model, underpinning the electromagnetic force, the weak forces, and quantum chromodynamics.

Fibre bundles, or as they were then known fibre spaces, were first introduced by H. Seifert in the early 1930’s [19] in the context of algebraic topology and were only applied to physics later. The independence of physicists’ and mathematicians’ approaches could not be demonstrated clearer than when, in 1931, Dirac discovered a theory of magnetic monopoles. H. Hopf discovered in the same year the *Hopf fibration* and it would not be until 1977 when Andrzej Trautman would identify the Dirac monopole with the Hopf fibration [10], unifying the ideas.

The first gauge theory is well known to all physicists: electromagnetism. The electric and magnetic fields can be written in terms of a scalar potential ϕ and a vector potential $\mathbf{A}$,

$$\begin{aligned}\mathbf{E} &= -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \\ \mathbf{B} &= \nabla \times \mathbf{A}.\end{aligned}$$

These potentials are known not to be unique, since the following transformations,

$$\begin{aligned}\phi &\mapsto \phi - \frac{\partial\chi}{\partial t}, \\ \mathbf{A} &\mapsto \mathbf{A} + \nabla\chi,\end{aligned}$$

leave the electric and magnetic fields unchanged. One might then say that the potentials are unique *up to gauge transformations*, that is to say, up to being part of a family of potential related to each other by *gauge transformations*. These transformations at first glance appear to be ad hoc, but by considering electromagnetism as a $U(1)$ -gauge theory in the setting of fibre bundles, specifically *principal* fibre bundles, one reproduces these results purely geometrically.

The aim of this work is to present the theory of principal bundles, gauge theory, connections, cohomology and equivariant bundles and to reinforce this theory with examples and applications. The various examples include both the trivial case and more interesting examples, chosen to build up understanding of the geometric constructions. The content in Chapter 5 is presented to become well acquainted with ideas in cohomology theory in order to provide a proof of the quantisation of magnetic charge.

We begin this dissertation with an overview of the differential geometry required to understand the succeeding chapters. The principal idea is that of the manifold, the playground for all the geometry explored. We introduce differential forms and vector fields, objects which will stay with us for the remainder of the dissertation, giving a firm foundation to build on. The chapter closes with a pragmatic approach to Lie groups and Lie algebras, giving the necessary calculational tools to work with Lie algebras, established within setting of the standard model of particle physics.

Chapter 3 introduces the reader first to the general theory of bundles, quickly specialising to principal bundles followed by the introduction of associated bundles, connections

and gauge theory. Connections on principal bundles are exactly the gauge fields from physics, as Yang noted [30], and are therefore of great importance to study in order to understand the geometric underpinning of particle physics. We introduce a topological invariant showing that the integral of the curvature of a principal bundle over a 2-sphere is quantised. Just like the connection, the curvature has a physical interpretation and is interpreted as the field strength of the gauge field. The main achievement of this chapter is the use of examples throughout, demonstrating the theory in the simplest case to get a grasp of the often-abstract nature of statements and definitions. We end the chapter with examples of gauge-invariant physics, showing a scalar electromagnetic theory and the differential forms approach to Maxwell's equations.

After covering the theory of principal bundles with the simplest examples in Chapter 3, Chapter 4 covers the Hopf fibration, a more interesting example of a principal bundle. We go through the defining properties of a principal bundle, showing that the Hopf fibration satisfies each one. Thereafter, we calculate a connection and the associated curvature form to calculate the topological invariant for this bundle. The chapter concludes with the gauge transformation of the Hopf fibration, interpreted as local rotations.

The following chapter is self-contained and covers geometry of a different flavour, cohomology. De Rham cohomology is introduced first, popularly studied alongside differential geometry since it is intrinsically entwined with differential forms and is the study of when closed forms are exact. We move then to present a brief introduction to presheaves and sheaves to prepare the reader to study the Čech cohomology. Čech cohomology is introduced next and is defined in terms of a sheaf and an open cover, the local data that can be used instead to describe a manifold. Familiar examples such as globally defined differential forms and the transformation of transition functions are derived as examples of Čech cohomology in action. Combining the mathematics presented throughout the chapter, the chapter culminates with the proof that magnetic charges lie within $\check{H}^2(\mathcal{U}, \mathbb{Z})$ and as a corollary of this result, we prove Proposition 3.33 directly with these being the main achievements of the chapter. Čech cohomology works on the level of open sets and topological data therefore we present the bundle reconstruction theorem with its constructive proof and provide a suitable example employing this construction.

The concluding chapter continues from groundwork laid in Chapters 2-5, taking principal bundles and adding more structure to produce the lesser studied H -equivariant principal bundles. We present definitions of equivariant principal bundles and equivariant connections on these bundles together with examples of such calculations. We study the case $H = \mathbb{Z}_2$ and work through the simplest examples of $\mathbb{Z}_2$ -equivariant principal $U(1)$ -bundles over $\mathbb{R}$ and $\mathbb{R}^n$, as well as demonstrate the structure of the associated complex line bundle of charge q . We find a general section of this bundle and show that depending on whether q is odd or even, the bundle exhibits either Dirichlet or Neumann boundary conditions at the fixed points of the $\mathbb{Z}_2$ -action. Guided by [6], we provide a proof that there are only two $\mathbb{Z}_2$ -equivariant principal $U(1)$ -bundles over $\mathbb{R}^n$. Along the way, we introduce classifying spaces from homotopy theory and equivariant cohomology from algebraic topology. This proof is one of the main achievements of the chapter, pulling from tools from all of the preceding chapters. Hereafter, we reintroduce the Hopf fibration and derive equivariant $\mathbb{Z}_2$ -actions, forming $\mathbb{Z}_2$ -equivariant Hopf bundles. To better study these equivariant bundles, we introduce an algebraic analogue to the Hopf fibration, the algebra of polynomial functions on $\mathbb{S}^3$. To this end, we show that this is an equivalent formulation and work with this algebra to show that it admits a useful decomposition and calculate the space of sections of the equivariant Hopf associated bundles. After deriving this space, we show an explicit example of an equivariant section. Sections are of significance to study because, like connections and the curvature, sections of associated bundles have a physical interpretation, they are the matter fields of the theory. We then return to the geometric picture and use methods such as trivialisation of the bundle and stereographic projection to reduce the bundle to a simpler form to study sections. Equivariant sections of the equivariant Hopf associated bundle are more interesting and therefore less simple to study than in the case over $\mathbb{R}^n$. We introduce the concept of functions with compact support to construct a specific example of an equivariant section of this bundle, an achievement due to the non-trivial nature of both the Hopf fibration and the $\mathbb{Z}_2$ -action chosen.

The scope of this dissertation is broad, with the main areas of study being principal bundle theory, gauge theory, Lie theory, de Rham, Čech and equivariant cohomology theories, sheaf theory, and the theory of equivariant principal bundles. We present these

ideas to a reasonable depth and provide examples throughout to aid comprehension. Though the work is purely mathematical, remarks rooting the mathematics within the context of physics are made throughout as motivation.

2 Preliminary Differential Geometry

Differential geometry is one of the pillars of mathematics and an indispensable tool in the modern study of physics, having applications throughout the field with the most influential example being general relativity, allowing one to view gravitation entirely geometrically. In this chapter, we will cover the key features of differential geometry to be able to understand the following chapters, assuming some rudimentary knowledge of topology. We recommend the following books [18], [20], [21] and [22] for more general presentations of differential geometry.

2.1 Manifolds and Structures on Manifolds

2.1.1 Manifolds

One of the most important and preliminary structures to be introduced to study geometry is the manifold. This generalises $\mathbb{R}^n$ to more exotic spaces, keeping one's intuition in Euclidean geometry since, locally, manifolds are indistinguishable to $\mathbb{R}^n$.

Definition 2.1. Let X and Y be topological spaces with a map $\varphi : X \rightarrow Y$ between them. φ is a homeomorphism if it is continuous with continuous inverse φ^{-1} .

If there exists such a map between two topological spaces, we say that the two spaces are homeomorphic to each other. We can now acquaint ourselves with smooth manifolds which will form the playgrounds for the rest of this dissertation.

Definition 2.2. A smooth n -dimensional manifold is a topological space M such that:

1. For each point $p \in M$, there exists an open neighbourhood U_α with $p \in U_\alpha \subseteq M$ and a homeomorphism $\psi_\alpha : U_\alpha \rightarrow V_\alpha \subseteq \mathbb{R}^n$ from U_α to an open subset $V_\alpha \subseteq \mathbb{R}^n$.
2. On all non-empty overlaps $U_\alpha \cap U_\beta \ni p$, the transition functions, defined by

$$\psi_\beta \circ \psi_\alpha^{-1} : \psi_\alpha(U_\alpha \cap U_\beta) \rightarrow \psi_\beta(U_\alpha \cap U_\beta),$$

are smooth maps.

From now on, when referring to a *manifold*, this shall implicitly mean a smooth n -dimensional manifold.

Definition 2.3. Let M be a manifold. The tuple (ψ_α, U_α) of an open subset $U_\alpha \subseteq M$ and a homeomorphism $\psi_\alpha : U_\alpha \rightarrow V_\alpha \subseteq \mathbb{R}^n$ is called a coordinate chart.

This first condition aligns with intuition: one is unable to tell from a map of a city that one is living on a sphere.¹ A manifold encapsulates this local flatness feature, whilst also giving rise to more global features.

Definition 2.4. A diffeomorphism is a homeomorphism $\varphi : M \rightarrow N$ between two manifolds such that both φ and φ^{-1} are smooth maps.

Example 2.5. The 2-sphere $\mathbb{S}^2$ is a smooth 2-dimensional manifold. $\mathbb{S}^2$ can be embedded in $\mathbb{R}^3$ by $x^2 + y^2 + z^2 = 1$. By considering two open subsets U_N, U_S defined as $\mathbb{S}^2$ without the South pole or North pole respectively, one can use stereographic projection from the South or North pole as the homeomorphism. In spherical coordinates, one may map the standard spherical coordinates (θ, φ) into $\mathbb{C} \cong \mathbb{R}^2$ by

$$\begin{aligned}\psi_N : (\theta, \varphi) &\mapsto z = \frac{\cos \theta/2}{\sin \theta/2} e^{i\varphi}, \\ \psi_S : (\theta, \varphi) &\mapsto w = \frac{\sin \theta/2}{\cos \theta/2} e^{i\varphi}.\end{aligned}$$

These maps define the smooth structure of the manifold. On the only overlap $\mathbb{S}^2 \setminus \{N, S\}$, the transition functions are defined as

$$\begin{aligned}\psi_S \circ \psi_N^{-1}(z) &= \frac{1}{z} = w, \\ \psi_N \circ \psi_S^{-1}(w) &= \frac{1}{w} = z.\end{aligned}$$

These are smooth maps on the overlaps. Therefore, $\mathbb{S}^2$ indeed has the structure of a smooth 2-dimensional manifold. ∇

2.1.2 Vector Fields

When one has the structure of a smooth manifold, one can start to place structures on the manifold. One of the most ubiquitous of these is the vector field, prevalent in all of physics.

¹Though approximately a sphere, the Earth is an oblate spheroid.

Definition 2.6. Let M be a manifold and $p \in M$. A tangent vector to M at p is a linear map $v : C^\infty(M) \rightarrow \mathbb{R}$, satisfying the Leibniz rule,

$$v(fg) = f(p)v(g) + v(f)g(p), \quad \forall f, g \in C^\infty(M).$$

Definition 2.7. The set of all tangent vectors to M at p is called the tangent space to M at p , denoted $T_p M$.

Remark 2.8. The tangent space has a natural vector space structure; for $v, w \in T_p M$, $a \in \mathbb{R}$, $f \in C^\infty(M)$, defined by $(v + w)(f) = v(f) + w(f)$, $(av)(f) = av(f)$. $\triangle$

A more geometric definition of a tangent vector uses the locally Euclidean nature of manifolds.

Lemma 2.9. Let M be a manifold and $p \in M$, let $U \subseteq M$ be an open neighbourhood about p with local coordinates $(x^1, \dots, x^n)$. A tangent vector to M at p is a linear combination

$$v = v^a(x) \left. \frac{\partial}{\partial x^a} \right|_p.$$

Proof. Found in [20, pp. 248-249]. $\square$

Remark 2.10. By $v^a(x) \left. \frac{\partial}{\partial x^a} \right|_p$, we mean

$$v^a(x) \left. \frac{\partial}{\partial x^a} \right|_p = \sum_{a=1}^n v^a(x) \left. \frac{\partial}{\partial x^a} \right|_p.$$

This is known as the Einstein summation convention. $\triangle$

When one has two manifolds M and N and a smooth map φ between them, one might ask about moving a vector on M to a vector on N , motivating the pushforward.

Definition 2.11. Let M and N be manifolds, $\varphi : M \rightarrow N$ be a smooth map, v be a tangent vector to M at p and $f \in C^\infty(N)$ an arbitrary function on N . The pushforward $\varphi_*(v)$ of v through φ is defined

$$\varphi_*(v)(f) = v(f \circ \varphi). \quad (2.1)$$

Remark 2.12. The pushforward is often also denoted as $d\varphi_p(v)(f) = v(f \circ \varphi)$ and then one has $d\varphi_p : T_p M \rightarrow T_{\varphi(p)} N$. $d\varphi_p$ is also called the differential of the map. $\triangle$

Remark 2.13. Morally, one can consider the pushforward as multiplication by the Jacobi matrix of the transformation. Using local coordinates x^a on $U \subseteq M$ and y^i on $V \subseteq N$, one may write $f \circ \varphi$ as $f(y(x))$, then

$$\begin{aligned}\varphi_* \left(\frac{\partial}{\partial x^a} \right) (f) &= \frac{\partial}{\partial x^a} f(y(x)), \\ &= \frac{\partial y^i}{\partial x^a} \frac{\partial f}{\partial y^i}.\end{aligned}$$

Since f is an arbitrary function, this gives the result

$$\varphi_* \left(\frac{\partial}{\partial x^a} \right) = \frac{\partial y^i}{\partial x^a} \frac{\partial}{\partial y^i}.$$

This is just multiplication by the Jacobian so one may interpret the pushforward in coordinates as such. Δ

A tangent vector is defined at a point on a manifold; we want to be able to discuss global geometric features, requiring the introduction of the vector field. Intuitively, one imagines a vector field as an assignment of a tangent vector v to each point $p \in M$. After having read Chapter 3, one may understand vector fields as *sections* of the tangent bundle $\pi : TM \rightarrow M$, however we will not dwell on this point.

Definition 2.14. Let M be a manifold. A vector field on M is a linear map $v : C^\infty(M) \rightarrow C^\infty(M)$ satisfying the Leibniz rule,

$$v(fg) = v(f)g + fv(g), \quad \forall f, g \in C^\infty(M).$$

Given a point $p \in M$, a vector field determines a tangent vector, $v_p(f) = v(f)(p)$.

2.1.3 Differential Forms

A further class of objects which can be considered on manifolds are differential forms. These were first studied by Alexis Claude Clairaut [16] and were of the form $A dx + B dy$. At the time, this was shorthand for

$$\frac{dy}{dx} = -\frac{A}{B}.$$

However, Clairaut studied this further and showed that a necessary and sufficient condition for $A dx + B dy$ to be the differential of a function is if

$$\frac{\partial A}{\partial y} = \frac{\partial B}{\partial x}.$$

In the literature and in what we shall discuss later, this condition is such that $Adx + Bdy$ be a *closed form*. The study of differential forms has expanded much since Clairaut and is no longer limited to the context of differential equations. Differential forms are at the heart of the *de Rham cohomology* discussed in Chapter 5, and are central to connections discussed throughout this work, which allow us to write down covariant derivatives and calculate curvatures. This brings us to what a really differential form is.

Definition 2.15. The exterior algebra $\Lambda^\bullet(\mathbb{R}^n)$ over $\mathbb{R}^n$ is the algebra generated by objects dx^a together with the wedge product $\wedge$ satisfying $dx^a \wedge dx^b = -dx^b \wedge dx^a$.

$\Lambda^\bullet(\mathbb{R}^n)$ forms a vector space with basis elements $1, dx^a, dx^{a_1} \wedge dx^{a_2}, \dots, dx^{a_1} \wedge \dots \wedge dx^{a_n}$ for $a_1 < \dots < a_n$. Given this basis, there is a natural decomposition of the exterior algebra into

$$\Lambda^\bullet(\mathbb{R}^n) = \bigoplus_{k=0}^n \Lambda^k(\mathbb{R}^n).$$

Definition 2.16. A differential form ω on $\mathbb{R}^n$ is an element

$$\omega \in C^\infty(\mathbb{R}^n) \otimes_{\mathbb{R}} \Lambda^\bullet(\mathbb{R}^n) =: \Omega^\bullet(\mathbb{R}^n). \quad (2.2)$$

The decomposition of the exterior algebra extends to a decomposition of differential forms $\Omega^k(\mathbb{R}^n)$. An element $\omega \in \Omega^k(\mathbb{R}^n)$ is called a k -form. Each k -form may be written as

$$\omega = \sum_{a_1 < \dots < a_k} f_{a_1 \dots a_k}(x) dx^{a_1} \wedge \dots \wedge dx^{a_k}. \quad (2.3)$$

A special class is the 0-forms, which are the smooth functions on $\mathbb{R}^n$. This algebra is equipped with a natural differential operator.

Definition 2.17. The exterior derivative d is a linear operator on differential forms,

$$d : \Omega^k(\mathbb{R}^n) \rightarrow \Omega^{k+1}(\mathbb{R}^n), \quad (2.4)$$

satisfying:

1. If $f \in \Omega^0(\mathbb{R}^n)$, then $df = \frac{\partial f}{\partial x^a} dx^a$.
2. For $\omega = \sum_{a_1 < \dots < a_k} f_{a_1 \dots a_k}(x) dx^{a_1} \wedge \dots \wedge dx^{a_k}$, $d\omega = \sum_{a_1 < \dots < a_k} df_{a_1 \dots a_k}(x) \wedge dx^{a_1} \wedge \dots \wedge dx^{a_k}$.

3. $d^2 = 0$ on any differential form.

In close analogue to the pushforward of vector fields, one may pull differential forms back through a map. As motivation, we shall demonstrate its construction using 0-forms. As we shall see, this is all we need define. Consider the following diagram:

$$\begin{array}{ccc} \mathbb{R}^m & \xrightarrow{\varphi} & \mathbb{R}^n \\ & \searrow f \circ \varphi & \downarrow f \\ & & \mathbb{R} \end{array}$$

Here we have a function $f \in C^\infty(\mathbb{R}^n)$ and a map $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^n$. By taking the composition, we find a function $\varphi^*(f) := f \circ \varphi \in C^\infty(\mathbb{R}^m)$, the pullback of f through φ . In order to see how this promotes to k -forms, we must first realise that $dx^a = d(x^a)$ by considering the coordinates themselves as functions. Secondly, we note that the pullback commutes with the exterior derivative so for f a function, $\varphi^*df = d(\varphi^*f)$. Using these two facts we can see the generalisation.

Definition 2.18. Let $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $\omega \in \Omega^k(\mathbb{R}^n)$, then

$$\varphi^*\omega = \sum_{a_1 < \dots < a_k} \varphi^*(f_{a_1 \dots a_k}) d\varphi^*(x^{a_1}) \wedge \dots \wedge d\varphi^*(x^{a_k}) \in \Omega^k(\mathbb{R}^m).$$

In fact for a manifold M , one might consider a map $\varphi : M \rightarrow \mathbb{R}^n$ and by the pullback, one would define a differential form on M .

Example 2.19. Consider $\omega = xdy - ydx \in \Omega^1(\mathbb{R}^2)$ and $f : \mathbb{S}^1 \rightarrow \mathbb{R}^2$ by $\theta \mapsto (x, y) = (\cos \theta, \sin \theta)$.

$$\begin{aligned} f^*\omega &= \cos \theta d \sin \theta - \sin \theta d \cos \theta, \\ &= \cos \theta \cos \theta d\theta + \sin \theta \sin \theta d\theta, \\ &= d\theta \in \Omega^1(\mathbb{S}^1). \end{aligned}$$

▽

With the strength of the pullback of differential forms from $\mathbb{R}^n$ to manifolds, we can define a differential form on an arbitrary manifold M .

Definition 2.20. Let M be a manifold with coordinate charts (ψ_α, U_α) . A globally defined differential form on M is a collection of forms $\{\omega_\alpha\} \in \Omega^k(\psi_\alpha(U_\alpha))$ on $V_\alpha = \psi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$ such that the consistency condition is satisfied on non-empty overlaps:

$$(\psi_\beta \circ \psi_\alpha^{-1})^* \omega_\beta = \omega_\alpha.$$

Remark 2.21. We had before that $\Omega^\bullet(\mathbb{R}^n) = C^\infty(\mathbb{R}^n) \otimes_{\mathbb{R}} \Lambda^\bullet(\mathbb{R}^n)$. Differential forms on arbitrary manifolds may take values in other spaces. After having read Chapter 3, one will understand the following statement pertaining to these differential forms. Let $E \rightarrow M$ be a smooth vector bundle. An E -valued differential k -form is written

$$\Omega^k(M, E) = \Gamma^\infty(E) \otimes_{\Omega^0(M)} \Omega^k(M).$$

This involves the tensor product of modules over the ring $\Omega^0(M) = C^\infty(M)$ which we shall not write about here. Furthermore, let V be a finite dimensional vector space, then

$$\Omega^k(M, V) \cong \Omega^k(M) \otimes_{\mathbb{R}} V.$$

These types of differential forms play important roles through this work. We will deal with such forms on a more practical and less abstract level and have thus presented their definitions here to be definitive. Δ

There are two further viewpoints on differential forms. The first concerns itself with the relationship between 1-forms and vector fields. One can consider 1-forms as comprising the space of linear functionals on vector fields. Given a 1-form $\omega \in \Omega^1(M)$ on a manifold M and a vector field v , one can define an inner product as $\omega(v) \in C^\infty(M)$. Written locally this can be seen as

$$(\omega_a dx^a) \left(v^b \frac{\partial}{\partial x^b} \right) = \omega_a v^b \delta_b^a = \omega_a v^a.$$

Here we have used that dx^a and $\partial/\partial x^b$ are dual basis vectors, meaning that $dx^a(\partial/\partial x^b) = \delta_b^a$. δ_b^a is the Kronecker delta

$$\delta_b^a = \begin{cases} 1, & \text{if } a = b, \\ 0, & \text{if } a \neq b. \end{cases}$$

This acts as an index notation identity matrix and, through the Einstein summation notation, swaps an index as seen $v^b \delta_b^a = v^a$. With the summation in the final result, one sees $\omega_a v^a$ as a generalisation of the dot product.

The second alternative viewpoint of differential forms is that they are objects which one might integrate. To this end, there are multiple ways of integrating differential forms on manifolds, through methods such as *integration over chains*. In this dissertation we will only require integration of forms in a specific setting and the appropriate method is one with more history: Stokes' theorem.

Theorem 2.22. *Let M be a compact smooth manifold with boundary ∂M and let ω be an $(n-1)$ -form on M . Then*

$$\int_M d\omega = \int_{\partial M} \omega.$$

The integral on the right hand side is taken with respect to the orientation induced by the inclusion map $i : \partial M \rightarrow M$.

Stokes' theorem is a fundamental result in differential geometry which will play a vital role in Chapter 5.

2.2 Lie Groups and Lie Algebras

We move tangentially to pay attention to groups, Lie groups, and Lie algebras. These structures will play just as an important role as manifolds will in this dissertation and their interplay is at the heart of this subject.

Definition 2.23. A group is a set G with an element e_G called the identity element and maps:

- Multiplication $\mu : G \times G \rightarrow G$, $\mu(g, h) = gh \in G$, $\forall g, h \in G$.
- Inverse $\sigma : G \rightarrow G$, $\sigma(g) = g^{-1}$ such that $gg^{-1} = g^{-1}g = e_G$.

We require that μ be associative $(gh)k = g(hk) \forall g, h, k \in G$ and that $ge = eg = g$.

An example of a group is the set of invertible matrices with real coefficients which we denote $GL_n(\mathbb{R})$

Example 2.24. Let $GL_n(\mathbb{R}) = \{M \in Mat_n(\mathbb{R}) : \det(M) \neq 0\}$ and let $A, B \in GL_n(\mathbb{R})$. Multiplication is defined by matrix multiplication. $\mu(A, B) = AB$. We need to check that $AB \in GL_n(\mathbb{R})$ by investigating its determinant. $\det(AB) = \det(A)\det(B) \neq 0$ since

$\det(A) \neq 0$ and $\det(B) \neq 0$. The inverse map is given by the matrix inverse. Since $\det(A^{-1}) = \det(A)^{-1}$, this is also non-zero and so $A^{-1} \in GL_n(\mathbb{R})$. Matrix multiplication is associative and the identity element is given by the identity matrix. $GL_n(\mathbb{R})$ thus satisfies the group axioms and is therefore a group. ∇

Definition 2.25. Let G be a group and let $H \subseteq G$ be a subset. H is a subgroup of G with the same maps μ and σ and same identity element $e_G \in H \subseteq G$ if $\mu : H \times H \rightarrow H$ and $\sigma : H \rightarrow H$.

If H is closed with respect to both μ and σ , then one writes $H \leq G$.

Definition 2.26. Let G be a group and N a subgroup of G . N is a normal subgroup if

$$gng^{-1} \in N \quad \forall g \in G.$$

This can be reread as N being a normal subgroup if $gng^{-1} = \tilde{n} \in N$ and $\tilde{n}$ need not necessarily be n . Any set G with maps satisfying the properties in Def. 2.23 forms a group. By adding more structure to this, one can form a more useful concept in differential geometry, the Lie group.

Definition 2.27. Let G be a group. G is a Lie group if it also has the structure of a manifold such that both multiplication and inversion are smooth maps.

Example 2.28. We consider now the set of matrices $SU(2) = \{M \in Mat_2(\mathbb{C}) : MM^\dagger = \mathbb{1}, \det(M) = 1\}$. This is a subset of $GL_2(\mathbb{C})$ with the usual matrix multiplication and matrix inversion. Any matrix in $SU(2)$ can be written in the form

$$M = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix},$$

with $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$.

Let $M_1, M_2 \in SU(2)$, then $\det(M_1 M_2) = \det(M_1) \det(M_2) = 1$ and $(M_1 M_2)(M_1 M_2)^\dagger = M_1(M_2 M_2^\dagger)M_1^\dagger = M_1 M_1^\dagger = \mathbb{1}$. Since $MM^\dagger = \mathbb{1}$, we have $M^\dagger = M^{-1} \in SU(2)$. Therefore, $SU(2)$ forms a subgroup of $GL_2(\mathbb{C})$ and is itself a group. It must also have the structure of a manifold to be a Lie group.

Any $SU(2)$ matrix can be determined by two complex numbers α and β , which can be written $\alpha = x_0 + ix_1$ and $\beta = x_2 + ix_3$. By the determinant condition, $|\alpha|^2 + |\beta|^2 =$

$(x_0)^2 + (x_1)^2 + (x_2)^2 + (x_3)^2 = 1$, this describes a diffeomorphism to the 3-sphere embedded in $\mathbb{R}^4$ and $SU(2)$ is thus a Lie group. ∇

One may compare groups and inspect whether they have the same structure and whether two groups are equivalent. Looking structure-wise, one considers group homomorphisms.

Definition 2.29. Let G and H be groups with multiplication maps μ_G and μ_H respectively. Let $\varphi : G \rightarrow H$. φ is a group homomorphism if $\forall g_1, g_2 \in G$,

1. $\varphi(\mu_G(g_1, g_2)) = \mu_H(\varphi(g_1), \varphi(g_2))$.
2. $\varphi(e_G) = e_H$.

Definition 2.30. Let G and H be groups and φ a group homomorphism. φ is a group isomorphism if it is bijective.

If there is an isomorphism between two group G and H , then G and H are isomorphic, written $G \cong H$.

Groups are generally non-abelian, however there are a special class of groups which are, which are the main groups used in this dissertation.

Definition 2.31. Let G be a group. G is an abelian group if $\mu(g_1, g_2) = \mu(g_2, g_1) \forall g_1, g_2 \in G$.

There is a more general concept of the “*abelian part*” of the group.

Definition 2.32. Let G be a group. The centre $Z(G)$ is the subgroup of G

$$Z(G) = \{z \in G : zg = gz, \forall g \in G\}.$$

The centre is always non-empty because $e_G \in Z(G)$. An abelian group is then the case when $Z(G) = G$ and the subgroups of any abelian group are automatically normal.

Groups can be subdivided by a subgroup into subsets whose unity form the group again.

Definition 2.33. Let G be a (Lie) group, H a (Lie) subgroup of G . The left coset of H in G with respect to some $g \in G$ is

$$gH = \{gh : h \in H\}.$$

Remark 2.34. One may define a right coset of H in G with respect to some element $g \in G$ analogously and this is written Hg . $\triangle$

Each element of G belongs to exactly one coset and consequently only H itself can be a subgroup since any other coset will not contain the identity element. This allows us to create new groups out of old ones by taking a suitable quotient.

Definition 2.35. Let G be a (Lie) group and N a normal (Lie) subgroup of G . The quotient group of G by N is

$$G/N = \{gN : g \in G\}.$$

One may think of G/N as equivalence classes formed by the cosets of N in G .

Groups do not just act on themselves, one may extend this to a group interacting with and acting on a manifold.

Definition 2.36. Let M be a manifold and G a Lie group. A left G -action on M is a map

$$L : G \times M \rightarrow M,$$

$$L(g, x) = gx,$$

such that $L(gh, x) = L(g, L(h, x))$ and $L(e, x) = x$.

A right group action is analogously defined $R : M \times G \rightarrow M$.

Remark 2.37. The notation $\mu(g, h) = gh$ was used for group multiplication and in a similar vein we will use $L(g, x) = g \cdot x$ and $R(x, g) = x \cdot g$ to ease legibility in denoting group actions. Furthermore, useful notations for group action include $L(g, x) = L_g(x)$ and $R(x, g) = R_g(x)$. All such notations will be used throughout this work. $\triangle$

The group action takes a point on a manifold to a (possibly) different point on the same manifold. In this way, one might imagine the group action carrying the point along the manifold forming a trajectory or orbit.

Definition 2.38. Let M be a manifold with a left G -action. The orbit $\mathcal{O}_x$ of a point $x \in M$ is

$$\mathcal{O}_x = \{y \in M : \exists g \in G \text{ with } y = g \cdot x\}.$$

Not all group actions are the same and one might further classify these actions.

Definition 2.39. Let M be a manifold with a left G -action.

- The group action is called free if there are no fixed points, $g \cdot x = x \Leftrightarrow g = e_G$.
- The group action is called transitive if any two points may be connected by the group action, $\forall x, y \in M, \exists g \in G$ such that $g \cdot x = y$.

Transitivity can be interpreted as there being a single orbit of G .

The final section of differential geometry we will explore before delving into the rich geometry of bundle theory is that of the Lie algebra of a Lie group.

Definition 2.40. A Lie algebra is a pair $(\mathfrak{g}, [\cdot, \cdot])$ consisting of a vector space $\mathfrak{g}$ and a bilinear map

$$\begin{aligned} [\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} &\rightarrow \mathfrak{g}, \\ (x, y) &\mapsto [x, y], \end{aligned}$$

called the Lie bracket, satisfying

1. Antisymmetry $[x, y] = -[y, x], \forall x, y \in \mathfrak{g}$.
2. Jacobi Identity $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0, \forall x, y, z \in \mathfrak{g}$.

We will, however, take a more pragmatic approach to Lie algebras and we only present matrix Lie algebras. Matrix Lie algebras have the matrix commutator

$$[A, B] = AB - BA,$$

as a Lie bracket, which satisfies antisymmetry and the Jacobi identity. To this end, we will show the calculation of matrix Lie algebras and will not delve further than this.

Definition 2.41. Let $X \in Mat_n(\mathbb{R})$. The exponential of a matrix is

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!} = \mathbb{1} + X + \frac{X^2}{2} + \dots,$$

The matrix exponential satisfies the following properties:

1. $e^{tX}e^{sX} = e^{(t+s)X}$ for $t, s \in \mathbb{R}$.

$$2. (e^X)^{-1} = e^{-X}.$$

$$3. e^{X^\dagger} = (e^X)^\dagger.$$

$$4. e^{X^T} = (e^X)^T$$

Because of the second property, $\exp(X)$ is always an invertible matrix.

There are several equivalent definitions of a Lie algebra of a Lie group. The most suitable for calculation purposes is from [14].

Definition 2.42. Let G be a Lie group. The Lie algebra $\mathfrak{g}$ of G is the tangent space at e_G ,

$$T_e G = \{\gamma'(0) : \gamma : (-\varepsilon, \varepsilon) \rightarrow G \text{ differentiable with } \gamma(0) = \mathbf{1}\}.$$

This definition together with matrix exponentiation is constructive and we show this explicitly.

Example 2.43. Consider the unitary group,

$$U(n) = \{M \in GL_n(\mathbb{C}) : MM^\dagger = \mathbf{1}\}.$$

Let $\gamma(t) = e^{tX}$ for some $X \in Mat_n(\mathbb{C})$ so $\gamma(0) = \mathbf{1}$. This choice of γ will work for any complex coefficient matrix, with $X \in Mat_n(\mathbb{R})$ to be chosen for real coefficient matrices.

We impose now the group condition,

$$(e^{tX})(e^{tX})^\dagger = \mathbf{1}.$$

Recalling that $\exp(tX)^\dagger = \exp(tX^\dagger)$ and taking the derivative at $t = 0$,

$$\left. \frac{d}{dt}(e^{tX})(e^{tX})^\dagger \right|_{t=0} = X e^{tX} e^{tX^\dagger} \Big|_{t=0} + e^{tX} X^\dagger e^{tX^\dagger} \Big|_{t=0} = X + X^\dagger = \left. \frac{d}{dt} \mathbf{1} \right|_{t=0} = 0.$$

This is the only condition on $U(n)$ so we found that the matrices comprising the Lie algebra of $U(n)$ are those such that

$$X = -X^\dagger,$$

these are anti-Hermitian matrices. The special case of $n = 1$ has $X \in Mat_1(\mathbb{C}) \cong \mathbb{C}$ so

$$X = -\overline{X} \implies X \in i\mathbb{R},$$

are purely imaginary numbers. This is a Lie algebra that will be important in Chapter 3 and when discussing the Lie algebra of a specific group, say $U(1)$, one writes $\mathfrak{u}(1)$. ∇

For completeness, we calculate the Lie algebras $\mathfrak{su}(n)$.

Example 2.44. Consider the group

$$SU(n) = \{M \in Mat_n(\mathbb{C}) : MM^\dagger = \mathbb{1}, \quad \det(M) = 1\}.$$

By choosing $\gamma(t) = e^{tX}$ again and imposing the unitary condition $MM^\dagger = \mathbb{1}$, we find again anti-Hermitian matrices by Ex. 2.43, however we must also impose the determinant condition. To do this we use Jacobi's formula,

$$\ln(\det(M)) = Tr(\ln(M)).$$

Imposing $M = \exp(tX)$ and $\det(M) = 1$ as the group condition, this becomes

$$\ln(\det(M)) = \ln(1) = 0 = Tr(\ln(e^{tX})) = Tr(tX) = tTr(X).$$

This must hold true for all t , thus we must have that $Tr(X) = 0$. The Lie algebra $\mathfrak{su}(n)$ is the space of $n \times n$ -trace free, anti-Hermitian matrices. ∇

The Lie algebras in Ex. 2.44 are important in physics especially in the cases $n = 2$ and $n = 3$. These Lie algebras admit bases, $\mathfrak{su}(2)$ has the Pauli matrices as a basis and $\mathfrak{su}(3)$ has the Gell-Mann λ -matrices as a basis. Together with $U(1)$, these three Lie groups play fundamental roles within particle physics and the modern understanding of the standard model.

3 Bundles, Connections and Gauge Theory

This chapter aims to introduce the reader to the mathematics and motivations underpinning bundle theory and the additional structures on bundles such as connections. Most definitions presented here are from [8] and [21], both of which will provide the interested reader an even more general overview of the subject surrounding gauge theory and mathematical physics as a whole. The general framework is illustrated through the simplest examples before we present the reader more advanced examples in Chapter 4.

3.1 Bundle Theory

After having studied manifolds, the next simplest idea one can conceive is that of two manifolds and a map between them. This simple, yet fundamental idea gives rise to many different structures and provides a rich framework for geometry. This idea is the very basis of bundle theory and we present its foundations here.

Definition 3.1. A bundle is a collection (E, B, π) of two smooth manifolds, E and B , and a smooth surjective map,

$$\pi : E \rightarrow B.$$

E is called the total space, B is called the base space and π is called the projection map.

When studying bundles with physics in mind, the base space B is often viewed as spacetime.

Since the projection map need only be surjective, it is only natural to ask about the preimage of this map. This turns out to be a very special question to ask and gives rise to the concept of a *fibre*.

Definition 3.2. The fibre over a point $x \in B$ is the preimage of the projection map π , written $F_x = \{\pi^{-1}(x)\}$.

Definition 3.3. Let M be a manifold and $U_\alpha \subseteq M$ open subsets of M . The collection $\mathcal{U} = \{U_\alpha\}$ is called an open cover if

$$M = \bigcup_{\alpha} U_{\alpha}.$$

Definition 3.4. A fibre bundle (E, B, π, F) is a bundle (E, B, π) together with a smooth manifold F such that $F_x \cong F$ for all $x \in B$. We require the fibre bundle to be locally trivialisable in the following sense: there exists an open cover $\{U_\alpha\}$ of B and diffeomorphisms

$$\psi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times F,$$

making the diagram

$$\begin{array}{ccc} \pi^{-1}(U_\alpha) & \xrightarrow{\psi_\alpha} & U_\alpha \times F \\ \pi \downarrow & \swarrow pr_1 & \\ U_\alpha & & \end{array}$$

commute, where $pr_1 : U_\alpha \times F \rightarrow U_\alpha$ is the projection onto the first factor. The tuple $(\{U_\alpha\}, \{\psi_\alpha\})$ is called a local trivialisation of the bundle.

A local trivialisation takes the form

$$\begin{aligned} \psi_\alpha : \pi^{-1}(U_\alpha) &\rightarrow U_\alpha \times F \\ p &\mapsto (\pi(p), \phi_\alpha(p)), \quad p \in \pi^{-1}(U_\alpha), \\ \phi_\alpha : \pi^{-1}(U_\alpha) &\rightarrow F. \end{aligned}$$

On overlaps $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \emptyset$, there are two functions ψ_α and ψ_β . One can consider the relationship between them and how much these differ on the overlaps. These are the transition functions.

Definition 3.5. Let (E, B, π, F) be a fibre bundle with a choice of local trivialisation $(\{U_\alpha\}, \{\psi_\alpha\})$. The corresponding transition functions are the maps

$$g_{\alpha\beta} : U_{\alpha\beta} \rightarrow \text{Aut}(F),$$

to the automorphism group of the fibre F , constructed by

$$g_{\alpha\beta}(x)(f) = \phi_\alpha(x) \circ \psi_\beta^{-1}(x, f).$$

These functions satisfy the cocycle conditions:

1. $g_{\alpha\alpha}(x) = id_F$,

$$2. \ g_{\alpha\beta}(x) = g_{\beta\alpha}^{-1}(x),$$

$$3. \ g_{\alpha\beta}(x)g_{\beta\gamma}(x) = g_{\alpha\gamma}(x), \forall x \in U_\alpha \cap U_\beta \cap U_\gamma = U_{\alpha\beta\gamma}.$$

Through transition functions, one may relate two functions of a local trivialisation on overlaps by

$$\psi_\alpha^{-1}(x, f) = \psi_\beta^{-1}(x, g_{\beta\alpha}(x)f). \quad (3.1)$$

In this way, one may imagine the transition functions as being the glue holding together the bundle when viewed locally.

Since it was natural to ask about a map between two manifolds and we chose a map from E to B , it is also just as natural to ask about a map from B to E as a complement to π . The consideration of such a map gives birth to the concept of a *section* of the fibre bundle.

Definition 3.6. Let (E, B, π, F) be a fibre bundle. A section of the fibre bundle is a smooth map,

$$s : B \rightarrow E,$$

satisfying $\pi \circ s(x) = x$, for all $x \in B$.

If a section is only defined locally it is called a local section,

$$s_\alpha : U_\alpha \rightarrow E.$$

The space of sections of a bundle is denoted $\Gamma^\infty(B, E)$.

Example 3.7. Let $E = B \times F$ for smooth manifolds B and F , then one has the projection map $pr_1 : B \times F \rightarrow B$, which is clearly surjective and smooth. One can use $U = B$ to build a trivialisation using the map $\psi = id$, the identity map.

$$\begin{array}{ccc} \pi^{-1}(U) = B \times F & \xrightarrow{id} & B \times F \\ pr_1 \downarrow & \swarrow pr_1 & \\ U = B & & \end{array}$$

▽

The simplest case of a fibre bundle is the trivial bundle above. This motivates the term *local trivialisation* since in the neighbourhood U_α , one has that the bundle can be locally written as a product space. In the case of the trivial bundle, we have a global trivialisation but this is far from the general case. Furthermore, this was only one choice of trivialisation and there are other choice one could make. Any map $B \rightarrow \text{Aut}(F)$ will give a trivialisation.

A fibre bundle as so far presented is very general; by adding additional structure to the fibre F , one may construct more sophisticated bundles such as vector bundles and principal bundles.

Definition 3.8. A vector bundle (E, B, π, V) is a fibre bundle such that each F_x carries the structure of a vector space and the local trivialisations $(\{U_\alpha\}, \{\psi_\alpha\})$ can be chosen such that for all $x \in U_\alpha \subseteq B$,

$$\psi_\alpha|_{\pi^{-1}(x)} : F_x \rightarrow \{x\} \times V \cong V,$$

is a linear isomorphism of vector spaces.

Remark 3.9. As a consequence of the local trivialisations being distinguished by the linear vector space isomorphism, the transition functions $g_{\alpha\beta}$ take values in $GL_n \subset \text{Aut}(V)$. Δ

Simple examples of vector bundles are when $V = \mathbb{C}$ or $\mathbb{R}$, know as complex or real line bundles. The transition functions of such vector bundles are $GL(1, \mathbb{C})$ or $GL(1, \mathbb{R})$, also known as $\mathbb{C} \setminus \{0\}$ and $\mathbb{R} \setminus \{0\}$. The concept of vector bundles may seem abstract, however they can produce very familiar structures.

Example 3.10. Let $(\mathbb{R}^{1,3} \times \mathbb{C}, \mathbb{R}^{1,3}, pr_1, \mathbb{C})$ be a complex line bundle. Let f be a smooth map

$$f : \mathbb{R}^{1,3} \rightarrow \mathbb{C},$$

such that it is square-integrable. Sections of this bundle are of the form

$$s(x) = (x, f(x)).$$

One may check that this section is compatible with the projection map,

$$pr_1 \circ s(x) = pr_1(x, f(x)) = x.$$

Therefore, this is a well-defined section of the bundle. Recalling that the base space is often viewed as spacetime, the map f can be identified as the wave function from quantum mechanics [29]. ∇

3.2 Principal Bundles

Throughout the rest of this chapter, we shall introduce the theory of principal bundles, connections, and curvature and present the simplest examples. In Chapter 4, this theory will be used again in a more interesting example. Within the scope of principal bundles, it is common to use P and M to denote the total and base spaces respectively, a convention we shall now adopt.

Definition 3.11. Let G be a Lie group. A principal G -bundle is a fibre bundle (P, M, π, F) with $F = G$ and a smooth right G -action on P , $R : P \times G \rightarrow P$, satisfying:

1. The group action preserves the fibres, i.e. $\pi(p \cdot g) = \pi(p)$.
2. The group action is free, $R_g p = p \cdot g = p \Leftrightarrow g = e$.
3. The group action is transitive on fibres, $\forall p, p' \in \pi^{-1}(x)$ with $x \in M$, there exists, due to freeness, a unique $g \in G$ such that $p' = p \cdot g$.
4. The bundle is locally trivialisable in a G -equivariant way, i.e. the following diagram commutes.

$$\begin{array}{ccc}
 \pi^{-1}(U_\alpha) \times G & \xrightarrow{R} & \pi^{-1}(U_\alpha) \\
 \psi_\alpha \times id_G \downarrow & & \downarrow \psi_\alpha \\
 U_\alpha \times G \times G & \xrightarrow{id_{U_\alpha} \times \mu} & U_\alpha \times G
 \end{array}$$

Remark 3.12. As a consequence of the transitive group action, fibres are generated by the orbit of a single element of the fibre. Δ

G -equivariance is best viewed on elements $(p, g) \in \pi^{-1}(U_\alpha) \times G$.

$$\begin{array}{ccc}
(p, g) & \xrightarrow{R} & (p \cdot g) \\
\downarrow \psi_\alpha \times id_G & & \downarrow \psi_\alpha \\
& & (\pi(p \cdot g), \phi_\alpha(p \cdot g)) \\
& & \downarrow \\
(\pi(p), \phi_\alpha(p), g) & \xrightarrow{id_{U_\alpha} \times \mu} & (\pi(p), \phi_\alpha(p) \cdot g)
\end{array}$$

Importantly, condition 1. of Def. 3.11 means that the group only operates within the fibres and condition 4. means that

$$\phi_\alpha(p \cdot g) = \phi_\alpha(p) \cdot g. \quad (3.2)$$

Remark 3.13. An important alternative definition of a principal bundle involves the base space M being a consequence of having a free and transitive group action on P . One instead defines a principal bundle as $\pi : P \rightarrow P/G$ and in fact there is an isomorphism $P/G \cong M$ [15, p. 59], which we will refer to later. $\triangle$

Example 3.14. Consider the collection $(\mathbb{S}^2 \times \mathbb{S}^1, \mathbb{S}^2, pr_1, U(1))$. We will show that this forms a principal $U(1)$ -bundle. Elements of $\mathbb{S}^2 \times \mathbb{S}^1$ can be written as $(x, e^{i\theta})$ and similarly, elements of $U(1)$ can be written as $e^{i\varphi}$ with $\theta, \varphi \in [0, 2\pi)$. Consider the right action of $U(1)$ on this manifold by

$$(x, e^{i\theta}) \cdot e^{i\varphi} = (x, e^{i(\theta+\varphi)}).$$

This action is free: Let $(x, e^{i\theta}) \cdot e^{i\varphi} = (x, e^{i\theta})$. We also have $(x, e^{i\theta}) \cdot e^{i\varphi} = (x, e^{i(\theta+\varphi)})$. We must have that $\theta + \varphi \equiv \theta \pmod{2\pi}$, then $\varphi \equiv 0 \pmod{2\pi}$. Since $\varphi \in [0, 2\pi)$, we have $\varphi = 0$, this corresponds to the $U(1)$ element $e^{i \cdot 0} = 1$, the identity element.

The action is transitive on fibres: Consider $p, p' \in \pi^{-1}(x)$ with $x \in \mathbb{S}^2$. Since the projection map is the first projection, its preimage is simply $\pi^{-1}(x) = pr_1^{-1}(x) = (x, g)$ with $g \in U(1)$. Therefore, p, p' have the forms $p = (x, g)$, $p' = (x, g')$. One can clearly see the relationship between these two points $p' = p \cdot (g^{-1}g')$. p and p' are arbitrary points therefore there is always $\tilde{g} = g^{-1}g'$ such that $p' = p \cdot \tilde{g}$.

Local trivialisations can be chosen: Since this collection is globally trivial, one may in fact choose a global trivialisation $\psi(p) = \psi(x, g) = (x, g)$, the identity map.

$$\begin{array}{ccc}
(x, g, g') & \xrightarrow{R} & (x, g \cdot g') \\
\downarrow \psi \times id_{U(1)} & & \downarrow \psi \\
& & (x, g \cdot g') \\
& & \downarrow \\
(x, g, g') & \xrightarrow{id_{\mathbb{S}^2} \times \mu} & (x, g \cdot g')
\end{array}$$

Since this is the trivial case, we expect that this diagram should commute as seen above; we also have no additional features such as equivariant maps arising from this commuting diagram. The collection $(\mathbb{S}^2 \times \mathbb{S}^1, \mathbb{S}^2, pr_1, U(1))$ does indeed constitute a principal $U(1)$ -bundle. ∇

3.2.1 Local Trivialisations of Principal Bundles

Given a principal G -bundle (P, M, π, G) and open cover $\{U_\alpha\}$, one can find a correspondence between local sections and local trivialisations. First, one notes that in a local trivialisation, G acts only on the G component. That is $\psi_\alpha(p \cdot g) = (\pi(p), \phi_\alpha(p) \cdot g)$. Since these maps are diffeomorphisms, there exist smooth inverses

$$\psi_\alpha^{-1} : U_\alpha \times G \rightarrow P|_{U_\alpha} = \pi^{-1}(U_\alpha),$$

whose images are in P and so G may act on the right. Taking an arbitrary point $p \in P$ and group element $g \in G$, we have as before that

$$\begin{aligned}
\psi_\alpha(p) &= (\pi(p), \phi_\alpha(p)), \\
\psi_\alpha(p \cdot g) &= (\pi(p), \phi_\alpha(p) \cdot g).
\end{aligned}$$

Using the inverse maps, we see that

$$\begin{aligned}
p &= \psi_\alpha^{-1}(\pi(p), \phi_\alpha(p)), \\
p \cdot g &= \psi_\alpha^{-1}(\pi(p), \phi_\alpha(p) \cdot g).
\end{aligned}$$

The local trivialisations for a principal bundle are such that they are compatible with the group action. One might instead consider first mapping to p , then applying the group action. This yields the equality

$$p \cdot g = \psi_\alpha^{-1}(\pi(p), \phi_\alpha(p) \cdot g) = \psi_\alpha^{-1}(\pi(p), \phi_\alpha(p)) \cdot g = p \cdot g.$$

Given this, one can play the same game considering $(x, g) \in U_\alpha \times G$.

$$\psi_\alpha^{-1}(x, g) = \psi_\alpha^{-1}(x, e \cdot g) = \psi_\alpha^{-1}(x, e) \cdot g =: s_\alpha(x) \cdot g. \quad (3.3)$$

Proposition 3.15. $s_\alpha(x) = \psi_\alpha^{-1}(x, e)$ defines a local section.

Proof. ψ_α is a diffeomorphism and therefore has a smooth inverse and so s_α is a smooth map. $s_\alpha(x) \in \pi^{-1}(x)$, therefore $\pi \circ s = id_M$. $\square$

Proposition 3.16. Local sections and local trivialisations are in a one-to-one correspondence.

Proof. We have already seen that, given a local trivialisation, one may define a local section. Conversely, given a local section $s : U \rightarrow P|_U$, let $p \in P|_U$, by the free and transitive properties of G on P , there exists a unique group element $\phi(p)$ such that $p = s(\pi(p)) \cdot \phi(p)$. One may then define a local trivialisation as $\psi(p) = (\pi(p), \phi(p)) \in U \times G$. $\square$

Given a principal bundle with $M = \bigcup_{\alpha \in \mathcal{I}} U_\alpha$, one may consider a point $x \in U_{\alpha\beta}$. A neighbourhood about this point can be written in a local trivialisation in two different ways, using either ψ_α or ψ_β . These two maps are related to each other through (3.1) and therefore correspond to canonical local sections, s_α and s_β , by $s_\alpha(x) = s_\beta(x) \cdot g_{\beta\alpha}(x)$. One may ask what happens under a change of local section by considering $\{\tilde{s}_\alpha\}$ with transition functions $\{\tilde{g}_{\alpha\beta}\}$. Because both sets of sections must be compatible with the same projection map, it must be that the images of the sections lie in the same fibre.

Due to the free and transitive properties of G 's action on P , there must exist a unique group element $h_\alpha(x) \in G$ with

$$h_\alpha : U_\alpha \rightarrow G,$$

such that $\tilde{s}_\alpha(x) = s_\alpha(x) \cdot h_\alpha(x)$. Using this relation one finds

$$\begin{aligned} s_\beta(x) \cdot h_\beta(x) &= \tilde{s}_\beta(x), \\ &= \tilde{s}_\alpha(x) \cdot \tilde{g}_{\alpha\beta}(x), \\ &= s_\alpha(x) \cdot h_\alpha(x) \cdot \tilde{g}_{\alpha\beta}(x), \\ &= s_\beta(x) \cdot g_{\beta\alpha}(x) \cdot h_\alpha(x) \cdot \tilde{g}_{\alpha\beta}(x). \end{aligned}$$

Since this holds for arbitrary sections s_β , one readily reads that

$$\tilde{g}_{\alpha\beta} = h_\alpha^{-1} \cdot g_{\alpha\beta} \cdot h_\beta. \quad (3.4)$$

3.2.2 Associated Bundles

Given a principal bundle and a *representation* of the group G on vector space V one may construct a vector bundle from this data. More concretely:

Definition 3.17. A representation of a group G on a finite dimension vector space V is a map

$$\varrho : G \rightarrow GL(V),$$

such that $\varrho(g \cdot h) = \varrho(g)\varrho(h)$, $\forall g, h \in G$.

Remark 3.18. One may equivalently use the left G -action notation

$$\varrho : G \times V \rightarrow V.$$

Both notations will be used in this dissertation. Δ

A useful example of a representation are the representations of $U(1)$ on $\mathbb{C}$.

Example 3.19. Given $n \in \mathbb{Z}$, one may define the following representation

$$\begin{aligned} \varrho_n : U(1) \times \mathbb{C} &\rightarrow \mathbb{C}, \\ (e^{i\varphi}, z) &\mapsto \varrho_n(e^{i\varphi}, z) = e^{in\varphi}z. \end{aligned}$$

∇

Definition 3.20. Let (P, M, π, G) be a principal G -bundle and V a vector space with a G -representation ϱ . The corresponding associated bundle $(P \times_{\varrho} V, M, \pi_V, V)$ is defined by $P \times_{\varrho} V := (P \times V) / \sim_G = P \times V / \sim$ such that $(p \cdot g, v) \sim (p, \varrho(g)v)$ and $\pi_V = \pi \circ pr_1$. This is a vector bundle with fibres isomorphic to V .

The new projection map is well defined due to the equivalence relation on the associated bundle. Consider

$$\pi_V(p \cdot g, \varrho(g^{-1})v) = \pi(p \cdot g) = \pi(p) = \pi_V(p, v).$$

Here we used

$$(p, u) \sim (p \cdot g, \varrho(g^{-1})u),$$

as an equivalent equivalence relation. One sees this by letting $u = \varrho(g)v$, then $v = \varrho(g^{-1})u$ and the result follows.

The fibres of this new bundle really are copies of V . One can always inspect this by checking the locally trivial structure of the bundle. That is, if the bundle can be written as a product of a subset of M and the fibre. Let $U \subseteq M$, then

$$(P \times_{\varrho} V)|_U \cong (U \times G) \times_{\varrho} V \stackrel{!}{\cong} U \times V.$$

The first isomorphism is clear since the principal G -bundle structure allows one to write P in a local trivialisation. The second is not so trivial and can be checked by considering a general point $((x, g), v) \in (U \times G) \times_{\varrho} V$. We have

$$((x, g), v) = ((x, e \cdot g), v) \sim ((x, e), \varrho(g)v) \cong (x, \varrho(g)v).$$

That is, due to the equivalence relation structure, the G component can always be removed locally, leaving only V as the new fibre.

Continuing with the trivial bundle, one may form its associated vector bundle with $\mathbb{C}$ as the vector space.

Example 3.21. Consider $((\mathbb{S}^2 \times \mathbb{S}^1) \times_{\varrho_n} \mathbb{C}, \mathbb{S}^2, \pi_V, U(1))$ with ϱ_n defined in Ex. 3.19. The projection map takes the form $\pi_V = pr_1 \circ pr_1^{\varrho_n}$, where $pr_1^{\varrho_n}$ maps onto the first component of the equivalence relation product. The structure of the equivalence relation is, with $x \in \mathbb{S}^2, e^{i\theta} \in \mathbb{S}^1, e^{i\varphi} \in U(1), w \in \mathbb{C}$,

$$((x, e^{i\theta} \cdot e^{i\varphi}), w) \sim ((x, e^{i\theta}), e^{in\varphi}w).$$

It is easy to see that the total space

$$(\mathbb{S}^2 \times \mathbb{S}^1) \times_{\varrho_n} \mathbb{C} \cong \mathbb{S}^2 \times \mathbb{C},$$

because

$$((z, 1 \cdot e^{i\theta}), w) \sim ((z, 1), e^{in\theta}w) \cong (z, e^{in\theta}w).$$

▽

3.3 Connections on Principal Bundles

An important structure that can be put onto principal bundles is that of a connection. Connections, as will be seen later, allow one to define concepts of curvature and covariant derivatives and form the basis for gauge theory. These concepts are important within modern physics from general relativity, to the standard model of particle physics, whether one wants either reference frame-invariant equations or gauge invariant actions.

3.3.1 Definitions

In the standard literature, there are two equivalent definitions of a connection. The more abstract definition is presented, however will not be used throughout this work. The second definition provides a concrete object with which to perform calculations and is, therefore, of more practical use. Connections allow one to separate each tangent space to P into two subspaces, the vertical and the horizontal subspaces which will now be demonstrated.

Definition 3.22. Let (P, M, π, G) be a principal G -bundle and let $p \in P$. The vertical subspace of $T_p P$ is the subspace $V_p P := \ker(\pi_* : T_p P \rightarrow T_{\pi(p)} M)$.

One might consider $V_p P$ to be the subspace of $T_p P$ that is tangent to G_x at p . In this way of thinking, $V_p P$ is constructed as follows:

Let $X \in \mathfrak{g}$ the Lie algebra and consider the right action of $\exp(\lambda X)$ on a point $p \in P$ parametrised by $\lambda \in \mathbb{R}$,

$$R_{\exp(\lambda X)} p = p \cdot \exp(\lambda X),$$

This is a curve through p , defined in P and since $\pi(p \cdot \exp(\lambda X)) = \pi(p)$, the curve lies in G_x as defined in Def. 3.22. We define a vector $\bar{X}$ by

$$\bar{X}(f) = \left. \frac{d}{d\lambda} f(p \cdot \exp(\lambda X)) \right|_{\lambda=0},$$

with $f : P \rightarrow \mathbb{R}$ an arbitrary smooth function. $\bar{X}$ is tangent to G_x at p so $\bar{X}$ is vertical. One may define a $\bar{X}$ at each point in P and so construct a vector field $\bar{X}$. This is called the *fundamental* vector field generated by X . The horizontal subspace $H_p P$ is a complement of $V_p P$. There are different choices for a horizontal subspace in contrast to the canonical choice for the vertical subspace. Making a choice is an extra datum, the connection.

Now we have a concept of horizontal and vertical subspaces, we can understand the concept of splitting the tangent space into these two subspaces through a connection.

Definition 3.23. A connection on a principal bundle (P, M, π, G) is a unique point-wise smooth decomposition of the tangent space $T_p P$ into the vertical subspace $V_p P$ and a horizontal subspace $H_p P$ such that

1. $T_p P = H_p P \oplus V_p P$.
2. $H_{p \cdot g} P = R_{g*} H_p P$ for arbitrary $p \in P$ and $g \in G$.

The decomposition is point-wise smooth in the following sense: a smooth vector field X on P is decomposed into smooth vector fields X^H and X^V as $X = X^H + X^V$. The second condition is that horizontal subspaces in the same fibre are related by the linear map pushforward R_{g*} so that all horizontal subspaces in the same fibre are generated by one. This is the geometric picture behind the connection and we will now move to consider a more tractable definition.

Definition 3.24. A connection form on a principal bundle (P, M, π, G) is a Lie algebra valued 1-form $\omega \in \Omega^1(P, \mathfrak{g})$ satisfying

1. $R_g^* \omega = \text{Ad}_{g^{-1}} \circ \omega, \forall g \in G$.
2. For any $X \in \mathfrak{g}$, let $\bar{X}$ be the corresponding fundamental vector field on P , then $\omega(\bar{X}) = X$.

Remark 3.25. In the first condition, the adjoint action is conjugation by g , $\text{Ad}_{g^{-1}} \circ \omega = g\omega g^{-1}$. In the case $G = U(1)$, which is a case of interest, G is an abelian group and the adjoint action is trivial, reducing condition 1. to $R_g^* \omega = \omega$. Δ

This definition is constructive as we shall see through the following example.

Example 3.26. Let $(\mathbb{S}^2 \times \mathbb{S}^1, \mathbb{S}^2, pr_1, U(1))$ be the trivial $U(1)$ -bundle over $\mathbb{S}^2$. Since the bundle is trivial, $U(1)$ only acts on the fibres. Let $p = (x, e^{i\varphi})$ be points in $\mathbb{S}^2 \times \mathbb{S}^1$. We start with the second condition and so must calculate the fundamental vector field

associated to a point. The Lie algebra of $U(1)$ is $\mathfrak{u}(1) = i\mathbb{R}$, so the parametrised action is $p \cdot e^{i\lambda} = (x, e^{i\lambda} e^{i\varphi})$, where the action as a matrix takes the form $\text{diag}(1, e^{i\lambda})$. Then, one has

$$\overline{X}(f) = \left. \frac{d}{d\lambda} f(p \cdot e^{i\lambda}) \right|_{\lambda=0} = \left. \frac{d}{d\lambda} (p^j (e^{i\lambda})_j^k) \right|_{\lambda=0} \frac{\partial f}{\partial p^k} = e^{i\varphi} \left. \frac{d}{d\lambda} e^{i\lambda} \right|_{\lambda=0} \frac{\partial f}{\partial (e^{i\varphi})} = \frac{\partial f}{\partial \varphi},$$

with summation notation used. Since we used $X = i$, we require a 1-form that acts as $\omega(\partial/\partial\varphi) = i$. The simplest choice that does this is $\omega = id\varphi$. We now need to check whether the first condition is satisfied, or if the form requires further restriction. Consider $g = e^{i\theta} \in U(1)$ and note that $d\varphi = -ie^{-i\varphi} de^{i\varphi}$, one has

$$R_g^* \omega = iR_g^*(-ie^{-i\varphi})d(R_g^* e^{i\varphi}) = i(-ie^{-i\varphi - i\theta})de^{i\varphi + i\theta} = i(-ie^{-i(\varphi + \theta)})ie^{i(\varphi + \theta)}d\varphi = id\varphi.$$

Both conditions are satisfied and we have calculated a connection 1-form for the trivial bundle, $\omega = id\varphi$. \(\nabla\)

3.3.2 Curvature

With the concept and concrete example of a connection established, we can consider the concept of curvature. Since a bundle has two manifolds in its construction, it makes sense to consider curvature "upstairs" on the total space, as well as curvature "downstairs" on the base space. We present the description in [8].

Definition 3.27. Let (P, M, π, G) be a principal bundle with connection form ω . The curvature 2-form $\Omega \in \Omega^2(P, \mathfrak{g})$ is defined by

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega]. \tag{3.5}$$

Remark 3.28. Let $\eta, \varphi \in \Omega^1(P, \mathfrak{g})$, then $[\eta, \varphi](X, Y) := \frac{1}{2}[\eta(X), \varphi(Y)] - \frac{1}{2}[\eta(Y), \varphi(X)]$. For curvature, one has $[\omega, \omega](X, Y) = \omega(X)\omega(Y) - \omega(Y)\omega(X)$. When G is abelian, $[\omega, \omega] = 0$ and the curvature simplifies to $\Omega = d\omega$ and depends linearly on ω . \(\Delta\)

Using this definition of curvature, one can define curvature locally on the base space. Let $\{U_\alpha\}$ be an open cover of M and let $s_\alpha : U_\alpha \rightarrow P|_{U_\alpha}$ be local sections with transition functions $\{g_{\alpha\beta}\}$ such that $s_\beta = s_\alpha \cdot g_{\alpha\beta}$. Define $A_\alpha := s_\alpha^* \omega$ and $\mathcal{F}_\alpha := s_\alpha^* \Omega$, then locally (3.27) reads

$$\mathcal{F}_\alpha = s_\alpha^* \Omega = s_\alpha^* \left(d\omega + \frac{1}{2}[\omega, \omega] \right) = dA_\alpha + \frac{1}{2}[A_\alpha, A_\alpha],$$

so the local data also obeys the same equation! It is shown in [8, p. 52] that in general, the transformation for the local curvature obeys

$$\mathcal{F}_\beta = Ad_{g_{\alpha\beta}^{-1}} \circ \mathcal{F}_\alpha, \quad (3.6)$$

implying that in the abelian case, $\mathcal{F}_\alpha = \mathcal{F}_\beta$ on $U_{\alpha\beta}$ meaning that $\mathcal{F}_\alpha$ is in fact the restriction of a globally defined form $\mathcal{F} \in \Omega^2(M, \mathfrak{g})$. Given the general transformation behaviour of the local curvature, together with local sections, together they yield well-defined global sections of the associated bundle $P \times_{Ad} \mathfrak{g}$. With $X, Y \in T_x M$ and $x \in M$,

$$[s_\beta, \mathcal{F}_\beta(X, Y)] = [s_\alpha \cdot g_{\alpha\beta}, Ad_{g_{\alpha\beta}^{-1}} \circ \mathcal{F}_\alpha(X, Y)] \sim [s_\alpha, \mathcal{F}_\alpha(X, Y)].$$

We have that, for fixed vector fields X and Y , $[s_\alpha, \mathcal{F}_\alpha(X, Y)]$ is the restriction of a globally defined section of $P \times_{Ad} \mathfrak{g}$, so the 2-forms $[s_\alpha, \mathcal{F}_\alpha]$ defined by $[s_\alpha, \mathcal{F}_\alpha](X, Y) := [s_\alpha, \mathcal{F}_\alpha(X, Y)]$ are actually the restrictions of a globally defined 2-form $\mathcal{F} \in \Omega^2(M, P \times_{Ad} \mathfrak{g})$ to U_α . This result follows from Thm. 3.30 below with $V = \mathfrak{g}$ and the adjoint representation. To understand this theorem, one must first be introduced to two specific types of differential forms.

Definition 3.29. Let (P, M, π, G) be a principal bundle and let $\omega \in \Omega^k(P, V)$ be a k -form on P with values in some vector space V with ϱ a representation of G . A form ω is called:

1. Horizontal if $\omega(X_1, \dots, X_k) = 0$ if at least one X_i is vertical.
2. G -equivariant if $R_g^* \omega = \varrho(g^{-1}) \omega, \forall g \in G$.

Forms that are horizontal and G -equivariant are denoted $\omega \in \Omega_{hor}^k(P, V)^G$. Using these two definitions, one can now understand the following theorem.

Theorem 3.30. Let (P, M, π, G) be a principal bundle and V be a vector space with ϱ a representation of G . The vector space $\Omega^k(M, P \times_{\varrho} V)$ of k -forms on M with values in the associated bundle $P \times_{\varrho} V$ is isomorphic to the vector space $\Omega_{hor}^k(P, V)^G$.

Proof. Found in [4, pp. 86-87]. □

Corollary 3.31. There is a 1 : 1 correspondence between sections of associated bundles and G -equivariant functions.

$$\Gamma^\infty(P \times_{\varrho} V) \cong C^\infty(P, V)^G \quad (3.7)$$

Lemma 3.32. *When G is abelian, $P \times_{Ad} \mathfrak{g} \cong M \times \mathfrak{g}$.*

Proof. For an abelian group, the adjoint action is trivial so for $(p, X) \in P \times_{Ad} \mathfrak{g}$ on the associated bundle, we have $(p \cdot g, X) \sim (p, \varrho(g)X) = (p, X)$. G does not therefore act on V . Hence,

$$P \times_{Ad} \mathfrak{g} = (P \times \mathfrak{g})/_G \cong P/_G \times \mathfrak{g} \cong M \times \mathfrak{g},$$

by Remark 3.13. □

This will be of importance later. Formally, the space $\Omega_{hor}^k(P, \mathfrak{g})^G$ is isomorphic to $\Omega^k(M, M \times \mathfrak{g})$ after using Lemma 3.32, however most people will write $\Omega^k(M, M \times \mathfrak{g}) = \Omega^k(M, \mathfrak{g})$.

3.3.3 A Topological Invariant

The curvature form on M in fact gives rise to a topological invariant with an interesting physical interpretation.

Proposition 3.33. *Let $(P, \mathbb{S}^2, \pi, U(1))$ be a principal $U(1)$ -bundle over $\mathbb{S}^2$ with $\mathcal{F}$ the global curvature form on $\mathbb{S}^2$. Then,*

$$\frac{1}{2\pi i} \int_{\mathbb{S}^2} \mathcal{F} \in \mathbb{Z}. \quad (3.8)$$

Proof. We defer the proof until Chapter 5. □

Proposition 3.34. *The integral in Prop. 3.33 is independent of the connection chosen.*

Proof. Given two connections ω and ω_0 , one may take their difference to form a horizontal, G -equivariant form $\tilde{A} = \omega - \omega_0 \in \Omega_{hor}^1(P, \mathfrak{g})^G$. We then have $\omega = \omega_0 + \tilde{A}$. On P we have $\tilde{\mathcal{F}} = d\omega = d\omega_0 + d\tilde{A}$, as $U(1)$ is abelian. Since $A \in \Omega_{hor}^1(P, \mathfrak{g})^G \cong \Omega^1(\mathbb{S}^2, \mathfrak{g})$ by Thm. 3.30, $d\tilde{A}$ is isomorphic to a 2-form dA on $\mathbb{S}^2$. Thus on $\mathbb{S}^2$, by writing $\mathcal{F}(\omega)$ to show the connection explicitly

$$\mathcal{F}(\omega) = \mathcal{F}(\omega_0) + dA,$$

with $dA \in \Omega^2(\mathbb{S}^2, \mathfrak{g})$. Thus,

$$\begin{aligned}\int_{\mathbb{S}^2} \mathcal{F}(\omega) &= \int_{\mathbb{S}^2} \mathcal{F}(\omega_0) + \int_{\mathbb{S}^2} dA, \\ &= \int_{\mathbb{S}^2} \mathcal{F}(\omega_0) + \int_{\partial\mathbb{S}^2} A, \\ &= \int_{\mathbb{S}^2} \mathcal{F}(\omega_0).\end{aligned}$$

with the second equality holding by Stokes' theorem and the third since the boundary of $\mathbb{S}^2$ is the empty set and hence an integral thereover vanishes. Since the choice of ω_0 and ω is arbitrary, the integral of the curvature is independent of this choice. $\square$

Example 3.35. Consider the trivial principal $U(1)$ -bundle over $\mathbb{S}^2$ with connection $\omega = d\varphi$ from Ex. 3.26. Calculating the curvature form for this connection is $\Omega = d\omega + 1/2[\omega, \omega] = d\omega$ since $U(1)$ is abelian. Then $\Omega = dd\varphi = 0$ and so $\mathcal{F} = 0$ on $\mathbb{S}^2$. For this bundle, we have $1/2\pi i \int_{\mathbb{S}^2} \mathcal{F} = 0 \in \mathbb{Z}$. ∇

Example 3.36. Consider a spacetime with a splitting $M = \mathbb{R} \times \Sigma$, representing a splitting into time and space. Locally, the curvature two form may be written as $\mathcal{F} = F_{\mu\nu} dx^\mu \wedge dx^\nu$. The Faraday tensor from electromagnetism arises as a curvature form and, locally, one may write $\mathcal{F} = E \wedge dt + B$ where E is a 1-form and B is a 2-form with basis $dx \wedge dy$, $dy \wedge dz$, $dz \wedge dx$. Taking a pullback from M to Σ to consider only space kills off dt , giving $\mathcal{F} = B$. Taking a further pullback from Σ to $\mathbb{S}^2$ kills off any normal components of $\mathcal{F}$ to Σ , giving

$$\frac{1}{2\pi i} \int_{\mathbb{S}^2} \mathcal{F} = \frac{1}{2\pi i} \int_{\mathbb{S}^2} B.$$

This integral represents the magnetic flux through a closed surface around a point, which vanishes in standard electromagnetism. This integral need not vanish and when it does not vanish: magnetic monopoles arise! ∇

With this interpretation of bundles over $\mathbb{S}^2$, the trivial example has a vanishing integral and so represents the trivial case of no magnetic flux and therefore no magnetic charge.

3.3.4 Covariant Differentials

Associated to the connection are the covariant differentials of the associated bundle. Within physics, these are intimately related to the covariant derivative. In the physics perspective,

the covariant derivative is introduced to allow gauge transformations and make physics gauge invariant. We introduce the covariant differential from the geometric perspective.

The covariant differential is a map given by the horizontal projection of the exterior derivative of the form,

$$\begin{aligned} d_\omega : \Omega_{hor}^k(P, V)^G &\rightarrow \Omega_{hor}^{k+1}(P, V)^G, \\ d_\omega \alpha &= \Pi_{hor}^*(d\alpha), \alpha \in \Omega_{hor}^k(P, V)^G. \end{aligned}$$

Proposition 3.37. *Let (P, M, π, G) be a principal G -bundle with connection form ω . Then $d_\omega \alpha = d\alpha + \varrho_*(\omega) \wedge \alpha$, for all $\alpha \in \Omega_{hor}^k(P, V)^G$.*

We prove in the case $k = 0$ and refer to [4] for the general result.

Proof. Let $\alpha \in \Omega^0(P, V)^G$ so $R_g^* \alpha = \varrho(g^{-1})\alpha$. By the above map, we have $d_\omega \alpha = \Pi_{hor}^*(d\alpha)$ with Π_{hor}^* the pullback horizontal projection map. Let u be a vector field with decomposition $u = u_V + \Pi_{hor}(u)$ into vertical and horizontal components. Then

$$d_\omega \alpha(u) = \Pi_{hor}^*(d\alpha)(u) = d\alpha(\Pi_{hor}(u)) = d\alpha(u - u_V).$$

Expanding this out,

$$d_\omega \alpha(u) = d\alpha(u) - d\alpha(u_V) = d\alpha(u) - \left(\frac{\partial \alpha}{\partial x^i} dx^i \right) \left(u_V^j \frac{\partial}{\partial x^j} \right) = d\alpha(u) - u_V^i \frac{\partial}{\partial x^i} \alpha = d\alpha(u) - u_V(\alpha).$$

The vertical component of u may be calculated by considering the fundamental vector field associated to $\omega(u) \in \mathfrak{g}$. Denoting this vector field as $\overline{X}^{\omega(u)}$ and with f an arbitrary differentiable function, one has

$$u_V(f) = \overline{X}^{\omega(u)}(f) = \frac{d}{d\lambda} f(u \cdot e^{\lambda \omega(u)}) \Big|_{\lambda=0} = \frac{\partial f}{\partial x^i} \frac{d}{d\lambda} u^i \cdot e^{\lambda \omega(u)} \Big|_{\lambda=0}.$$

Considering with α instead of f ,

$$u_V(\alpha) = \frac{d}{d\lambda} u(\alpha) \cdot e^{\lambda \omega(u)} \Big|_{\lambda=0} = \frac{d}{d\lambda} R_{\exp \lambda \omega(u)}^* \alpha \Big|_{\lambda=0}.$$

However since α is G -equivariant, one may replace this pullback map by the inverse representation as follows,

$$u_V(\alpha) = \frac{d}{d\lambda} \varrho(e^{-\lambda \omega(u)}) \alpha \Big|_{\lambda=0} = -\varrho_*(\omega)(u)\alpha.$$

Combining all this, one has

$$d_\omega \alpha(u) = d\alpha(u) + \varrho_*(\omega)(u)\alpha.$$

Omitting the arbitrary vector field, one has

$$d_\omega \alpha = d\alpha + \varrho_*(\omega)\alpha.$$

□

For the exterior derivative, one has $d^2 = 0$, we shall now see this does not hold with the exterior covariant differential. Let $\alpha \in \Omega_{hor}^k(P, V)^G$,

$$\begin{aligned} d_\omega d_\omega \alpha &= d_\omega(d\alpha + \varrho_*(\omega) \wedge \alpha), \\ &= d^2\alpha + d(\varrho_*(\omega) \wedge \alpha) + \varrho_*(\omega) \wedge (d\alpha + \varrho_*(\omega) \wedge \alpha), \\ &= \varrho_*(d\omega) \wedge \alpha - \varrho_*(\omega) \wedge d\alpha + \varrho_*(\omega) \wedge d\alpha + \varrho_*(\omega) \wedge \varrho_*(\omega) \wedge \alpha, \\ &= \varrho_*(d\omega + \omega \wedge \omega) \wedge \alpha, \\ &= \varrho_*(d\omega + \frac{1}{2}[\omega, \omega]) \wedge \alpha, \\ &= \varrho_*(\Omega) \wedge \alpha. \end{aligned}$$

One may then write, by abuse of notation,

$$d_\omega^2 = \varrho_*(\Omega),$$

or simply

$$d_\omega^2 = \Omega.$$

The exterior covariant differential gives rise to the curvature!

3.3.5 Covariant Derivatives on Associated Bundles

Since the exterior covariant derivative takes horizontal, G -equivariant forms to other horizontal, G -equivariant forms, one may consider applying Thm. 3.30. One can concretely get an understanding in the cases $k = 0$ and $k = 1$.

Let (P, M, π, G) be a principal G -bundle and (V, ϱ) a vector space with G -representation. One has

$$d_\omega : C^\infty(P, V)^G \rightarrow \Omega_{hor}^1(P, V)^G,$$

and by Cor. 3.31, one can relate covariant derivatives of G -equivariant functions to the covariant derivatives of sections of the corresponding associated bundle, which we denote

$$\nabla^\omega : \Gamma^\infty(P \times_\varrho V) \rightarrow \Omega^1(M, P \times_\varrho V).$$

Example 3.38. Consider a principal $U(1)$ -bundle $(P, M, \pi, U(1))$ and vector space $\mathbb{C}$ with $U(1)$ -representation ϱ_n . Let $X \in \mathfrak{u}(1)$ and $v \in \mathbb{C}$. The pushforward of the representation is

$$\varrho_{n*}(X)(v) = \left. \frac{d}{d\lambda} \varrho_n(e^{\lambda X}, v) \right|_{\lambda=0}.$$

Explicitly,

$$\left. \frac{d}{d\lambda} \varrho_n(e^{i\lambda}, v) \right|_{\lambda=0} = \left. \frac{d}{d\lambda} v e^{in\lambda} \right|_{\lambda=0} = in v.$$

Let s be a $U(1)$ -equivariant function on P with values in V . The covariant derivative is therefore

$$d_\omega s = ds + in\omega s. \tag{3.9}$$

$U(1)$ -bundles represent electromagnetism by definition. Each choice of $n \in \mathbb{Z}$ is a different representation of $U(1)$ on $\mathbb{C}$ and we interpret n as the electric charge of the bundle. In operator notation, compare the quantum electrodynamics covariant derivative from particle physics,

$$d_\omega = d + in\omega,$$

$$D_\mu = \partial_\mu + iq\omega_\mu.$$

It is clear that these represent the same physical situation! In physics ω_μ is referred to as the gauge potential, here it is the connection. Physically, the connection represents the gauge field of a physical theory. ▽

Remark 3.39. Up to a sign, the electron has charge 1 and quarks have charges of either $1/3$ or $2/3$, due to the electron being discovered first. Since $n \in \mathbb{Z}$, the "unit" charge of the electron would actually be the case $n = 3$! △

3.4 Gauge Theory

Modern physical theories such as the standard model are based on the principle of gauge invariance, investigating the group of transformations that leave the underlying Lagrangian invariant. Geometrically, these are the automorphisms on the total space which leave the base space untouched.

Definition 3.40. A principal G -bundle automorphism is a G -equivariant diffeomorphism $f : P \rightarrow P$.

As a consequence of Remark 3.13, there exists a diffeomorphism $\bar{f}$ such that the following diagram commutes

$$\begin{array}{ccc} P & \xrightarrow{f} & P \\ \pi \downarrow & & \downarrow \pi \\ M & \xrightarrow{\bar{f}} & M \end{array}$$

Definition 3.41. A gauge transformation of a principal bundle (P, M, π, G) is a principal G -bundle automorphism f such that $\pi = \pi \circ f$, i.e. $\bar{f} = id_M$. We denote by $\mathcal{G}(P)$ the gauge group.

Proposition 3.42. *There is a 1 : 1 correspondence between Ad-equivariant $C^\infty(P, G)$ functions and gauge transformations.*

Proof. Let (P, M, π, G) be a principal bundle and f a gauge transformation. Then $f(p) = p \cdot g(p)$ for some unique $g(p)$, $g : P \rightarrow G$ since gauge transformations map within the same fibre and by the free and transitive group action on fibres. Consider then $f(p \cdot h)$ for $h \in G$. One may use first the equivariance of the map, then fibre-preserving structure and group properties such that

$$f(p \cdot h) = f(p) \cdot h = p \cdot g(p) \cdot h,$$

or simply the fibre-preserving structure and group properties such that

$$f(p \cdot h) = p \cdot h \cdot g(p \cdot h).$$

These must be equivalent and so

$$g(p) \cdot h = h \cdot g(p \cdot h),$$

or equivalently

$$g(p \cdot h) = h^{-1} \cdot g(p) \cdot h, \quad (3.10)$$

g must therefore be an Ad -equivariant, smooth function from P to G . $\square$

Example 3.43. For an abelian group G , any function $g : M \rightarrow G$ defines a gauge transformation by defining $f(p) := p \cdot g(\pi(p))$. Then $f(p \cdot g') = p \cdot g' \cdot g(\pi(p)) \stackrel{!}{=} p \cdot g(\pi(p)) \cdot g' = f(p) \cdot g'$, where the equality $\stackrel{!}{=}$ holds due to G being abelian. One may construct a gauge transformation in this way for any group if one restricts $g : M \rightarrow Z(G)$, where $Z(G) = \{h \in G : g \cdot h = h \cdot g, \forall g \in G\}$ is the centre of the group. This is the subgroup of elements of G which commute with all other elements. ∇

3.4.1 Gauge Transformations of Connections

Given a gauge transformation f of a principal bundle and a connection form ω on this bundle, one may transform the connection as $f^*(\omega)$, which is still a connection [8, pp. 69-70].

Proposition 3.44. *Let $f \in \mathcal{G}(P)$ be a gauge transformation and $\tilde{f}$ the corresponding Ad -equivariant function due to Prop. 3.42. Then we can express the gauge transformation of a connection as*

$$f^*(\omega) = \tilde{f}\omega\tilde{f}^{-1} + \tilde{f}d\tilde{f}^{-1}.$$

Proof. Proof found in [4, pp. 112-113] $\square$

3.4.2 Gauge Transformations in Associated Bundles

One may extend the notion of gauge transformations to the associated bundle by considering the simplest extension possible $f \rightarrow f \times id_V$. We shall see that this preserves the structure. Consider the following diagram.

$$\begin{array}{ccc} P \times_{\varrho} V & \xrightarrow{f \times id_V} & P \times_{\varrho} V \\ & \searrow \pi_V & \swarrow \pi_V \\ & M & \end{array}$$

Recall that the associated bundle has the equivalence class $[p, v] \sim [p \cdot g, \varrho(g^{-1})v]$ and observing the gauge transformation, $[p, v] \xrightarrow{f \times id_V} [f(p), v]$. This is well defined since one may

consider instead classes of the form $[p \cdot g, v] \sim [p, \varrho(g)v]$ such that $[p \cdot g, v] \xrightarrow{f \times id_V} [f(p \cdot g), v] \stackrel{!}{=} [f(p) \cdot g, v] \sim [f(p), \varrho(g)v]$ where $\stackrel{!}{=}$ holds by the G -equivariance of gauge transformations. However, one may instead start with $[p \cdot g, v] \sim [p, \varrho(g)v] \xrightarrow{f \times id_V} [f(p), \varrho(g)v]$. One obtains the same result whether one first uses the associated bundle structure or first the gauge transformation structure. This shows that gauge transformations preserve the structure of the associated bundle! In this construction, one may also consider gauge-transformed sections $\tilde{s} = (f \times id_V) \circ s$ due to the following diagram commuting.

$$\begin{array}{ccc}
 P \times_{\varrho} V & \xrightarrow{f \times id_V} & P \times_{\varrho} V \\
 \searrow \pi_V & & \swarrow \pi_V \\
 & M & \\
 \nearrow s & & \nwarrow \tilde{s}
 \end{array}$$

Example 3.45. Consider a $U(1)$ -associated bundle with vector space $\mathbb{C}$ and representation ϱ_n . One then finds $[p, v] \xrightarrow{f \times id} [f(p), v] = [p \cdot g(p), v] = [p, \varrho_n(g(p))v]$. Due to G -equivariance, $g(p)$ only really depends on $x = \pi(p)$ so then one could choose $[f(p), v] = [p, e^{in\Lambda(x)}v]$, when the logarithm exists, representing a local phase rotation by charge n . ∇

3.4.3 Gauge Invariant Physics

If one considers a bundle with metric g and an orientation on the base space, one may introduce the Hodge star map. This is a linear map whose action is defined through basis vectors of $\Omega^k(M)$.

Definition 3.46. Let M be an oriented manifold of dimension n with metric tensor g . The Hodge $*$ is a linear map,

$$* : \Omega^k(M) \rightarrow \Omega^{n-k}(M),$$

with action on basis vectors

$$* (dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_k}) = \frac{\sqrt{|g|}}{(n-k)!} \varepsilon^{\mu_1 \dots \mu_k}_{\nu_{k+1} \dots \nu_n} dx^{\nu_{k+1}} \wedge \cdots \wedge dx^{\nu_n} \quad (3.11)$$

The special case of this is when $k = 0$ and the action is on the basis vector 1. This is called the *volume element*, defined as

$$* 1 = \frac{\sqrt{|g|}}{n!} \varepsilon_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_n} = \sqrt{|g|} dx^1 \wedge \cdots \wedge dx^n =: vol_M. \quad (3.12)$$

One may now introduce the idea of a Lagrangian with the idea of a gauge invariant action in mind, which allows the coupling of matter fields and gauge fields.²

Example 3.47. Let $(P, M, \pi, U(1))$ be a principal $U(1)$ -bundle with vector space $\mathbb{C}$, representation ϱ_n and connection form ω . Due to Thm. 3.30, one can find forms on M with values in the corresponding associated bundle. In the case $k = 0$, these are sections of the associated bundle $\Phi \in \Omega^0(M, P \times_{\varrho_n} \mathbb{C})$. One may introduce a covariant derivative ∇^ω and therefore form a Klein-Gordon-like Lagrangian,

$$\mathcal{L} = \overline{\nabla^\omega \Phi} \wedge * \nabla^\omega \Phi + f(\overline{\Phi} \Phi) \cdot \text{vol}_M. \quad (3.13)$$

The sections Φ are interpreted as the matter fields in the theory and these are coupled to the gauge fields due to the covariant derivative since, by expansion, one introduces terms of the form $\omega \Phi$, showing direct interaction between the matter and gauge fields. f is an arbitrary function for the potential term which, when multiplied by the volume form, produces a top form in line with the kinetic term. This could be interpreted as a scalar electromagnetic theory due to its $U(1)$ -gauge invariance. ∇

One can also reproduce classical electromagnetism, not just a scalar electromagnetic theory in this language.

Example 3.48. Consider a principal $U(1)$ -bundle with the curvature form $\mathcal{F}$ on the base space M of dimension 4. Using just this data, one can produce the sourceless Maxwell equations by the variation of the following Maxwell action,

$$S_{Maxwell} = \int -\frac{1}{2} \mathcal{F} \wedge \star \mathcal{F}$$

$U(1)$ is abelian so we have $\mathcal{F} = dA$. We take the variation up to $O(\delta A)$,

$$\begin{aligned} \delta S[A] &= S[A + \delta A] - S[A], \\ &= -\frac{1}{2} \int dA \wedge \star dA + d\delta A \wedge \star dA + dA \wedge \star d\delta A - dA \wedge \star dA, \\ &= -\frac{1}{2} \int d\delta A \wedge \star dA + dA \wedge \star d\delta A, \\ &= -\frac{1}{2} \int d\delta A \wedge \star \mathcal{F} + \mathcal{F} \wedge \star d\delta A. \end{aligned}$$

²Gauge fields are the connections on principal bundles, when referenced in physics literature.

One can define an inner product on differential p -forms. For $\alpha, \beta \in \Omega^p(M)$,

$$(\alpha, \beta) = \int \alpha \wedge \star \beta = \int \beta \wedge \star \alpha = (\beta, \alpha),$$

which is symmetric as inner products must be. dA is a 2-form and since we are in dimension 4, we have that $\star \mathcal{F}$ is also a 2-form so we can apply this.

$$-\frac{1}{2} \int d\delta A \wedge \star \mathcal{F} + \mathcal{F} \wedge \star d\delta A = - \int d\delta A \wedge \mathcal{F}.$$

By integrating by parts with vanishing boundary conditions we have,

$$\delta S = \int \delta A \wedge d \star \mathcal{F} \stackrel{!}{=} 0$$

δA is arbitrary, therefore the equations of motion are satisfied when $d \star \mathcal{F} = 0$. This is usually written $\star d \star \mathcal{F} = 0$, which we can do since $\star 0 = 0$. This produces Gauss' law and Ampère's law. $\mathcal{F} = dA$, therefore it is a closed form, which produces the final two of Maxwell's equations. The four Maxwell equations without sources in this language reduce down to

$$\star d \star \mathcal{F} = 0,$$

$$d\mathcal{F} = 0.$$

▽

4 The Hopf Fibration

We will now consider the well-known non-trivial example of a principal bundle, the Hopf fibration, in order to illustrate the theory from Chapter 3. We will demonstrate that it is indeed a principal bundle, derive a connection and its (non-trivial) curvature form and see that it gives rise to a non-trivially charged magnetic monopole.

4.1 The Hopf Fibration as a Principal Bundle

The Hopf fibration is the collection $(\mathbb{S}^3, \mathbb{S}^2, \pi, U(1))$ where $\mathbb{S}^3$ is parametrised by two complex coordinates (z_1, z_2) such that $|z_1|^2 + |z_2|^2 = 1$. One may do this by imagining $\mathbb{S}^3$ being embedded in $\mathbb{R}^4 \cong \mathbb{C}^2$. The projection map is then defined as

$$\pi(z_1, z_2) = (|z_1|^2 - |z_2|^2, 2z_1\bar{z}_2) \in \mathbb{R} \times \mathbb{C}. \quad (4.1)$$

Observing the norm of these coordinates,

$$\begin{aligned} (|z_1|^2 - |z_2|^2)^2 + |2z_1\bar{z}_2|^2 &= |z_1|^4 + |z_2|^4 - 2|z_1|^2|z_2|^2 + 4|z_1|^2|z_2|^2, \\ &= |z_1|^4 + |z_2|^4 + 2|z_1|^2|z_2|^2, \\ &= (|z_1|^2 + |z_2|^2)^2, \\ &= 1. \end{aligned}$$

These are coordinates of the form (x, z) in $\mathbb{R} \times \mathbb{C} \cong \mathbb{R}^3$ satisfying $x^2 + |z|^2 = 1$, which defines a sphere in $\mathbb{R}^3$ - it maps onto $\mathbb{S}^2$. The $U(1)$ -action on $\mathbb{S}^3$ is defined as

$$(z_1, z_2) \cdot e^{i\varphi} = (z_1 e^{i\varphi}, z_2 e^{i\varphi}). \quad (4.2)$$

We need to check that $U(1)$ is indeed the structure group of this collection of data by inspecting the preimage of the now well-defined projection map.

Proposition 4.1. *Fibres of π are isomorphic to $U(1)$.*

Proof. One needs to check how points in the same fibre are related. Let $\pi(z_1, z_2) = \pi(w_1, w_2)$ such that $(|z_1|^2 - |z_2|^2, 2z_1\bar{z}_2) = (|w_1|^2 - |w_2|^2, 2w_1\bar{w}_2)$, then one observes that $z_i = w_i\alpha$ with $|\alpha|^2 = 1$ so then

$$\pi(z_1, z_2) = (|z_1|^2 - |z_2|^2, 2z_1\bar{z}_2) = (|\alpha|^2|w_1|^2 - |\alpha|^2|w_2|^2, 2|\alpha|^2w_1\bar{w}_2) = \pi(w_1, w_2).$$

Consider a point $(z_1, z_2) \in \mathbb{S}^3$ with $\pi(z_1, z_2) = (x, z) \in \mathbb{S}^2$, it has preimage

$$\pi^{-1}(x, z) = \{(w_1, w_2) \in \mathbb{S}^3 : (w_1, w_2) = (\alpha z_1, \alpha z_2), \text{ for } \alpha \in \mathbb{C}, |\alpha|^2 = 1\},$$

which is isomorphic to $U(1)$. □

Proposition 4.2. *The projection map is invariant under the group action.*

Proof. Let $g \in U(1)$ and $(z_1, z_2) \in \mathbb{S}^3$, then

$$\begin{aligned} \pi(z_1 \cdot g, z_2 \cdot g) &= (|z_1 g|^2 - |z_2 g|^2, 2z_1 g \bar{z}_2 \bar{g}), \\ &= (|z_1|^2 - |z_2|^2, 2z_1 \bar{z}_2), \\ &= \pi(z_1, z_2). \end{aligned}$$

□

The projection map is well-defined and has fibres isomorphic to the unit circle, we can now check that the group action is free, and transitive on these fibres.

Proposition 4.3. *The group action is free, and transitive on the fibres.*

Proof. Let $(z_1, z_2) \in \mathbb{S}^3$ be reference points such that $\pi(z_1, z_2) = (x, z)$. The fibre of π over (x, z) is

$$\pi^{-1}(x, z) = \{(w_1, w_2) \in \mathbb{S}^3 : \exists \alpha \in U(1), \text{ such that } (w_1, w_2) = (\alpha z_1, \alpha z_2)\}.$$

We note $(0, 0) \notin \mathbb{S}^3$. The group action is clearly free and by the omission of the origin, this group action is clearly transitive on fibres. □

Due to these propositions, the Hopf fibration is indeed a principal $U(1)$ -bundle.

4.2 Connection and Curvature

We can use Def. 3.24 to directly construct a connection 1-form as in Ex. 3.26. Consider a point on $\mathbb{S}^3$ in full by 4 coordinates $u = (z_1, z_2, \bar{z}_1, \bar{z}_2)$ with a group action $u \cdot e^{i\lambda}$ in matrix form given by

$$\begin{pmatrix} e^{i\lambda} & 0 & 0 & 0 \\ 0 & e^{i\lambda} & 0 & 0 \\ 0 & 0 & e^{-i\lambda} & 0 \\ 0 & 0 & 0 & e^{-i\lambda} \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ \bar{z}_1 \\ \bar{z}_2 \end{pmatrix}.$$

By using index notation for condition 2. in Def. 3.24 with $X = i \in \mathfrak{u}(1) = i\mathbb{R}$, we have, for f an arbitrary smooth function,

$$\overline{X}(f) = \frac{d}{d\lambda} (e^{i\lambda})_j^k u^j \Big|_{\lambda=0} \frac{\partial f}{\partial u^k} = \left(iz_1 \frac{\partial}{\partial z_1} + iz_2 \frac{\partial}{\partial z_2} - i\bar{z}_1 \frac{\partial}{\partial \bar{z}_1} - i\bar{z}_2 \frac{\partial}{\partial \bar{z}_2} \right) f.$$

One might therefore consider $\omega = \bar{z}_1 dz_1 + \bar{z}_2 dz_2$, then

$$\begin{aligned} \omega(\overline{X}) &= (\bar{z}_1 dz_1 + \bar{z}_2 dz_2) \left(iz_1 \frac{\partial}{\partial z_1} + iz_2 \frac{\partial}{\partial z_2} - i\bar{z}_1 \frac{\partial}{\partial \bar{z}_1} - i\bar{z}_2 \frac{\partial}{\partial \bar{z}_2} \right), \\ &= i(|z_1|^2 + |z_2|^2), \\ &= i. \end{aligned}$$

The second criterion is satisfied. Recall that since $U(1)$ is abelian, the first criterion simply reads $R_g^* \omega = \omega$.

$$\begin{aligned} R_g^* \omega &= \bar{z}_1 e^{-i\lambda} d(z_1 e^{i\lambda}) + \bar{z}_2 e^{-i\lambda} d(z_2 e^{i\lambda}), \\ &= e^{-i\lambda} e^{i\lambda} (\bar{z}_1 dz_1 + \bar{z}_2 dz_2), \\ &= \omega. \end{aligned}$$

Both criteria hold and ω is a valid connection 1-form!

From (3.27), the curvature 2-form on $\mathbb{S}^3$ can be easily calculated due to $U(1)$ being an abelian group.

$$\Omega = d\omega = d\bar{z}_1 dz_1 + d\bar{z}_2 dz_2. \quad (4.3)$$

The curvature form on $\mathbb{S}^2$ is now more involved to calculate. We shall derive a form to use in calculating the integral in (3.8). This integration can be simplified by removing the North pole from $\mathbb{S}^2$, giving $\mathbb{S}^2 \setminus \{N\}$, and then restricting the bundle as in the following diagram.

$$\begin{array}{ccc} \mathbb{S}^3 & \xrightarrow{\pi} & \mathbb{S}^2 \\ \uparrow & & \uparrow \\ \mathbb{S}^3|_{\pi^{-1}(\mathbb{S}^2 \setminus \{N\})} & \xrightarrow{\pi} & \mathbb{S}^2 \setminus \{N\} \\ & \searrow \pi' & \downarrow \gamma \\ & & \mathbb{C} \\ & \swarrow s & \end{array}$$

Instead of integrating over $\mathbb{S}^2$, we integrate over $\mathbb{S}^2 \setminus \{N\}$ by the exclusion of one point. This does not affect the value of the integral because a set of a single element has measure zero. One defines $\pi' = \gamma \circ \pi$ and the map γ represents stereographic projection from the north pole, mapping the sphere to the complex plane, which is then easy to integrate over. In terms of the coordinates (x, z) on $\mathbb{S}^2$, the stereographic projection is defined,

$$\gamma(x, z) = \frac{z}{1 - x}. \quad (4.4)$$

Through stereographic projection, we have defined

$$\pi'(z_1, z_2) = \frac{2z_1\bar{z}_2}{1 - |z_1|^2 + |z_2|^2}. \quad (4.5)$$

This is a new projection map from which we can calculate a section to calculate a curvature form.

Let $w = \pi'(z_1, z_2)$. A section here is a map $s : \mathbb{C} \rightarrow \mathbb{S}^3|_{\pi^{-1}(\mathbb{S}^2 \setminus \{N\})}$, $s(w) = (\alpha, \beta) \in \mathbb{S}^3|_{\pi^{-1}(\mathbb{S}^2 \setminus \{N\})}$ such that

1. $\pi'(s(w)) = w$.
2. $|\alpha|^2 + |\beta|^2 = 1$.

This first criterion requires

$$w = \pi'(\alpha, \beta) = \frac{2\alpha\bar{\beta}}{1 - |\alpha|^2 + |\beta|^2},$$

but since $|\beta|^2 = 1 - |\alpha|^2$,

$$w = \frac{2\alpha\bar{\beta}}{2|\beta|^2} = \frac{\alpha}{\beta} \implies \alpha = w\beta.$$

We must now enforce that $|\alpha|^2 + |\beta|^2 = 1$ so

$$|\alpha|^2 + |\beta|^2 = |w\beta|^2 + |\beta|^2 = |\beta|^2(|w|^2 + 1) = 1 \implies \beta = \frac{1}{\sqrt{1 + |w|^2}}.^3$$

This allows us to fully define our section as

$$s(w) = \frac{1}{\sqrt{1 + |w|^2}}(w, 1). \quad (4.6)$$

³This is up to a phase factor which we have chosen to set to unity.

Returning to $\Omega = d\bar{z}_1 dz_1 + d\bar{z}_2 dz_2$, we can form a local curvature form on $\mathbb{C} \cong \mathbb{S}^2 \setminus \{N\}$ by $\mathcal{F} = s^* \Omega$. The usual coordinate on $\mathbb{C}$ is z and by using that $s(z) \in \mathbb{S}^3|_{\pi^{-1}(\mathbb{S}^2 \setminus \{N\})}$, one may write

$$z_1 = z(1 + |z|^2)^{-\frac{1}{2}}, \quad z_2 = (1 + |z|^2)^{-\frac{1}{2}}.$$

By using these coordinates, one finds

$$\mathcal{F} = s^* \Omega = -\frac{dz d\bar{z}}{(1 + |z|^2)^2}, \quad (4.7)$$

with the full calculation in Appendix A. We integrate, introducing polar coordinates,

$$\begin{aligned} \frac{1}{2\pi i} \int_{\mathbb{S}^2} \mathcal{F} &= \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{-1}{(1 + |z|^2)^2} dz d\bar{z}, \\ &= \frac{1}{2\pi i} \int_0^\infty \int_0^{2\pi} \frac{2ri}{(1 + r^2)^2} d\varphi dr, \\ &= \int_0^\infty \frac{2r}{(1 + r^2)^2} dr, \\ &= 1. \end{aligned}$$

The result is, as by Prop. 3.33, an integer and represents a magnetic monopole of unit charge!

4.3 Gauge Transformations of the Hopf Fibration

We have already seen in Ex. 3.43 that any function $g : M \rightarrow U(1)$ defines a gauge transformation, taking the form $f(p) = p \cdot g(\pi(p))$. Taking coordinates (z_1, z_2) on $\mathbb{S}^3$ and $\pi(z_1, z_2) = (|z_1|^2 - |z_2|^2, 2z_1 \bar{z}_2) = (x, z)$ as usual, we apply this gauge transformation

$$\begin{array}{ccc} (z_1, z_2) & \xrightarrow{f} & (z_1 g(\pi(p)), z_2 g(\pi(p))) \\ \pi \downarrow & & \downarrow \pi \\ & & (|z_1|^2 |g(\pi(p))|^2 - |z_2|^2 |g(\pi(p))|^2, 2z_1 g(\pi(p)) \bar{z}_2 \overline{g(\pi(p))}) \\ \pi \downarrow & & \\ (|z_1|^2 - |z_2|^2, 2z_1 \bar{z}_2) & \xrightarrow{id} & = (|z_1|^2 - |z_2|^2, 2z_1 \bar{z}_2) \end{array}$$

Because whether we perform f or simply project down, we arrive at the same result, i.e. $\bar{f} = id$ can be chosen, this is a gauge transformation of the Hopf bundle. That is, one may perform local rotations in the total space without affecting the base space.

Given a connection form ω on the Hopf fibration, one can perform a gauge transformation on the connection. This simplifies because $U(1)$ is abelian and so the connection transforms as:

$$\omega \rightarrow \omega + g^{-1}dg.$$

5 The Cohomological Viewpoint

We use now this section to lay down foundations of bundles using local data through Čech cohomology to prove the quantisation of magnetic charge. Along the way, we will cover the basics of both the de Rham and Čech cohomologies and we will see the bundle reconstruction theorem.

5.1 De Rham Cohomology

De Rham cohomology deals with quantifying closed and exact differential forms in a systematic way by considering equivalence classes of *cohomologous* closed forms.

Definition 5.1. Let M be a manifold and d the exterior derivative, satisfying $d^2 = 0$, then

$$\Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \dots$$

defines the de Rham complex.

Remark 5.2. Sometimes each exterior derivative is given an index such that $d_i : \Omega^i(M) \rightarrow \Omega^{i+1}(M)$. This becomes useful when considering the image and kernel of this map. $\triangle$

In this language, one may define closed and exact forms in terms of these vector spaces.

Definition 5.3. The kernel of d is called the vector space of closed forms,

$$\Omega_{cl}^p(M) = Z^p(M) = \{\omega \in \Omega^p(M) | d\omega = 0\} = \ker d_p. \quad (5.1)$$

The image of d is called the vector space of exact forms,

$$B^p(M) = \{\omega \in \Omega^p(M) | \exists \tau \in \Omega^{p-1}(M) \text{ such that } \omega = d\tau\} = \text{Im } d_{p-1}. \quad (5.2)$$

One can see from this definition, combined with $d^2 = 0$, that exact forms are closed; they form a subspace. This lends itself to the idea of considering a quotient of the two spaces, called the p^{th} de Rham cohomology of M .

Definition 5.4. The p^{th} de Rham cohomology of a manifold M is the vector space of equivalence classes formed by the quotient of closed forms modulo exact forms,

$$H_{dR}^p(M) = Z^p(M) / B^p(M). \quad (5.3)$$

Elements of the de Rham cohomology are equivalence classes of elements which are *cohomologous*, i.e. closed forms which differ by an exact form,

$$H_{dR}^p(M) \ni [\omega] = \{\omega' | \exists \mu \in \Omega^{p-1}(M) \text{ such that } \omega = \omega' + d\mu\}.$$

Remark 5.5. If one finds that $Z^k(M) \cong B^k(M)$ for some k , then $H_{dR}^k(M) = 0$ and so $d\omega = 0 \implies \exists \mu \in \Omega^{k-1}(M)$ such that $\omega = d\mu$, that is that closed implies exact. $\triangle$

It would be useful to know in which cases closed implies exact, which brings us to Poincaré's lemma.

Theorem 5.6 (Poincaré's lemma.). *Let M be a manifold and $U \subset M$ a contractible open subset. Then $H_{dR}^k(U) = \{0\}$ for all $k \geq 1$. In other words, any closed k -form on U is exact [17].*

Any set that is homeomorphic to an open n -ball is contractible so in particular, if one covers, say, $\mathbb{S}^2$ by two open sets which meet at the equator, both of these open sets are homeomorphic to open discs (open 2-ball) and Poincaré's lemma applies.

5.2 Sheaves of Abelian Groups

Before introducing Čech cohomology, it is appropriate to prepare the relevant prerequisites. All smooth manifolds have an underlying topological space structure and hence all the following statements apply to smooth manifold. The statements refer to open subsets of topological spaces (or manifolds) and thus refer only to the topology of such a space. We follow closely the notes in Chapter 6 of [3] to present this primer.

Definition 5.7. Let X be a topological space. A presheaf $\mathcal{G}$ of sets on X is a rule which assigns to each open subset $U \subset X$ a set $\mathcal{G}(U)$ and to each inclusion $V \subset U$ a restriction map

$$\varrho_V^U : \mathcal{G}(U) \rightarrow \mathcal{G}(V),$$

such that $\varrho_U^U = id_{\mathcal{G}(U)}$ and for $W \subset V \subset U$,

$$\varrho_W^U = \varrho_W^V \circ \varrho_V^U.$$

Elements s of $\mathcal{G}(U)$ are called sections of $\mathcal{G}$ over U . One often denotes the restriction as $\varrho_V^U(s) = s|_V$ and if $W \subset V \subset U$, $s|_W = (s|_V)|_W$.

Definition 5.8. Let X be a topological space and let A be a set. The constant presheaf with value A is the presheaf that assigns the set A to each $U \subset X$, such that $\varrho_V^U = id_A$ for all restriction maps.

We move to more specific presheaves now, considering abelian presheaves. The goal is in proving Prop. 3.33, which is specific to $U(1)$ -bundles, with $U(1)$ being an abelian group.

Definition 5.9. Let X be a topological space. A presheaf of abelian groups on X is a presheaf $\mathcal{G}$ such that for each open subset $U \subset X$, the set $\mathcal{G}(U)$ is endowed with the structure of an abelian group and such that all restriction maps ϱ_V^U are homomorphisms of abelian groups in the sense that for $s, t \in \mathcal{G}(U)$,

$$\varrho_V^U(s + t) = \varrho_V^U(s) + \varrho_V^U(t).$$

Recall the notation $U_{\alpha_1 \dots \alpha_k} = \bigcap_{i=1}^k U_{\alpha_i}$.

Definition 5.10. Let X be a topological space, let $\mathcal{G}$ a presheaf of sets on X and let $U \subset X$ with open cover $\{U_\alpha\}$. We say that $\mathcal{G}$ is a sheaf if it satisfies the additional property. Given a collection of sections $\{s_\alpha \in \mathcal{G}(U_\alpha)\}$ such that for all α, β ,

$$s_\alpha|_{U_{\alpha\beta}} = s_\beta|_{U_{\alpha\beta}},$$

there exists a unique section $s \in \mathcal{G}(U)$ such that $s_\alpha = s|_{U_\alpha}$.

Definition 5.11. Let X be a topological space and $\mathcal{G}$ a presheaf of abelian groups on X . We say that $\mathcal{G}$ is a sheaf of abelian groups if the underlying presheaf of sets is a sheaf.

Example 5.12. Let M be a manifold with $U \subset M$ an open subset with open cover $\{U_\alpha\}$ and let G be an abelian Lie group. The sheaf of functions with values in an abelian Lie group G also known as the constant sheaf of an abelian Lie group G , denoted $\underline{G}$, is defined as

$$\mathcal{G}(U) = C^\infty(U, G).$$

This assigns to each open subset an abelian group with group structure defined by point-wise multiplication. Given sections $s_\alpha \in \mathcal{G}(U_\alpha)$ such that for all α and β we have $s_\alpha|_{U_{\alpha\beta}} = s_\beta|_{U_{\alpha\beta}}$, we define a global section in the following way. For each $x \in M$, there is at least one U_α such that $x \in U_\alpha$. Define $s(x) = s_\alpha(x)$, however the choice of s_α may not be unique on overlaps. Because of the gluing condition $s_\alpha|_{U_{\alpha\beta}} = s_\beta|_{U_{\alpha\beta}}$, this is consistent as these functions agree on overlaps. The smooth structure is clear since smoothness is a local property. For every $x \in U_\alpha$, there is an open neighbourhood about x within U_α and locally s looks like s_α , which is smooth. ∇

Remark 5.13. An interesting case of Ex. 5.12 is when G is a discrete group since any discrete group is a 0-dimensional Lie group. In this case, smooth is replaced by locally constant and we denote the sheaf by $\underline{G}^{dis}$. Such a sheaf will be important in the Čech-de Rham theorem later.

Example 5.12 also holds with the following exchanges:

$$\begin{aligned} \text{Manifold} &\leftrightarrow \text{Topological Space,} \\ \text{Lie Group} &\leftrightarrow \text{Topological Group,} \\ \text{Smooth} &\leftrightarrow \text{Continuous.} \end{aligned}$$

Δ

In the following section, we will also deal with sheaves of differential forms, called cotangent sheaves [12], and functions with values in an abelian group G .

5.3 Čech Cohomology

Čech cohomology looks at open covers of manifolds and sheaves of abelian groups. By taking more and more overlaps of these open subsets and considering how elements of these abelian sheaves look, one uncovers familiar equations and transformations arising naturally.

Definition 5.14. Let M be a manifold with open cover $\mathcal{U} = \{U_\alpha\}$. The open cover is called a good open cover if:

1. Each U_α is contractible.

2. Finite intersections $\bigcap_{\alpha} U_{\alpha}$ are either empty or contractible.

Definition 5.15. Let M be a manifold with a good open cover $\mathcal{U}$. The face maps d_i are defined by subset inclusion:

$$\bigsqcup_{\alpha} U_{\alpha} \begin{array}{c} \xleftarrow{d_0} \\ \xleftarrow{d_1} \end{array} \bigsqcup_{\alpha\beta} U_{\alpha\beta} \begin{array}{c} \xleftarrow{d_0} \\ \xleftarrow{d_1} \\ \xleftarrow{d_2} \end{array} \bigsqcup_{\alpha\beta\gamma} U_{\alpha\beta\gamma} \begin{array}{c} \xleftarrow{d_0} \\ \xleftarrow{d_1} \\ \xleftarrow{d_2} \\ \xleftarrow{d_3} \end{array} \dots \quad (5.4)$$

Combinatorially there are this many maps because $U_{\alpha\beta} \subset U_{\alpha}$ and $U_{\alpha\beta} \subset U_{\beta}$, therefore one may choose a map into either U_{α} or U_{β} , so two face maps arise. Similarly, one may map from $U_{\alpha\beta\gamma}$ to $U_{\alpha\beta}$, $U_{\beta\gamma}$, or $U_{\gamma\alpha}$ etc. The face maps may therefore be neatly written

$$d_j : \bigsqcup_{\alpha_1 \dots \alpha_k} U_{\alpha_1 \dots \alpha_k} \longrightarrow \bigsqcup_{\dots \alpha_{j-1} \alpha_{j+1} \dots} \bigcap_{i=1, i \neq j}^k U_{\alpha_i}. \quad (5.5)$$

One may consider local forms on these disjoint unions and use the pullback of the face maps,

$$\prod_{\alpha} \Omega^{\bullet}(U_{\alpha}) \begin{array}{c} \xrightarrow{d_0^*} \\ \xrightarrow{d_1^*} \end{array} \prod_{\alpha\beta} \Omega^{\bullet}(U_{\alpha\beta}) \begin{array}{c} \xrightarrow{d_0^*} \\ \xrightarrow{d_1^*} \\ \xrightarrow{d_2^*} \end{array} \prod_{\alpha\beta\gamma} \Omega^{\bullet}(U_{\alpha\beta\gamma}) \begin{array}{c} \xrightarrow{d_0^*} \\ \xrightarrow{d_1^*} \\ \xrightarrow{d_2^*} \\ \xrightarrow{d_3^*} \end{array} \dots \quad (5.6)$$

Because this deals with open sets of a manifold, this can be realised again in the language of sheaves. Let $U_{\alpha\beta} \subset U_{\alpha}$, one has the inclusion map

$$\iota_{U_{\alpha}}^{U_{\alpha\beta}} : U_{\alpha\beta} \rightarrow U_{\alpha}.$$

The face maps are these choices of inclusion map and by considering the sheaf of differential forms $\mathcal{G}$ and taking the pullback of these inclusion maps, one has

$$(\iota_{U_{\alpha}}^{U_{\alpha\beta}})^* : \mathcal{G}(U_{\alpha}) \rightarrow \mathcal{G}(U_{\alpha\beta}). \quad (5.7)$$

Then (5.7) is exactly (5.6) written in terms of sheaves.

Definition 5.16. Let M be a manifold with good open cover $\mathcal{U}$ and let $\mathcal{G}$ be a sheaf of abelian groups. The Čech complex is defined by

$$\check{C}^k(\mathcal{U}, \mathcal{G}) = \prod_{\alpha_1 \dots \alpha_k} \mathcal{G}(U_{\alpha_i}),$$

with differential

$$\delta : \check{C}^k(\mathcal{U}, \mathcal{G}) \rightarrow \check{C}^{k+1}(\mathcal{U}, \mathcal{G}), \quad (5.8)$$

defined as

$$\delta = \sum_{i=1}^{k+1} (-1)^i d_i^*.$$

Remark 5.17. As a result of the face maps satisfying the *simplicial identities*, one has that $\delta^2 = 0$ and this is proven in [11]. Δ

Remark 5.18. d_i^* should not be confused with the exterior derivative used in the de Rham cohomology. Δ

As in de Rham cohomology, one constructs the Čech cohomology by considering the quotient of the kernel of δ by its image.

Definition 5.19. Let M be a manifold with good open cover $\mathcal{U}$ and let $\mathcal{G}$ be a sheaf of abelian groups. The k^{th} Čech cohomology with respect to $\mathcal{U}$ with values in $\mathcal{G}$ is

$$\check{H}^k(\mathcal{U}, \mathcal{G}) = \frac{\ker \delta : \check{C}^k(\mathcal{U}, \mathcal{G}) \rightarrow \check{C}^{k+1}(\mathcal{U}, \mathcal{G})}{\operatorname{Im} \delta : \check{C}^{k-1}(\mathcal{U}, \mathcal{G}) \rightarrow \check{C}^k(\mathcal{U}, \mathcal{G})}. \quad (5.9)$$

We call elements $s \in \check{C}^k(\mathcal{U}, \mathcal{G})$ satisfying $\delta s = 0$ Čech k -cocycles.

5.3.1 Familiar Structures from Principal Bundle Theory

We will now take a more hands-on approach to Čech cohomology and directly compute using δ and see what recognisable results transpire.

Example 5.20. First we consider a collection of p -forms $\omega = \{\omega_\alpha \in \Omega^p(U_\alpha)\}$. The action of the Čech differential is as follows

$$(\delta\omega)_{\alpha\beta} = \omega_\beta|_{U_{\alpha\beta}} - \omega_\alpha|_{U_{\alpha\beta}}. \quad (5.10)$$

A Čech cocycle is the case when $(\delta\omega)_{\alpha\beta} = 0$, or equivalently when

$$\omega_\beta|_{U_{\alpha\beta}} = \omega_\alpha|_{U_{\alpha\beta}}.$$

When this happens, the collection $\{\omega_\alpha\}$ defines a globally-defined differential form. ∇

So global differential forms on a manifold are simply a special case of a Čech 0-cocycle. In fact, it is not just globally-defined differential forms that come to light in a Čech cohomological construction. By restricting Ex. 5.20 to 0-forms with values in $U(1)$ $C^\infty(U_\alpha, U(1))$, we have

$$\prod_\alpha C^\infty(U_\alpha, U(1)) \xrightarrow[\xrightarrow{d_1^*}]{d_0^*} \prod_{\alpha\beta} C^\infty(U_{\alpha\beta}, U(1)) \xrightarrow[\xrightarrow{d_2^*}]{d_1^*} \prod_{\alpha\beta\gamma} C^\infty(U_{\alpha\beta\gamma}, U(1)) \cdots \quad (5.11)$$

One may call elements of these spaces *circle-valued* functions.

Example 5.21. In a similar vein to considering a collection of forms, consider now a collection of circle-valued functions $\mathbf{g} = \{g_{\alpha\beta} \in C^\infty(U_{\alpha\beta}, U(1))\}$ on overlaps $U_{\alpha\beta}$. $U(1)$ is usually written in multiplicative notation and we therefore also write the differential multiplicatively

$$(\delta\mathbf{g})_{\alpha\beta\gamma} = g_{\beta\gamma}|_{U_{\alpha\beta\gamma}} \cdot g_{\alpha\gamma}^{-1}|_{U_{\alpha\beta\gamma}} \cdot g_{\alpha\beta}|_{U_{\alpha\beta\gamma}}. \quad (5.12)$$

A Čech 1-cocycle is now written

$$g_{\beta\gamma}|_{U_{\alpha\beta\gamma}} \cdot g_{\alpha\gamma}^{-1}|_{U_{\alpha\beta\gamma}} \cdot g_{\alpha\beta}|_{U_{\alpha\beta\gamma}} = 1. \quad (5.13)$$

One may recognise (5.13) from Def. 3.5 being the cocycle condition on the transition functions. One may thus identify transition functions as being Čech 1-cocycles, $\{g_{\alpha\beta}\} \in \check{C}^1(\mathcal{U}, \underline{U(1)})$, with $\underline{U(1)}$ representing $U(1)$ -valued functions as in Ex. 5.12. ∇

Example 5.22. A third familiar expression emerges through the consideration of a collection of circle-valued functions $\mathbf{h} = \{h_\alpha \in C^\infty(U_\alpha, U(1))\}$ so that

$$(\delta\mathbf{h})_{\alpha\beta} = h_\beta|_{U_{\alpha\beta}} \cdot h_\alpha^{-1}|_{U_{\alpha\beta}}.$$

Since the differential is now on the double-overlaps, one could consider a product,

$$(\mathbf{g}\delta\mathbf{h})_{\alpha\beta} = g_{\alpha\beta} \cdot h_\beta|_{U_{\alpha\beta}} \cdot h_\alpha^{-1}|_{U_{\alpha\beta}},$$

and since $U(1)$ is an abelian group, this can be rewritten in the more recognisable form

$$(\mathbf{g}\delta\mathbf{h})_{\alpha\beta} = h_\alpha^{-1}|_{U_{\alpha\beta}} \cdot g_{\alpha\beta} \cdot h_\beta|_{U_{\alpha\beta}} \stackrel{def}{=} \tilde{g}_{\alpha\beta}, \quad (5.14)$$

recovering the transformation of the transition functions in (3.4), which we can now call gauge transformations of the transition functions. ∇

Remark 5.23. Writing the product Ex. 5.22 using additive notation and naming it $\mathbf{f}$, we have

$$\mathbf{f} = \mathbf{g} + \delta \mathbf{h}.$$

Equivalently,

$$\mathbf{f} - \mathbf{g} = \delta \mathbf{h}.$$

$\mathbf{f}$ and $\mathbf{g}$ differ by a Čech cocycle $\delta \mathbf{h}$ and so are cohomologous. In multiplicative notation, $\mathbf{g}\delta \mathbf{h}$ is cohomologous to $\mathbf{g}$. $\triangle$

5.4 Application to Curvature

We now move to apply cohomology theory to the setting of principal bundles and consider the case for electromagnetism set out in [5]. By definition, the physical interpretation of $U(1)$ -bundles is electromagnetism and as seen in Prop. 3.33, there is a quantisation condition. It is this condition that we will investigate through cohomological methods and arguments. We first consider the short exact sequence,

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\iota} \underline{\mathbb{R}} \xrightarrow{e^{2\pi i \cdot}} \underline{U(1)} \longrightarrow 0, \quad (5.15)$$

where an underline represents smooth functions with values in $\mathbb{R}, \mathbb{Z}$ etc. This gives rise to the long exact sequence,

$$\begin{array}{ccccccc} & & & & \cdots & & \\ & & & & \downarrow & & \\ & & & & \downarrow & & \\ \rightarrow & \check{H}^1(\mathcal{U}, \underline{\mathbb{Z}}) & \longrightarrow & \check{H}^1(\mathcal{U}, \underline{\mathbb{R}}) & \longrightarrow & \check{H}^1(\mathcal{U}, \underline{U(1)}) & \\ & & & & & & \\ \rightarrow & \check{H}^2(\mathcal{U}, \underline{\mathbb{Z}}) & \longrightarrow & \check{H}^2(\mathcal{U}, \underline{\mathbb{R}}) & \longrightarrow & \check{H}^2(\mathcal{U}, \underline{U(1)}) & \\ & & & & & & \\ & & & & & & \\ \rightarrow & \cdots & & & & & \end{array} \quad (5.16)$$

By [5], $\check{H}^p(\mathcal{U}, \underline{\mathbb{R}}) = 0$ for $p > 0$, which induces the isomorphism

$$\check{H}^p(\mathcal{U}, \underline{U(1)}) \cong \check{H}^{p+1}(\mathcal{U}, \underline{\mathbb{Z}}). \quad (5.17)$$

This result is very pertinent when considering Prop. 3.33 since this is already relating $U(1)$ to the integers. In fact this is exactly the quantisation of magnetic charge. As a corollary we will prove Prop. 3.33 more directly.

5.4.1 Direct Calculation

Now, let us directly apply the Čech and de Rham cohomologies to the case of $\mathcal{F} \in \Omega_{Cl}^2(M)$ for some manifold M with good open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in \mathcal{I}}$. The interplay of the two cohomology theories is what we will observe to give rise to the desired result.

Given the good open cover, one may restrict $\mathcal{F}$ to these contractible open sets. By Thm. 5.6, Poincaré's lemma, there must exist some $A_\alpha \in \Omega^1(U_\alpha)$ such that

$$\mathcal{F}|_\alpha = dA_\alpha. \quad (5.18)$$

From this, one may consider the difference $(A_\beta - A_\alpha)|_{U_{\alpha\beta}}$ however, since $\mathcal{F}$ is well defined, we have that $d(A_\beta - A_\alpha)|_{U_{\alpha\beta}} = 0$. Therefore, by Poincaré's lemma, there must again exist some $\Lambda_{\alpha\beta} \in \Omega^0(U_{\alpha\beta})$ such that

$$(A_\beta - A_\alpha)|_{U_{\alpha\beta}} = d\Lambda_{\alpha\beta}. \quad (5.19)$$

Moving from the de Rham picture to the Čech picture, on these functions, we have

$$(\delta\Lambda)_{\alpha\beta\gamma} = \Lambda_{\beta\gamma}|_{U_{\alpha\beta\gamma}} - \Lambda_{\alpha\gamma}|_{U_{\alpha\beta\gamma}} + \Lambda_{\alpha\beta}|_{U_{\alpha\beta\gamma}}. \quad (5.20)$$

By applying the exterior derivative, one has

$$\begin{aligned} d(\delta\Lambda)_{\alpha\beta\gamma} &= d\Lambda_{\beta\gamma}|_{U_{\alpha\beta\gamma}} - d\Lambda_{\alpha\gamma}|_{U_{\alpha\beta\gamma}} + d\Lambda_{\alpha\beta}|_{U_{\alpha\beta\gamma}}, \\ &= (A_\gamma - A_\beta - A_\gamma + A_\alpha + A_\beta - A_\alpha)|_{U_{\alpha\beta\gamma}}, \\ &= 0. \end{aligned}$$

So by a third use of Poincaré's lemma, there must exist some real numbers $n_{\alpha\beta\gamma}$ satisfying

$$\Lambda_{\beta\gamma}|_{U_{\alpha\beta\gamma}} - \Lambda_{\alpha\gamma}|_{U_{\alpha\beta\gamma}} + \Lambda_{\alpha\beta}|_{U_{\alpha\beta\gamma}} = n_{\alpha\beta\gamma} \in \mathbb{R}. \quad (5.21)$$

Finally by taking the Čech differential, we arrive at

$$\begin{aligned}
(\delta n)_{\alpha\beta\gamma\delta} &= n_{\beta\gamma\delta}|_{U_{\alpha\beta\gamma\delta}} - n_{\alpha\gamma\delta}|_{U_{\alpha\beta\gamma\delta}} + n_{\alpha\beta\delta}|_{U_{\alpha\beta\gamma\delta}} - n_{\alpha\beta\gamma}|_{U_{\alpha\beta\gamma\delta}}, \\
&= (\Lambda_{\gamma\delta} - \Lambda_{\beta\delta} + \Lambda_{\beta\gamma} - \Lambda_{\gamma\delta} + \Lambda_{\alpha\delta} - \Lambda_{\alpha\gamma} \\
&\quad + \Lambda_{\beta\delta} - \Lambda_{\alpha\gamma} + \Lambda_{\alpha\beta} - \Lambda_{\beta\gamma} + \Lambda_{\alpha\gamma} - \Lambda_{\alpha\beta})|_{U_{\alpha\beta\gamma\delta}}, \\
&= 0,
\end{aligned}$$

a Čech 2-cocycle! This point of view started with the a closed 2-form and allowed us to identify it with a Čech 2-cocycle. This is indicative of a theorem within cohomology, the Čech-de Rham theorem.

Theorem 5.24. (*Čech-de Rham Theorem [24]*) *For a manifold M with good open cover $\mathcal{U}$, there is an isomorphism*

$$\check{H}^p(\mathcal{U}, \mathbb{R}^{dis}) \cong H_{dR}^p(M). \quad (5.22)$$

By Thm. 5.24, a relation between closed 2-forms and Čech 2-cocycles was always to be expected.

If we change tack slightly and start with the other set of data defining a principal bundle, the transition functions satisfying the cocycle conditions in Def. 3.5, or equivalently (5.13), considered as Čech 1-cocycles, we have as follows.

For a principal $U(1)$ -bundle, its transition functions are $U(1)$ -valued and may therefore be written as

$$g_{\alpha\beta} = e^{2\pi i \Lambda_{\alpha\beta}}, \quad (5.23)$$

for some real-valued functions $\Lambda_{\alpha\beta}$, since the logarithm exists on contractible open subsets. This lifts the cocycle condition to be the statement

$$e^{2\pi i (\Lambda_{\beta\gamma} - \Lambda_{\alpha\gamma} + \Lambda_{\alpha\beta})}|_{U_{\alpha\beta\gamma}} = 1,$$

which admits solutions of the form

$$(\Lambda_{\beta\gamma} - \Lambda_{\alpha\gamma} + \Lambda_{\alpha\beta})|_{U_{\alpha\beta\gamma}} = n_{\alpha\beta\gamma} \in \mathbb{Z}, \quad (5.24)$$

since $\exp(2\pi i \cdot n_{\alpha\beta\gamma}) = 1$. Again, one may compute $(\delta n)_{\alpha\beta\gamma\delta}$ and one will find that

$$(\delta n)_{\alpha\beta\gamma\delta} = 0, \quad (5.25)$$

a Čech 2-cocycle over the integers. This calculation shows that when one applies $U(1)$ gauge theory instead of simply differential forms, a discrete results appears which on the outset would have no reason to be discrete. By identifying $\check{C}^2(\mathcal{U}, \mathbb{R})$ with the differential form approach and $\check{C}^2(\mathcal{U}, \mathbb{Z})$ with the gauge theoretic approach one can consider the following short exact sequence in terms of constant presheaves,

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\iota} \mathbb{R} \xrightarrow{e^{2\pi i \cdot}} U(1) \longrightarrow 0. \quad (5.26)$$

This gives rise the long exact sequence,

$$\begin{array}{c} \cdots \\ \downarrow \\ \rightarrow \check{H}^1(\mathcal{U}, \mathbb{Z}) \longrightarrow \check{H}^1(\mathcal{U}, \mathbb{R}) \longrightarrow \check{H}^1(\mathcal{U}, U(1)) \\ \downarrow \\ \rightarrow \check{H}^2(\mathcal{U}, \mathbb{Z}) \longrightarrow \check{H}^2(\mathcal{U}, \mathbb{R}) \longrightarrow \check{H}^2(\mathcal{U}, U(1)) \\ \downarrow \\ \rightarrow \cdots \end{array} \quad (5.27)$$

Unlike the previous similar long exact sequence, $\check{H}^p(\mathcal{U}, \mathbb{R}) \neq 0$ therefore one may have a map

$$\check{C}^2(\mathcal{U}, \mathbb{Z}) \rightarrow \check{C}^2(\mathcal{U}, \mathbb{R}).$$

Such a map is clearly not surjective so $U(1)$ gauge theories do not comprise all closed forms: they form a subclass of all closed forms, giving rise to the quantisation of charge.

5.5 Proof of Prop. 3.33

By [27], curvature forms for $U(1)$ -bundles can be normalised in such a way that they have integral periods.⁴ Motivated by either [26] or that $1/2\pi i$ works for the two curvature integrals we performed, we choose $2\pi i$ as this normalising factor. We choose to write the normalised curvature form as $F = 1/2\pi i \mathcal{F}$ and so on $\mathbb{S}^2$, this is written $F \in \Omega_{Cl}^2(\mathbb{S}^2)_{\mathbb{Z}}$.

⁴The concept of integral period is pertinent to the third type of cohomology theory, singular cohomology, which we do not venture to explore. This cohomology relates to de Rham cohomology through the de Rham theorem. One considers p -simplices Δ^p where p is the degree of a differential form ω to integrate. Then taking any smooth map $\sigma : \Delta^p \rightarrow M$, a differential form has integral periods if $\int_{\Delta^p} \sigma^* \omega \in \mathbb{Z}$. In this particular case, we would have $\int_{\Delta^2} \sigma^* \mathcal{F} \in \mathbb{Z}$ for any $\sigma : \Delta^2 \rightarrow \mathbb{S}^2$.

Proof of Prop. 3.33. Given a principal $U(1)$ -bundle over $\mathbb{S}^2$ with open sets $\{U_+, U_-\}$ such that $U_+ \cap U_- = \text{equator}$, $U_+ \cup U_- = \mathbb{S}^2$ and the boundary of each is $\mathbb{S}^1$. Then we have

$$\begin{aligned}
\int_{\mathbb{S}^2} F &\stackrel{1.}{=} \int_{U_+} F|_{U_+} + \int_{U_-} F|_{U_-}, \\
&\stackrel{2.}{=} \int_{U_+} dA_+ + \int_{U_-} dA_-, \\
&\stackrel{3.}{=} \int_{\mathbb{S}^1} A_+ - A_-, \\
&\stackrel{4.}{=} \int_{\mathbb{S}^1} d \log g_{-+}, \\
&\stackrel{5.}{=} \int_{\mathbb{S}^1 \setminus \{0\}} d \log g_{-+}, \\
&\stackrel{6.}{=} \Lambda_{-+}(1) - \Lambda_{-+}(0).
\end{aligned}$$

Where the first equality holds by decomposing F into its local data. The second holds due to Poincaré's lemma. The third equality holds due to Stokes' theorem and because integration on the two open sets are in opposite orientations. The fourth holds by considering the Čech argument in (5.19). The fifth equality holds because $\{0\}$ is a set of measure zero in $\mathbb{S}^1$ and so the integral's value remains unchanged. The final equality holds by considering a parametrisation of a path along the circle by $\varphi \in [0, 1)$.

On the interval $\mathbb{S}^1 \setminus \{0\}$, one can find Λ_{-+} such that $g_{-+} = e^{2\pi i \Lambda_{-+}}$. This defines

$$\begin{aligned}
g_{-+} : \mathbb{S}^1 \setminus \{0\} &\rightarrow U(1), \\
\varphi &\mapsto e^{2\pi i \Lambda_{-+}(\varphi)}.
\end{aligned}$$

Since g_{-+} is a function on the circle, we want the translation $\varphi \rightarrow \varphi + 1$ to leave g_{-+} invariant. Such a map requires that

$$g_{-+}(\varphi + 1) = e^{2\pi i \Lambda_{-+}(\varphi+1)} \stackrel{!}{=} e^{2\pi i \Lambda_{-+}(\varphi)}.$$

This demands that

$$\Lambda_{-+}(\varphi + 1) - \Lambda_{-+}(\varphi) = n \in \mathbb{Z}. \quad (5.28)$$

This should hold for all $\varphi \in [0, 1]$ and in particular for $\varphi = 0$, leaving

$$\Lambda_{-+}(1) - \Lambda_{-+}(0) = n \in \mathbb{Z}. \quad (5.29)$$

That is to say

$$\frac{1}{2\pi i} \int_{\mathbb{S}^2} \mathcal{F} = \int_{\mathbb{S}^2} F = \Lambda_{-+}(1) - \Lambda_{-+}(0) = n \in \mathbb{Z}. \quad (5.30)$$

□

5.6 Bundle Reconstruction Theorem

We present briefly now the bundle reconstruction theorem. This theorem is very much of the same flavour as the rest of this chapter as it allows one to construct principal bundles given the data of an open cover and transition functions.

Theorem 5.25. *Let M be a manifold with $\{U_\alpha\}$ an open cover of M and let G be a Lie group. Given mappings*

$$g_{\alpha\beta} : U_{\alpha\beta} \rightarrow G,$$

satisfying

$$g_{\alpha\gamma} = g_{\alpha\beta} \cdot g_{\beta\gamma}, \text{ on } U_{\alpha\beta\gamma}, \quad (5.31)$$

one can construct a principal fibre bundle (P, M, π, G) with transition functions $g_{\alpha\beta}$.

The proof of this theorem is constructive and as such we will follow the proof outlined in [22, p. 52], uncovering the principal bundle.

From (5.31), one reads that $g_{\alpha\alpha}(x) = e_G \in G$ and also that $g_{\alpha\beta}(x) \cdot g_{\beta\alpha}(x) = e_G$, for all $x \in U_{\alpha\beta}$. For each index α , let

$$X_\alpha = U_\alpha \times G, \quad (5.32)$$

and let

$$X = \bigsqcup_{\alpha} X_\alpha = \{(\alpha, x, g) : \alpha \text{ an index, } x \in U_\alpha, g \in G\}. \quad (5.33)$$

Each X_α is a smooth manifold and as a disjoint union of X_α , X is naturally a smooth manifold. We introduce an equivalence relation $\sim$ on X by

$$(\alpha, x, g) \sim (\beta, x, g_{\alpha\beta}(x) \cdot g) \quad (5.34)$$

for $x \in U_{\alpha\beta}$. We then define the total space to be

$$P = X / \sim, \quad (5.35)$$

giving a space of equivalence classes denoted $[\alpha, x, g]$. The G -action is defined as one would expect,

$$(\alpha, x, g) \cdot h = (\alpha, x, g \cdot h),$$

and one can see that G acts freely on the right on P . The projection map is simply,

$$\pi : P \rightarrow M,$$

$$(\alpha, x, g) \mapsto x,$$

$$\pi = pr_2.$$

The projection map must be G -invariant, preserving the fibre structure. Let $u, v \in P$ and let $\pi(u) = \pi(v)$, then by representation $u = (\alpha, x, g)$, $v = (\beta, y, h)$ this means $\pi(u) = \pi(v)$ implies $x = y \in U_{\alpha\beta}$. By the equivalence relation, one may write $u = (\beta, x, g_{\alpha\beta}(x) \cdot g)$ and we see $\pi(u) = \pi(v)$ if and only if $u = v \cdot c$ for some $c \in G$. It is easy to verify that $c = h^{-1} \cdot g_{\alpha\beta}^{-1}(x) \cdot g$. Therefore $\pi(u \cdot c) = \pi(u)$ and the projection map is G -invariant, preserving the fibre structure. P also has the structure of a smooth manifold as argued in [22, p. 52]. Therefore (P, M, π, G) as defined above defined a smooth principal bundle. If one defines

$$\psi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times G, \tag{5.36}$$

by

$$\psi_\alpha([\alpha, x, g]) = (x, g),$$

then $g_{\alpha\beta}$ are the transition functions corresponding to $\{U_\alpha\}$.

Example 5.26. Let $G = U(1)$ and $M = \mathbb{S}^2$, covered by two open subsets $\{U_N, U_S\}$, defined

$$U_N = \{(\theta, \varphi) \in \mathbb{S}^2 : 0 \leq \theta < \pi/2 + \varepsilon, 0 \leq \varphi < 2\pi\},$$

$$U_S = \{(\theta, \varphi) \in \mathbb{S}^2 : \pi/2 - \varepsilon < \theta \leq \pi, 0 \leq \varphi < 2\pi\}.$$

Taking $\varepsilon \rightarrow 0^+$, the overlap U_{NS} tends to the unit circle $\mathbb{S}^1$. Transition functions on this overlap are simply

$$g_{NS} : U_{NS} \rightarrow U(1).$$

A choice of such transition functions would be

$$g_{NS}(\varphi) = e^{in\varphi}, \quad n \in \mathbb{Z}, \varphi \in [0, 2\pi).$$

We must also have that

$$g_{SN}(\varphi) = e^{-in\varphi}, \quad g_{NN}(\varphi) = 1.$$

We construct the space

$$X = (U_N \times U(1)) \bigsqcup (U_S \times U(1)),$$

and denote an equivalence relation $\sim$ on X as follows. Let $g \in U(1)$, $x \in U_{NS}$,

$$(N, x, g) \sim (S, x, ge^{in\varphi}),$$

writing equivalence classes as $[\alpha, x, g]$. The total space, depending on one's choice for n , is then,

$$P_n = \bigsqcup_{\alpha=N,S} U_\alpha \times U(1) / \sim.$$

The projection map is $\pi([\alpha, x, g]) = x$ and one may locally trivialise this bundle by

$$\psi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times U(1),$$

$$\psi_\alpha([\alpha, x, g]) = (x, g).$$

▽

6 Equivariant Bundles

The study of equivariant bundles is the study of principal bundles with additional structure, a finite group H acting on both the base and total spaces. We substantiate this.

Definition 6.1. An H -equivariant principal G -bundle is a principal G -bundle (P, M, π, G) with left H -actions $L : H \times P \rightarrow P$ and $\lambda : H \times M \rightarrow M$ such that the left and right actions commute and that the projection map is H -equivariant. That is, that the following diagram commutes

$$\begin{array}{ccccc}
 H \times P \times G & \xrightarrow{id_H \times R} & H \times P & \xrightarrow{id_H \times \pi} & H \times M \\
 L \times id_G \downarrow & & L \downarrow & & \downarrow \lambda \\
 P \times G & \xrightarrow{R} & P & \xrightarrow{\pi} & M
 \end{array}$$

One can verify, by choosing $(h, p, g) \in H \times P \times G$ and following the commutative diagram that:

- $(h \cdot p) \cdot g = h \cdot (p \cdot g)$, the two actions are compatible.
- $\pi(h \cdot p) = h \cdot \pi(p)$, π is H -equivariant.

In this chapter, we will be working with two equivariant bundle constructions. The first is a trivial bundle with $M = \mathbb{R}^n$, $G = U(1)$, $H = \mathbb{Z}_2$. The second is the $\mathbb{Z}_2$ -equivariant Hopf bundle. We now introduce the first in the simplest case of $M = \mathbb{R}$.

Example 6.2. Consider a principal $U(1)$ -bundle over $\mathbb{R}$. All principal bundles over $\mathbb{R}$ are trivial by [20, p. 222] so we only need consider $P = \mathbb{R} \times U(1)$ with projection map $\pi = pr_1$. The group $\mathbb{Z}_2$ we consider as the set of elements $\mathbb{Z}_2 = \{+1, -1\}$ with multiplication as the group operation. An obvious choice for the $\mathbb{Z}_2$ -action on P would be

$$(\pm 1) \cdot (x, e^{i\varphi}) = (\pm x, e^{i\varphi}).$$

This acts on the one $\mathbb{R}$ coordinate, with (-1) exchanging x and $-x$. This action does not touch the fibres at all, however, is this the only admissible action? Let's investigate.

We require that the actions commute since $h \cdot (p \cdot g) = (h \cdot p) \cdot g$ for equivariance, furthermore, since $(+1)$ is the identity element of this group, we only need concern ourselves

with the (-1) element. Looking at the fibres we have

$$(-1) \cdot e^{i\varphi} = ((-1) \cdot e_G) \cdot e^{i\varphi} =: e^{i\tilde{\varphi}} \cdot e^{i\varphi}.$$

A property of $\mathbb{Z}_2$ is that $(-1)^2 = +1$, so consider

$$e_G = (-1)^2 \cdot e_G = (-1) \cdot ((-1) \cdot e_G) = (-1) \cdot e^{i\tilde{\varphi}} = ((-1) \cdot e_G) \cdot e^{i\tilde{\varphi}} = (e^{i\tilde{\varphi}})^2.$$

This means we have only two possible candidates for $e^{i\tilde{\varphi}}$. The first $e^{i\tilde{\varphi}} \cdot e^{i\varphi} = e^{i\varphi}$ and the second $e^{i\tilde{\varphi}} \cdot e^{i\varphi} = -e^{i\varphi}$. We will consider each action as pertaining to characteristically different bundles. ∇

A natural question that arises when studying geometry is "when are two spaces equivalent", when are they isomorphic? An isomorphism, by name, should be an invertible map between the two spaces that preserves all structures. This motivates our definition.

Definition 6.3. Two H -equivariant principal G -bundles P and Q are called isomorphic if there exist smooth maps $\varphi : P \rightarrow Q$ and $\varphi^{-1} : Q \rightarrow P$ such that $\varphi \circ \varphi^{-1} = id_Q$, $\varphi^{-1} \circ \varphi = id_P$ and for all elements $h \in H$, $g \in G$ and points $p \in P$, $q \in Q$, we have

- $\varphi(h \cdot p) = h \cdot \varphi(p)$, $\forall h \in H$, $p \in P$.
- $\varphi(p \cdot g) = \varphi(p) \cdot g$, $\forall g \in G$, $p \in P$.
- $\varphi^{-1}(h \cdot q) = h \cdot \varphi^{-1}(q)$, $\forall h \in H$, $q \in Q$.
- $\varphi^{-1}(q \cdot g) = \varphi^{-1}(q) \cdot g$, $\forall g \in G$, $q \in Q$.

This definition in words translates to φ being a map from P to Q that is equivariant with respect to both H and G and as such, preserving the compatibility of both group actions.

With principal bundles in mind, one might question how connections in this regime might look.

Definition 6.4. An equivariant connection on an H -equivariant principal G -bundle is a $\mathfrak{g}$ -valued 1-form $\omega \in \Omega^1(P, \mathfrak{g})$ such that:

1. $R_g^* \omega = Ad_{g^{-1}} \circ \omega$, $\forall g \in G$.

2. $\omega(\overline{X}) = X$, for $X \in \mathfrak{g}$ and $\overline{X}$ its associated fundamental vector field.

3. $L_h^* \omega = \omega$, $\forall h \in H$.

Such a connection is a connection on the principal bundle which is invariant with respect to H [23]. We see this now in action.

Example 6.5. We consider $\mathbb{Z}_2$ -equivariant principal $U(1)$ -bundles over $\mathbb{R}^n$, defining the $\mathbb{Z}_2$ -action in analogue to bundles over $\mathbb{R}$. Let $\mathbf{x} \in \mathbb{R}^n$ and let $(\pm 1) \cdot \mathbf{x} = \pm \mathbf{x} := (x^1, \dots, \pm x^i, \dots, x^n)$, the $\mathbb{Z}_2$ -action affects only one coordinate. The total space is $\mathbb{R}^n \times U(1)$ and there are two choices of $\mathbb{Z}_2$ -action on the fibre as in Ex. 6.2.

Since $U(1)$ is abelian, a connection must be invariant with respect to both $\mathbb{Z}_2$ and $U(1)$ and must as usual, produce from a fundamental vector field, its Lie algebra element. The Lie algebra of $U(1)$ is $\mathfrak{u}(1) = i\mathbb{R}$. Take an arbitrary function f , then the fundamental vector field is defined as

$$\overline{X}(f) = \left. \frac{d}{d\lambda} f(p \cdot e^{\overline{X}\lambda}) \right|_{\lambda=0}.$$

As before, the $U(1)$ -action on the trivial bundles is only on the fibres. Then with $p = (\mathbf{x}, e^{i\varphi})$, we have

$$\overline{X}(f) = \left. \frac{d}{d\lambda} f(p \cdot e^{i\lambda}) \right|_{\lambda=0} = e^{i\varphi} \left. \frac{d}{d\lambda} e^{i\lambda} \right|_{\lambda=0} \frac{\partial}{\partial e^{i\varphi}} f = \frac{\partial}{\partial \varphi} f.$$

Given this fundamental vector field, one could choose $\omega = id\varphi := e^{-i\varphi} de^{i\varphi}$. Though this is not the only choice, one could equally extend this connection to $\omega = id\varphi + iA(\mathbf{x})$ with $A \in \Omega^1(\mathbb{R}^n)$. We have that

$$\omega \left(\frac{\partial}{\partial \varphi} \right) = i$$

as required. Now we check the invariance with respect to both groups.

$$\begin{aligned} R_{e^{i\theta}}^* (e^{-i\varphi} de^{i\varphi} + iA(\mathbf{x})) &= e^{-i(\theta+\varphi)} de^{i(\theta+\varphi)} + iA(\mathbf{x}), \\ &= e^{-i\varphi} e^{-i\theta} e^{i\theta} de^{i\varphi} + iA(\mathbf{x}), \\ &= e^{-i\varphi} de^{i\varphi} + iA(\mathbf{x}), \\ &= \omega. \end{aligned}$$

Only the action of the non-trivial element (-1) need be considered. Both actions on the fibres can be summed up by $\pm$ and we can consider both at the same time

$$(-1)^* (e^{-i\varphi} de^{i\varphi} + iA(\mathbf{x})) = \pm e^{-i\varphi} d(\pm e^{i\varphi}) + iA(-\mathbf{x}) = e^{-i\varphi} de^{i\varphi} + iA(-\mathbf{x}) \stackrel{!}{=} \omega.$$

We get the restriction on A that $A(-\mathbf{x}) = A(\mathbf{x})$. If this is satisfied, then we have confirmed that ω satisfies the three criteria for an equivariant connection on an H -equivariant principal G -bundle. ∇

6.1 Associated Bundles

As discussed in the non-equivariant case, the interesting structures are those corresponding to real life physics, the connections and the sections of associated bundles. We will now discuss the latter of the two. We saw in Ex. 3.47 that the sections represented matter fields in a scalar electromagnetic theory. The construction of an H -equivariant associated bundle is intuitive.

Definition 6.6. Let (P, M, π, G, H) be an H -equivariant principal G -bundle and V a vector space with ϱ a representation of G on V . The equivariant associated bundle is the bundle $P \times_{\varrho} V \xrightarrow{\pi_V} M$ such that $(p \cdot g, v) \sim (p, \varrho(g)v)$.

The total space $P \times_{\varrho} V$ carries its own H -action, defined as

$$h \cdot [(p, v)] = [(h \cdot p, v)].$$

We explore this now through example.

6.1.1 Trivial Associated Complex Line Bundle

Let $(\mathbb{R}^n \times U(1), \mathbb{R}^n, pr_1, U(1), \mathbb{Z}_2)$ be a $\mathbb{Z}_2$ -equivariant principal $U(1)$ -bundle over $\mathbb{R}^n$ with $\mathbb{Z}_2$ -action as in Ex. 6.5. This forms two bundles with the two distinct $\mathbb{Z}_2$ -actions:

$$\begin{aligned} \text{Trivial} : \pm \cdot (\mathbf{x}, e^{i\varphi}) &= (\pm \mathbf{x}, e^{i\varphi}), \\ \text{Non-trivial} : \pm \cdot (\mathbf{x}, e^{i\varphi}) &= (\pm \mathbf{x}, \pm e^{i\varphi}). \end{aligned}$$

From this data, along with $\mathbb{C}$ and the representation ϱ_q , we can form associated complex line bundles $P \times_{\varrho_q} \mathbb{C}$ for both actions. Since P is globally trivial, its associated bundle is too, as seen in Ex. 3.21. $(\mathbb{R}^n \times U(1)) \times_{\varrho_q} \mathbb{C} \cong \mathbb{R}^n \times \mathbb{C}$. We investigate now the non-trivial action as this is the most interesting case. Furthermore, we need only consider the action of (-1) too.

Consider a section of this associated bundle,

$$\begin{array}{ccc}
P \times_{\varrho_q} \mathbb{C} & \xleftarrow{\cong} & \mathbb{R}^n \times \mathbb{C} \\
\pi_V \downarrow & \nearrow pr_1 & \uparrow \\
\mathbb{R}^n & \xleftarrow{\sigma} &
\end{array}$$

Clearly, a section must take the form

$$\begin{aligned}
\sigma : \mathbb{R}^n &\rightarrow \mathbb{R}^n \times \mathbb{C} \\
\sigma : \mathbf{x} &\mapsto (\mathbf{x}, s(\mathbf{x})),
\end{aligned}$$

with $s(\mathbf{x}) \in C^\infty(\mathbb{R}^n, \mathbb{C})$ and sections are characterised wholly by this function.

We want to be able to consider the transformations of these sections under the $\mathbb{Z}_2$ -action since these should represent physical matter fields. One considers transformations as general bundle transformations of the form:

$$\begin{array}{ccc}
P \times_{\varrho_q} \mathbb{C} & \xleftarrow{(-1)^{-1}} & P \times_{\varrho_q} \mathbb{C} \\
(-1) \cdot \sigma \uparrow & & \uparrow \sigma \\
\mathbb{R}^n & \xrightarrow{(-1)} & \mathbb{R}^n
\end{array}$$

The motivation behind this diagram is that there is a $\mathbb{Z}_2$ -action on both the total and base spaces of the equivariant associated bundle and we want to consider the (-1) -action and the restrictions this compatibility places on sections. In this case, $(-1)^{-1} = (-1)$ and the $\mathbb{Z}_2$ -action translates on the space $(\mathbb{R}^n \times U(1)) \times_{\varrho_q} \mathbb{C}$ in the following way by noting all points on this space can be written $(\mathbf{x}, e_G, z)$,

$$(-1) \cdot (\mathbf{x}, e_G, z) = (-\mathbf{x}, -e_G, z) = (-\mathbf{x}, e^{i\pi}, z) \sim (-\mathbf{x}, e_G, e^{iq\pi} z) \cong (-\mathbf{x}, e^{iq\pi} z).$$

The inherited action on the new fibres is multiplication by $e^{iq\pi}$. We can now put this into our commutative diagram,

$$\begin{array}{ccc}
(+\mathbf{x}, e^{iq\pi} s(-\mathbf{x})) & \xleftarrow{(-1)^{-1}} & (-\mathbf{x}, s(-\mathbf{x})) \\
(-1) \cdot \sigma \uparrow & & \uparrow \sigma \\
\mathbf{x} & \xrightarrow{(-1)} & -\mathbf{x}
\end{array}$$

From this, one reads the transformation of sections as

$$(-1) \cdot \sigma(\mathbf{x}) = (+\mathbf{x}, e^{iq\pi} s(-\mathbf{x})) \stackrel{!}{=} \sigma(\mathbf{x}). \quad (6.1)$$

These sections are determined wholly by the complex-valued function, therefore we can treat the equivariance condition as

$$e^{iq\pi}s(-\mathbf{x}) = s(\mathbf{x}). \quad (6.2)$$

For sections, equivariance is the condition of being Ad -invariant - this arises since we can rewrite (6.1) as

$$(-1)^{-1} \circ \sigma \circ (-1)(\mathbf{x}) = \sigma(\mathbf{x}), \quad (6.3)$$

with the adjoint structure clearly showing. We have come to the following definition.

Definition 6.7. Equivariant sections of an H -equivariant associated bundle are Ad -invariant sections with respect to the H -action.

6.1.2 Boundary Conditions

Our choice of $\mathbb{Z}_2$ -action only affects one component in $\mathbb{R}^n$ so $\mathbf{x}$ must be split into normal and tangential parts to the $\mathbb{Z}_2$ -action, $\mathbf{x} = (\mathbf{x}_{\parallel}, x_{\perp})$ with action

$$(-1) \cdot (\mathbf{x}_{\parallel}, x_{\perp}) = (\mathbf{x}_{\parallel}, -x_{\perp}). \quad (6.4)$$

This is very much in keeping with intuition - consider the Cartesian plane, the $\mathbb{Z}_2$ -action is $(x, y) \rightarrow (-x, y)$, only affecting one coordinate.

Recall the equivariance condition of the section in terms of the defining complex-valued function s . We have,

$$(-1) \cdot s(\mathbf{x}_{\parallel}, x_{\perp}) = e^{iq\pi}s(\mathbf{x}_{\parallel}, -x_{\perp}) \stackrel{!}{=} s(\mathbf{x}_{\parallel}, x_{\perp}) \quad (6.5)$$

The boundary of this flip is $x_{\perp} = 0$, $(-1) \cdot s(\mathbf{x}_{\parallel}, 0) = e^{iq\pi}s(\mathbf{x}_{\parallel}, 0)$. If the phases disagree, then both sides must vanish, $s(\mathbf{x}_{\parallel}, 0) = 0$. More concretely, since $q \in \mathbb{Z}$, we have $s(\mathbf{x}_{\parallel}, 0)$ is either an odd or even function, depending on whether q is odd or even.

If q is odd, then $s(\mathbf{x}_{\parallel}, 0) = -s(\mathbf{x}_{\parallel}, 0)$ so it must be that $s(\mathbf{x}_{\parallel}, 0) = 0$, describing a Dirichlet boundary condition.

If q is even, then $s(\mathbf{x}_{\parallel}, 0)$ is an even function and we have,

$$\left. \frac{d}{dx_{\perp}} s(\mathbf{x}_{\parallel}, x_{\perp}) \right|_{x_{\perp}=0} = \left. \frac{d}{dx_{\perp}} s(\mathbf{x}_{\parallel}, -x_{\perp}) \right|_{x_{\perp}=0} = - \left. \frac{d}{dx_{\perp}} s(\mathbf{x}_{\parallel}, x_{\perp}) \right|_{x_{\perp}=0},$$

yielding

$$\left. \frac{d}{dx_{\perp}} s(\mathbf{x}_{\parallel}, x_{\perp}) \right|_{x_{\perp}=0} = 0, \quad (6.6)$$

a Neumann boundary condition.

This unexpected feature is that the characteristic behaviour of equivariant sections, physically representing charged scalar fields, is dependent on whether it is oddly or evenly charged. If physics were represented by such an equivariant associated bundle, this would have the implication that, say, up and down quarks would behave completely differently since an up quark would be the case $q = 2$ and a down quark when $q = -1$!

6.2 Equivariant Cohomology

We saw in the previous section that there were two $\mathbb{Z}_2$ -actions on the total space that we could consider. We found this by explicit calculation, but it turns out that there is a more sophisticated method of working this out through equivariant cohomology.

According to Brylinski in [6] Prop. A1, we have

Proposition 6.8. *The H -equivariant $U(1)$ -bundles over M are classified by the equivariant cohomology group $H_H^2(M, \mathbb{Z})$.*

We analyse the case $H = \mathbb{Z}_2$, $M = \mathbb{R}^n$ and will follow the method outlined by Tu in [28], however this first requires a brief encounter with *classifying spaces*.

6.2.1 Classifying Spaces

Cohomology is homotopy invariant, it only *cares* what a topological space is up to homotopy equivalence. Because of this, we can use combinatorial methods to probe equivariant cohomology.

We start with simplices, the building blocks of polyhedra.

Definition 6.9. Let $p_0, p_1, \dots, p_n$ be geometrically independent point in $\mathbb{R}^m$ with $n \leq m$. The n -simplex is

$$\Delta^n = \{x \in \mathbb{R}^n : x = \sum_{i=0}^n c_i p_i, c_i \geq 0, \sum_{i=0}^n c_i = 1\}.$$

Simplices are often written in terms of just their coordinates p_i ; the 0-simplex is written $\langle p_0 \rangle$, the 1-simplex written as $\langle p_0 p_1 \rangle$ etc. These form rather more familiar shapes: Δ^0 is the point; Δ^1 is the line segment; Δ^2 the triangle and Δ^3 the tetrahedron.

We now apply this to a topological space X by introducing X^n .

Definition 6.10. For a topological space X , X^n is the set of continuous maps from the n -simplex to X ,

$$X^n = \{\lambda : \Delta^n \rightarrow X\}.$$

From this definition, one can consider an infinite sequence of sets,

$$X^0 \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} X^1 \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} X^2 \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \\ \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} \dots \quad (6.7)$$

called a simplicial set. One may map topological spaces into simplicial sets, and it turns out that topological spaces modulo homotopy equivalence are equivalent to simplicial sets. This is however through geometric realisation, which I will not explore here. For further reading on this topic, see [25].

Taking this simplicial perspective, we will call BG the classifying space and make it a simplicial set associated with our group G ,

$$BG = \{\star\} \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} G \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} G^2 \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \\ \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} \dots \quad (6.8)$$

where $\{\star\}$ is a single point. There is also a total space denoted EG , forming a principal G -bundle $EG \xrightarrow{\pi} BG$ [15, p. 78].

As in Chapter 5, we call the maps between the spaces face maps. We can make these more explicit. Let $\{\star\} \in G$, the two maps $G \rightarrow \{\star\}$ are the same map, simply mapping all points in G to $\{\star\}$. Looking at $G \times G \rightarrow G$, we have three choices,

$$\begin{aligned} (g_1, g_2) &\rightarrow g_2, \\ (g_1, g_2) &\rightarrow g_1 \cdot g_2, \\ (g_1, g_2) &\rightarrow g_1. \end{aligned}$$

This pattern generalises to $G \times G \times G \rightarrow G \times G$ and higher. A subtle note is that since G is a group, it is associative and satisfies the simplicial identities, [25].

There are also the degeneracy maps,

$$\{\star\} \longrightarrow G \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} G^2 \begin{array}{c} \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} G^3 \begin{array}{c} \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} \cdots, \quad (6.9)$$

which are so called because, in the case of groups, one inserts the identity. Consider the two second maps, $G \rightarrow G \times G$,

$$g_1 \rightarrow (g_1, e_G),$$

$$g_1 \rightarrow (e_G, g_1).$$

One may use this perspective to study manifolds by replacing the manifold with a so-called simplicial cover via the Čech perspective simplicial set,

$$\check{M} = \bigsqcup_{\alpha} U_{\alpha} \begin{array}{c} \longleftarrow \\ \longleftarrow \end{array} \bigsqcup_{\alpha\beta} U_{\alpha\beta} \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} \bigsqcup_{\alpha\beta\gamma} U_{\alpha\beta\gamma} \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} \cdots, \quad (6.10)$$

$$BG = \{\star\} \begin{array}{c} \longleftarrow \\ \longleftarrow \end{array} G \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} G^2 \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} \cdots \quad (6.11)$$

One can map between the Čech simplicial cover and the classifying space. The map $\bigsqcup_{\alpha} U_{\alpha} \rightarrow \{\star\}$ is trivial so we look at the next one up,

$$\bigsqcup_{\alpha\beta} U_{\alpha\beta} \rightarrow G. \quad (6.12)$$

This map must be compatible both with the face and the degeneracy maps and it must also preserve the face and degeneracy maps. This yields functions one may label as $g_{\alpha\beta}$, satisfying the cocycle conditions - the transition functions!

6.2.2 Analysis of $\mathbb{Z}_2$ -Equivariant Principal $U(1)$ -Bundles over $\mathbb{R}^n$

Equivariant cohomology is related to classifying spaces; we are now going to explore the $M = \mathbb{R}^n$, $H = \mathbb{Z}_2$ case. From Brylinski [6] the space we need to calculate is $H_{\mathbb{Z}_2}^2(\mathbb{R}^n, \mathbb{Z})$. From a few observations, we can simplify this expression,

$$\begin{aligned} H_{\mathbb{Z}_2}^2(\mathbb{R}^n, \mathbb{Z}) &\stackrel{1.}{\cong} H_{\mathbb{Z}_2}^2(\{\star\}, \mathbb{Z}), \\ &\stackrel{2.}{\cong} H^2(E\mathbb{Z}_2/\mathbb{Z}_2, \mathbb{Z}), \\ &\stackrel{3.}{\cong} H^2(B\mathbb{Z}_2, \mathbb{Z}). \end{aligned}$$

1. $\mathbb{R}^n$ is a contractible space. Because cohomology is homotopy invariant, one may contract $\mathbb{R}^n$ to a point and consider this instead.
2. By [28], we study the space $M \times EG/G \cong \{\star\} \times E\mathbb{Z}_2/\mathbb{Z}_2 \cong E\mathbb{Z}_2/\mathbb{Z}_2$.
3. $EG/G \cong BG \implies E\mathbb{Z}_2/\mathbb{Z}_2 \cong B\mathbb{Z}_2$ by Remark 3.13.

By [1], we find that $B\mathbb{Z}_2 = \mathbb{RP}^\infty$, the real projective space of countable dimension. In order to calculate $H^2(B\mathbb{Z}_2, \mathbb{Z})$, we follow the following simple recipe: first find the homology groups of $\mathbb{RP}^\infty$, then relate this to $H^2(B\mathbb{Z}_2, \mathbb{Z})$ by the *universal coefficient theorem*.

The homology of $\mathbb{RP}^\infty$ by [2] is given by

$$H_p(\mathbb{RP}^\infty, \mathbb{Z}) = \begin{cases} \mathbb{Z}_2, & p = 1, 3, 5, \dots \\ 0, & p = 2, 4, 5 \dots \\ \mathbb{Z}, & p = 0 \end{cases} \quad (6.13)$$

Everything is now set up to use the universal coefficient theorem [13, p. 195]. The theorem is very general but we present it here tailored to our specific circumstances.

Theorem 6.11 (Universal Coefficient Theorem). *Let X be a topological space, then*

$$0 \rightarrow \text{Ext}(H_{n-1}(X, \mathbb{Z}), \mathbb{Z}) \rightarrow H^n(X, \mathbb{Z}) \rightarrow \text{Hom}(H_n(X, \mathbb{Z}), \mathbb{Z}) \rightarrow 0. \quad (6.14)$$

$\text{Hom}(A, B)$ is the space of homomorphisms from A to B .

We want to explore the case $X = \mathbb{RP}^\infty$ and $n = 2$. We then have by (6.13) that $H_1(\mathbb{RP}^\infty) = \mathbb{Z}_2$, moreover, $\text{Ext}(H_{n-1}(X, \mathbb{Z}), \mathbb{Z})$ is calculated in [13, p. 195] as $\text{Ext}(\mathbb{Z}_n, \mathbb{Z}) = \mathbb{Z}_n$, therefore $\text{Ext}(H_{n-1}(X, \mathbb{Z}), \mathbb{Z}) = \text{Ext}(H_1(\mathbb{RP}^\infty, \mathbb{Z}), \mathbb{Z}) = \text{Ext}(\mathbb{Z}_2, \mathbb{Z}) = \mathbb{Z}_2$. By (6.13), $H_2(\mathbb{RP}^\infty, \mathbb{Z}) = 0$ so $\text{Hom}(H_2(\mathbb{RP}^\infty, \mathbb{Z}), \mathbb{Z}) = \text{Hom}(0, \mathbb{Z}) = 0$. This simplifies our use of the universal coefficient theorem to

$$0 \rightarrow \mathbb{Z}_2 \rightarrow H^2(\mathbb{RP}^\infty, \mathbb{Z}) \rightarrow 0 \rightarrow 0.$$

Since this is a short exact sequence, it follows that

$$H^2(\mathbb{RP}^\infty, \mathbb{Z}) \cong \mathbb{Z}_2. \quad (6.15)$$

Therefore, we have classified the $\mathbb{Z}_2$ -equivariant principal $U(1)$ -bundles over $\mathbb{R}^n$ to be $\mathbb{Z}_2$, that is, there are only two of them - the two we previously found!

6.3 Equivariant Hopf Fibration

The Hopf fibration can also be made $\mathbb{Z}_2$ -equivariant by the addition of an appropriate left $\mathbb{Z}_2$ -action. An immediate thought would be to have a $\mathbb{Z}_2$ -action on $\mathbb{S}^2$ which acts by swapping the Northern and Southern hemispheres, stabilising the equator, however such an action is not compatible with the $U(1)$ -action on $\mathbb{S}^3$. Instead we will consider a suitable action on $\mathbb{S}^3$ and descend this action to $\mathbb{S}^2$. There are both left and right actions involved, as such we will write $\triangleright$ to denote the left actions and $\triangleleft$ to denote the right action to aid legibility and provide clarity. Consider the action of the non-trivial $\mathbb{Z}_2$ -element acting on $\mathbb{S}^3$ as follows,

$$(-1) \triangleright (z_1, z_2) = (\alpha z_2, \beta z_1). \quad (6.16)$$

For this action to be $\mathbb{Z}_2$, we require that $(-1)^2 = +1$, therefore,

$$(-1) \triangleright ((-1) \triangleright (z_1, z_2)) = (\alpha\beta z_1, \alpha\beta z_2) = (z_1, z_2),$$

leading to $\alpha\beta = 1$. Furthermore, this must be on $\mathbb{S}^3$, therefore,

$$|\beta z_2|^2 + |\alpha z_1|^2 = |\beta|^2 |z_2|^2 + |\alpha|^2 |z_1|^2 = 1 = |z_2|^2 + |z_1|^2.$$

Both α and β are constants and for this to hold, by comparing coefficients, we have $|\alpha|^2 = 1$ and $|\beta|^2 = 1$. Both are phases and, when combined with $\alpha\beta = 1$, must rotate in opposite directions. This is a family of group actions, parameterised by some angle ψ , which one might call $\mathbb{Z}_2^\psi$ to distinguish individual actions. We therefore have found the actions defined

$$\mathbb{Z}_2^\psi : \mathbb{Z}_2 \times \mathbb{S}^3 \rightarrow \mathbb{S}^3, \quad (6.17)$$

$$((-1), (z_1, z_2)) \mapsto (e^{i\psi} z_2, e^{-i\psi} z_1). \quad (6.18)$$

We check that this action is compatible with the $U(1)$ -action by checking on individual coordinates.

$$\begin{aligned} ((-1) \triangleright z_1) \triangleleft e^{i\varphi} &= (e^{i\psi} z_2) \triangleleft e^{i\varphi} = e^{i\psi} (z_2 \triangleleft e^{i\varphi}) = e^{i(\psi+\varphi)} z_2, \\ (-1) \triangleright (z_1 \triangleleft e^{i\varphi}) &= (-1) \triangleright (e^{i\varphi} z_1) = e^{i\varphi} ((-1) \triangleright z_1) = e^{i(\varphi+\psi)} z_2. \end{aligned}$$

One can verify this holds on z_2 as well, thus the two actions are compatible. These choices of $\mathbb{Z}_2$ -action descend very naturally onto $\mathbb{S}^2$ via the coordinates (x, z) defined by the projection map. On $\mathbb{S}^2$ this is:

$$\begin{aligned} (-1) \triangleright x &= (-1) \triangleright (|z_1|^2 - |z_2|^2) = |e^{i\psi} z_2|^2 - |e^{-i\psi} z_1|^2 = |z_2|^2 - |z_1|^2 = -x, \\ (-1) \triangleright z &= (-1) \triangleright (2z_1 \bar{z}_2) = 2e^{i\psi} z_2 e^{i\psi} \bar{z}_1 = e^{2i\psi} \bar{z}. \end{aligned}$$

This is a non-trivial family of $\mathbb{Z}_2$ -actions induced on $\mathbb{S}^2$. By introducing the coordinates $z = x^1 + ix^2$, $x = x^3$ one can see how this works in Cartesian space by embedding $\mathbb{S}^2$ in $\mathbb{R}^3$ via $(x^1)^2 + (x^2)^2 + (x^3)^2 = 1$. Then,

$$(-1) \triangleright (x^1, x^2, x^3) = (x^1 \cos 2\psi + x^2 \sin 2\psi, x^1 \sin 2\psi - x^2 \cos 2\psi, -x^3). \quad (6.19)$$

Geometrically this is a reflection along the x^3 -axis and a reflection and rotation in the (x^1, x^2) -plane. The easiest way to see this is to take an example. Consider the $\mathbb{Z}_2^0$ -action,

$$(-1) \triangleright (x^1, x^2, x^3) = (x^1, -x^2, -x^3),$$

under this $\mathbb{Z}_2$ -action, there are points that are stabilised, which must be of the form $(x^1, 0, 0)$. However there are only two such points, $(\pm 1, 0, 0)$, lying on the $Re(z)$ -axis. This action is simpler to visualise. Consider $\mathbb{S}^2$ embedded in $\mathbb{R}^3$ with a plane along $x^3 = 0$ and a plane along x^1 -axis, the $\mathbb{Z}_2$ -action is a reflection along both of these axes. Inspecting (6.19), one may see that $\psi = 0, \pi/2, \pi$, and $3\pi/2$ all give actions with stabilised points, either lying on the x^1 -axis or on the x^2 -axis, with the first and third actions giving the same stabilised points $(\pm 1, 0, 0)$ and second and fourth giving the same stabilised points $(0, \pm 1, 0)$. On $\mathbb{S}^2$ itself, these are all points on the equator, being aligned either on the $Re(z)$ -axis or the $Im(z)$ -axis respectively.

6.3.1 Algebraic Approach

A both interesting and useful perspective to take with the equivariant Hopf fibration is by using an algebraic representation as outlined in [7]. This treatment of the Hopf fibration has at its core the algebra of polynomial functions on $\mathbb{S}^3$.

Definition 6.12. The algebra of polynomial functions on $\mathbb{S}^3$ is denoted $\mathcal{A} = \mathcal{O}(\mathbb{S}^3)$ and is defined,

$$\mathcal{O}(\mathbb{S}^3) = \text{Polynomial}(z_1, z_2, \bar{z}_1, \bar{z}_2) / \langle z_1 \bar{z}_1 + z_2 \bar{z}_2 - 1 \rangle.$$

Plainly put, it is the algebra formed by equivalence classes of polynomials in $z_1, z_2, \bar{z}_1, \bar{z}_2$ as variables that differ by polynomial functions generated by $\langle z_1\bar{z}_1 + z_2\bar{z}_2 - 1 \rangle$.

Example 6.13. A trivial example of such equivalence classes would be

$$z_1^2\bar{z}_1 \equiv -z_1z_2\bar{z}_2 + z_1 \pmod{z_1\bar{z}_1 + z_2\bar{z}_2 - 1}.$$

This holds since one can take the difference of the left and right hand sides as

$$z_1^2\bar{z}_1 + z_1z_2\bar{z}_2 - z_1 = z_1(z_1\bar{z}_1 + z_2\bar{z}_2 - 1).$$

As in modular arithmetic, any polynomial of the form $p(z_1, z_2, \bar{z}_1, \bar{z}_2)(z_1\bar{z}_1 + z_2\bar{z}_2 - 1) \equiv 0$ in this algebra. ∇

This motivates the definition of equivalence classes.

Definition 6.14. Any $p, q \in \text{Polynomial}(z_1, z_2, \bar{z}_1, \bar{z}_2)$ are congruent in $\mathcal{A}$ if $p = q + r(z_1\bar{z}_1 + z_2\bar{z}_2 - 1)$ for any $r \in \text{Polynomial}(z_1, z_2, \bar{z}_1, \bar{z}_2)$.

The algebra $\mathcal{A}$ of polynomial functions on $\mathbb{S}^3$ has a $U(1)$ -action via the pullback.

Definition 6.15. The $U(1)$ -action on $\mathcal{A}$ is defined as follows with $\triangleright$ representing a left action,

$$\begin{aligned} U(1) \times \mathcal{A} &\rightarrow \mathcal{A}, & e^{i\varphi} \triangleright z_1 &= e^{i\varphi} z_1, & e^{i\varphi} \triangleright z_2 &= e^{i\varphi} z_2, \\ & & e^{i\varphi} \triangleright \bar{z}_1 &= e^{-i\varphi} \bar{z}_1, & e^{i\varphi} \triangleright \bar{z}_2 &= e^{-i\varphi} \bar{z}_2. \end{aligned}$$

One can view $\langle z_1, z_2, \bar{z}_1, \bar{z}_2 \rangle$ as a generating set of the algebra and because of this, the group action being defined on the generating set means the group action is defined on the whole algebra.

Proposition 6.16. *The $U(1)$ -action on $\mathcal{A}$ is a left action.*

Proof. Consider the following sequence of smooth functions between smooth manifolds M , N , and R and let $h \in C^\infty(R)$.

$$\begin{array}{ccccc} M & \xrightarrow{f} & N & \xrightarrow{g} & R \\ & \searrow & & \nearrow & \\ & & g \circ f & & \end{array}$$

By definition of the pullback, $g^*h = h \circ g$, this can be extended so then one has $f^*g^*h = (h \circ g) \circ f = h \circ (g \circ f) = (g \circ f)^*h$, so the pullback swaps the order of functions.

Right group actions satisfy that $R_{gg'} = R_{g'}R_g$, since when considering the Hopf fibration and a point $p \in \mathbb{S}^3$, $R_{g'}R_gp = R_{g'}(pg) = pgg' = R_{gg'}p$. If one considers the pullback action, to apply in the algebraic approach, one has $R_{gg'}^* = (R_{g'}R_g)^* = R_g^*R_{g'}^*$ since as we just observed, the pullback swaps the order. Because of this order swapping, the pullback action behaves just like a left action since a left action obeys $L_{gg'} = L_gL_{g'}$. Writing $R^* = L$, these two statements are the same and the pullback right action is thus a left action. $\square$

Example 6.17. Group action on $z_1^2 z_2 - z_2 \bar{z}_1$,

$$e^{i\varphi} \triangleright (z_1^2 z_2 - z_2 \bar{z}_1) = e^{i3\varphi} z_1^2 z_2 - z_2 \bar{z}_1. \quad (6.20)$$

∇

This is expected because the action is via pullback, so $e^{i\varphi} \triangleright h(z) = h(e^{i\varphi} z)$. Due to this left action, $\mathcal{A}$ has a natural and useful decomposition.

Proposition 6.18. *The algebra $\mathcal{A}$ admits decomposition into the $U(1)$ -representations encountered in Ex. 3.19,*

$$\mathcal{A} = \bigoplus_{n \in \mathbb{Z}} A^n, \quad A^n = \{a \in \mathcal{A} : e^{i\varphi} \triangleright a = e^{in\varphi} a, \forall e^{i\varphi} \in U(1)\}. \quad (6.21)$$

Proof. Let $p \in \text{Polynomial}(z_1, z_2, \bar{z}_1, \bar{z}_2)$ and $C_{\alpha\beta\gamma\delta} \in \mathbb{C}$, then one may write

$$p = \sum_{\alpha=0}^{n_1} \sum_{\beta=0}^{n_2} \sum_{\gamma=0}^{n_3} \sum_{\delta=0}^{n_4} C_{\alpha\beta\gamma\delta} z_1^\alpha z_2^\beta \bar{z}_1^\gamma \bar{z}_2^\delta.$$

$U(1)$ acts on p as

$$e^{i\varphi} \triangleright p = \sum_{\alpha=0}^{n_1} \sum_{\beta=0}^{n_2} \sum_{\gamma=0}^{n_3} \sum_{\delta=0}^{n_4} C_{\alpha\beta\gamma\delta} e^{i(\alpha+\beta-\gamma-\delta)\varphi} z_1^\alpha z_2^\beta \bar{z}_1^\gamma \bar{z}_2^\delta.$$

Let $a \in \mathcal{A}$, then it is in an equivalence class that may be written $a = q + p(z_1 \bar{z}_1 + z_2 \bar{z}_2 - 1)$ for some $p, q \in \text{Polynomial}(z_1, z_2, \bar{z}_1, \bar{z}_2)$. We note that the group action on $(z_1 \bar{z}_1 + z_2 \bar{z}_2 - 1)$ is trivial since in each term, there is an equal number of z_i and $\bar{z}_j$ so that the group actions cancel. Because of this, the group action on a is as follows writing the sums in shorthand,

$$e^{i\varphi} \triangleright a = \sum_{\substack{\alpha, \beta, \gamma, \delta, \\ \alpha', \beta', \gamma', \delta'}} e^{i(\alpha' + \beta' - \gamma' - \delta')\varphi} q + e^{i(\alpha + \beta - \gamma - \delta)\varphi} p(z_1 \bar{z}_1 + z_2 \bar{z}_2 - 1).$$

One might consider $e^{i\varphi} \triangleright a$ to be comprised of two parts, one in $\bigoplus_{\alpha', \beta', \gamma', \delta'} A^{\alpha' + \beta' - \gamma' - \delta'}$ and another in $\bigoplus_{\alpha, \beta, \gamma, \delta} A^{\alpha + \beta - \gamma - \delta}$. However since the degree of the polynomials in each variable is arbitrary, there will be over-counting in the indices, furthermore due to the arbitrariness, each sum varies over the whole of $\mathbb{Z}$. Owing to these two facts, we can combine this sum as

$$a \in \bigoplus_{n \in \mathbb{Z}} A^n,$$

which holds $\forall a \in \mathcal{A}$. Thus, $a \in \mathcal{A} \implies a \in \bigoplus_{n \in \mathbb{Z}} A^n$ and if $a \in \bigoplus_{n \in \mathbb{Z}} A^n$, then $a \in \mathcal{A}$. Both statements hold $\forall a \in \mathcal{A}, \forall a \in \bigoplus_{n \in \mathbb{Z}} A^n$, therefore we conclude $\mathcal{A} = \bigoplus_{n \in \mathbb{Z}} A^n$. $\square$

Since we want to make the Hopf fibration equivariant, we need to define the $\mathbb{Z}_2^\psi$ -actions on $\mathcal{A}$. These actions will be right actions since they are left actions in the geometric approach. By defining the actions of the non-trivial element (-1) on the generating set, we can define $\mathbb{Z}_2^\psi$ -actions on the whole algebra.

Definition 6.19. The right $\mathbb{Z}_2^\psi$ -actions are defined as follows,

$$\begin{aligned} \mathbb{Z}_2^\psi : \mathcal{A} \times \mathbb{Z}_2 &\rightarrow \mathcal{A}, & z_1 \triangleleft (-1) &= e^{i\psi} z_2, & z_2 \triangleleft (-1) &= e^{-i\psi} z_1, \\ \bar{z}_1 \triangleleft (-1) &= e^{-i\psi} \bar{z}_2, & \bar{z}_2 \triangleleft (-1) &= e^{i\psi} \bar{z}_1, \end{aligned}$$

Geometrically, the $U(1)$ and $\mathbb{Z}_2^\psi$ -actions commute by the definition of an equivariant bundle. We should check that this holds in the algebraic picture too, we need to check

$$e^{i\varphi} \triangleright (z_1 \triangleleft (-1)) \stackrel{!}{=} (e^{i\varphi} \triangleright z_1) \triangleleft (-1). \quad (6.22)$$

Taking the left hand side, we see that

$$e^{i\varphi} \triangleright (z_1 \triangleleft (-1)) = e^{i\varphi} \triangleright (e^{i\psi} z_2) = e^{i\psi} (e^{i\varphi} \triangleright z_2) = e^{i(\psi + \varphi)} z_2.$$

Similarly, for the right hand side of the equation,

$$(e^{i\varphi} \triangleright z_1) \triangleleft (-1) = (e^{i\varphi} z_1) \triangleleft (-1) = e^{i\varphi} (z_1 \triangleleft (-1)) = e^{i(\varphi + \psi)} z_2$$

These two agree, and a quick calculation will confirm that the actions are compatible and commute on the whole generating set, so the actions are compatible on the whole of the algebra.

The right $U(1)$ -action descends down to the base space of the Hopf fibration.

Definition 6.20. The base space is the space of $U(1)$ -invariant polynomials,

$$\mathcal{B} := A^{U(1)-inv} = \{a \in \mathcal{A} : e^{i\varphi} \triangleright a = a, \forall e^{i\varphi} \in U(1)\} = A^0 \cong \mathcal{O}(\mathbb{S}^2)$$

The final isomorphism is shown in Appendix B

Proposition 6.21.

$$\triangleleft : \mathcal{B} \times \mathbb{Z}_2^\psi \rightarrow \mathcal{B}. \quad (6.23)$$

Proof. Since the $U(1)$ and $\mathbb{Z}_2^\psi$ -actions commute, one may find the $U(1)$ -invariant elements of $\mathcal{A}$, then apply the $\mathbb{Z}_2^\psi$ -actions. This is equivalent to applying the $\mathbb{Z}_2^\psi$ -actions to $\mathcal{A}$, then looking at the $U(1)$ -invariant elements of $\mathcal{A}$. In this way, the $\mathbb{Z}_2^\psi$ -actions are only on $\mathcal{B}$ and the actions stay within $\mathcal{B}$. $\square$

The descent of the $\mathbb{Z}_2^\psi$ -actions onto $\mathcal{B}$ is not the only descent, in fact, due to the decomposition in terms of the $U(1)$ -action, there are $\mathbb{Z}_2^\psi$ -actions on each $A^n \in \mathcal{A}$.

Sections of associated bundles are the natural objects to study, and since our algebra deals with polynomial functions, we can use Thm. 3.30 in the case $k = 0$. This means we are looking at $C^\infty(P, V)^{G-equiv} \cong \Gamma^\infty(P \times_\rho V)$. In the algebraic sense, our algebra $\mathcal{A}$ takes the place of P and we choose our vector space V to be the complex numbers with group action $e^{i\varphi} \triangleright z = e^{iq\varphi} z$, which we will denote $\mathbb{C}_{(q)}$. Sections of such a bundle represent scalar fields of charge q . By using the decomposition of $\mathcal{A}$, we have

$$\begin{aligned} (\mathcal{A} \otimes \mathbb{C}_{(q)})^{U(1)-equiv} &= \left(\bigoplus_{n \in \mathbb{Z}} A^n \otimes \mathbb{C}_{(q)} \right)^{U(1)-equiv}, \\ &= A^q. \end{aligned}$$

This follows from the observation that

$$(\mathcal{A} \otimes \mathbb{C}_{(q)})^{U(1)-equiv} = \{a \in \mathcal{A} : e^{i\varphi} \triangleright a = e^{iq\varphi} a\},$$

which is simply A^q .

Now we have the space where sections of the Hopf associated line bundle live, we can promote these to equivariant sections to look at physics on a $\mathbb{Z}_2$ -equivariant Hopf fibration by requiring further that the sections be $\mathbb{Z}_2$ -invariant. These will then be q -charged scalar fields on the equivariant Hopf bundle.

We can in fact write down some simple examples of these equivariant sections, depending on the $\mathbb{Z}_2$ -action chosen. As throughout this section, we have been working with the $\mathbb{Z}_2^\psi$ -actions and we note first that z_1 and z_2 have charge $q = 1$, whereas $\bar{z}_1$ and $\bar{z}_2$ have charge $q = -1$. Therefore, any element $a \in A^q$ must have q -more z_1 and z_2 than $\bar{z}_1$ and $\bar{z}_2$ in each product term. Furthermore, since $\mathcal{A}$ is an algebra over $\mathbb{C}$, one can multiply its elements by constants in $\mathbb{C}$, in particular, one can multiply elements by phases.

In A^1 , we could consider $z_1 \in A^1$ as sections of the Hopf bundle, however this is not $\mathbb{Z}_2^\psi$ -invariant. An example of a $\mathbb{Z}_2^\psi$ -invariant section would be

$$a = e^{-i\psi/2} z_1 + e^{i\psi/2} z_2 \in A^1.$$

One observes,

$$a \triangleleft (-1) = e^{-i\psi/2} e^{i\psi} z_2 + e^{i\psi/2} e^{-i\psi} z_1 = e^{i\psi/2} z_2 + e^{-i\psi/2} z_1 = a.$$

For each ψ , we have found an equivariant-section representing a matter field of charge 1.

6.3.2 Geometric Approach

We return now to the geometric picture and we focus on the equivariant Hopf associated line bundles since the equivariant sections of such bundles physically represent matter fields. The equivariant Hopf associated line bundle is

$$\mathbb{S}^3 \times_{\varrho_q} \mathbb{C} \xrightarrow{\pi_V} \mathbb{S}^2.$$

As before, $\pi_V = \pi \circ pr_1$. An equivariant section $s(x, z, \bar{z})$ of this bundle is one such that the following diagram commutes.

$$\begin{array}{ccc} \mathbb{S}^3 \times_{\varrho_q} \mathbb{C} & \xrightarrow{(-1)} & \mathbb{S}^3 \times_{\varrho_q} \mathbb{C} \\ \uparrow s & & \uparrow s \\ \mathbb{S}^2 & \xrightarrow{(-1)} & \mathbb{S}^2 \end{array}$$

(-1) represents the action of the non-trivial element of $\mathbb{Z}_2$ as usual. One might read this either as an equivariance condition $s \circ (-1) = (-1) \circ s$ or as an invariance condition $(-1) \circ s \circ (-1) = s$ since $(-1) = (-1)^{-1}$ in $\mathbb{Z}_2$.

In order to consider such sections, we trivialise the Hopf principal bundle by considering an open subset of $\mathbb{S}^2$. Before, we used $\mathbb{S}^2 \setminus \{N\}$ and $\mathbb{S}^2 \setminus \{S\}$, however neither of these are $\mathbb{Z}_2$ -invariant so we instead take the double-punctured sphere $\mathbb{S}^2 \setminus \{N, S\}$. By stereographic projection we have the isomorphism

$$\mathbb{S}^2 \setminus \{N, S\} \cong \mathbb{C} \setminus \{0\}.$$

We recall the stereographic projection map defining this isomorphism

$$\begin{aligned} \gamma : \mathbb{S}^2 \setminus \{N, S\} &\rightarrow \mathbb{C} \setminus \{0\}, \\ (x, z, \bar{z}) &\mapsto w = \frac{z}{1-x}. \end{aligned}$$

This forms a new principal bundle,

$$\mathbb{S}^3|_{\tilde{\pi}^{-1}(\mathbb{C} \setminus \{0\})} \xrightarrow{\tilde{\pi}} \mathbb{C} \setminus \{0\},$$

induced by the restriction

$$\begin{array}{ccc} \mathbb{S}^3 & \xrightarrow{\pi} & \mathbb{S}^2 \\ \uparrow & & \uparrow \\ \mathbb{S}^3|_{\pi^{-1}(\mathbb{S}^2 \setminus \{N, S\})} & \xrightarrow{\pi} & \mathbb{S}^2 \setminus \{N, S\} \\ & \searrow \tilde{\pi} = \gamma \circ \pi & \downarrow \gamma \\ & & \mathbb{C} \setminus \{0\} \end{array}$$

The projection map is defined in terms of the original projection map and the stereographic projection. Explicitly this is

$$\tilde{\pi}(z_1, z_2) = \frac{z}{1-x} = \frac{2z_1\bar{z}_2}{1-|z_1|^2+|z_2|^2} = \frac{2z_1\bar{z}_2}{1-(|z_1|^2+|z_2|^2)+2|z_2|^2} = \frac{z_1}{z_2}$$

We can now trivialise the restricted Hopf bundle with $\mathbb{C} \setminus \{0\}$ as the base space by local trivialisation ψ .

$$\begin{array}{ccc} \mathbb{S}^3|_{\tilde{\pi}^{-1}(\mathbb{C} \setminus \{0\})} & \xrightarrow{\psi} & \mathbb{C} \setminus \{0\} \times U(1) \\ \downarrow \tilde{\pi} & \swarrow pr_1 & \\ \mathbb{C} \setminus \{0\} & & \end{array}$$

It is this trivialised bundle whose associated line bundle we will form and investigate. By (4.6), we already have a section that is defined on this space so by Prop. 3.15 we have a canonical local trivialisation. The section is

$$s : \mathbb{C} \setminus \{0\} \rightarrow \mathbb{S}^3|_{\tilde{\pi}^{-1}(\mathbb{C} \setminus \{0\})},$$

$$w \mapsto \frac{1}{\sqrt{1+|w|^2}}(w, 1).$$

This is equivalent to our local trivialisation $s(w) \cdot g = \psi^{-1}(w, g)$ and we need now only find ψ such that $\psi \circ \psi^{-1} = \psi^{-1} \circ \psi = id$. One can do this by definition of a general local trivialisation $\psi(p) = (\pi(p), \phi(p))$ with $\phi(p) \in G$. In our case we write $\psi(z_1, z_2) = (w, g) \in \mathbb{C} \setminus \{0\} \times U(1)$ with $w = z_1/z_2$. Then,

$$\psi^{-1} \circ \psi(z_1, z_2) = \psi^{-1}(w, g) = \frac{1}{\sqrt{1+|w|^2}}(wg, g) = \frac{1}{\sqrt{1+\left|\frac{z_1}{z_2}\right|^2}}\left(\frac{z_1}{z_2}g, g\right) \stackrel{!}{=} (z_1, z_2).$$

In order for this to be satisfied, one must have $g = z_2\sqrt{1+|z_1/z_2|^2}$. This is indeed an element of $U(1)$:

$$g\bar{g} = z_2\sqrt{1+\left|\frac{z_1}{z_2}\right|^2}\bar{z}_2\sqrt{1+\left|\frac{z_1}{z_2}\right|^2} = |z_2|^2\left(1+\left|\frac{z_1}{z_2}\right|^2\right) = |z_2|^2 + |z_1|^2 = 1.$$

One checks also that $\psi \circ \psi^{-1}(w, g) = (w, g)$:

$$\begin{aligned} \psi \circ \psi^{-1}(w, g) &= \psi\left(\frac{wg}{\sqrt{1+|w|^2}}, \frac{g}{\sqrt{1+|w|^2}}\right), \\ &= \left(\frac{wg}{\sqrt{1+|w|^2}} \frac{\sqrt{1+|w|^2}}{g}, \frac{g}{\sqrt{1+|w|^2}} \sqrt{1+\left|\frac{wg}{\sqrt{1+|w|^2}} \frac{\sqrt{1+|w|^2}}{g}\right|^2}\right), \\ &= (w, g). \end{aligned}$$

Because these hold, we have found a local trivialisation,

$$\psi(z_1, z_2) = \left(\frac{z_1}{z_2}, z_2\sqrt{1+\left|\frac{z_1}{z_2}\right|^2}\right). \quad (6.24)$$

We are now able to define a $\mathbb{Z}_2$ -action on this trivialised space defined by the following diagram commuting.

$$\begin{array}{ccc}
\mathbb{Z}_2 \times \mathbb{C} \setminus \{0\} \times U(1) & \xrightarrow{\psi^{-1}} & \mathbb{Z}_2 \times \mathbb{S}^3 \\
(-1) \downarrow & & \downarrow (-1) \\
\mathbb{C} \setminus \{0\} \times U(1) & \xleftarrow{\psi} & \mathbb{S}^3
\end{array}$$

In this way, we define the left $\mathbb{Z}_2$ -action by

$$(-1) \cdot (w, g) = \psi \left((-1) \cdot \psi^{-1}(w, g) \right). \quad (6.25)$$

We will now specialise from a general $\mathbb{Z}_2$ -action to the $\mathbb{Z}_2^\psi$ -actions found before. Inspecting this explicitly,

$$\begin{aligned}
(-1) \cdot (w, g) &= \psi \left((-1) \cdot \left(\frac{wg}{\sqrt{1+|w|^2}}, \frac{g}{\sqrt{1+|w|^2}} \right) \right), \\
&= \psi \left(\frac{ge^{i\psi}}{\sqrt{1+|w|^2}}, \frac{wge^{-i\psi}}{\sqrt{1+|w|^2}} \right), \\
&= \left(e^{i2\psi} \frac{1}{w}, e^{-i\psi} \frac{wg}{\sqrt{1+|w|^2}} \sqrt{1 + \frac{1}{|w|^2}} \right), \\
&= \left(e^{i2\psi} \frac{1}{w}, e^{-i\psi} \frac{w}{|w|} g \right).
\end{aligned}$$

Therefore, the $\mathbb{Z}_2^\psi$ -actions on $\mathbb{C} \setminus \{0\} \times U(1)$ are

$$(-1) \cdot (w, g) = \left(e^{i2\psi} \frac{1}{w}, e^{-i\psi} \frac{w}{|w|} g \right). \quad (6.26)$$

The actions on the first factor are the expected $\mathbb{Z}_2^\psi$ -actions on $\mathbb{C} \setminus \{0\}$ since $w = z_1/z_2$ and the $\mathbb{Z}_2^\psi$ -actions will result in $(-1) \cdot w = e^{i\psi} z_2/(e^{-i\psi} z_1) = e^{i2\psi}(1/w)$. However, the action on the fibres is non-trivial; they depend on w as well. The associated line bundle of this trivialisation is

$$(\mathbb{C} \setminus \{0\} \times U(1)) \times_{\varrho_q} \mathbb{C} \cong \mathbb{C} \setminus \{0\} \times \mathbb{C}.$$

Just as in Ex. 3.21, by definition of the associated bundle, one can express any point on $(\mathbb{C} \setminus \{0\} \times U(1)) \times_{\varrho_q} \mathbb{C}$ as (w, e_G, z) for some $w \in \mathbb{C} \setminus \{0\}$ and $z \in \mathbb{C}$. We consider the $\mathbb{Z}_2^\psi$ -actions on (w, e_G, z) to see how the actions lift to the associated bundle.

$$\begin{aligned}
(-1) \cdot (w, e_G, z) &= \left(e^{i2\psi} \frac{1}{w}, e^{-i\psi} \frac{w}{|w|}, z \right), \\
&\sim \left(e^{i2\psi} \frac{1}{w}, e_G, \left(e^{-i\psi} \frac{w}{|w|} \right)^q z \right), \\
&\cong \left(e^{i2\psi} \frac{1}{w}, \left(e^{-i\psi} \frac{w}{|w|} \right)^q z \right) \in \mathbb{C} \setminus \{0\} \times \mathbb{C}.
\end{aligned}$$

With this defined, we are now in a position to consider the equivariant sections, the matter fields, in this geometry. The bundle in question has reduced to

$$\mathbb{C} \setminus \{0\} \times \mathbb{C} \xrightarrow{pr_1} \mathbb{C} \setminus \{0\}, \quad (6.27)$$

thanks to the local trivialisation and structure of the associated line bundle. A section of such a bundle is simply a map,

$$\begin{aligned} s : \mathbb{C} \setminus \{0\} &\rightarrow \mathbb{C} \setminus \{0\} \times \mathbb{C}, \\ w &\mapsto (w, \varphi(w)). \end{aligned}$$

A section is wholly defined by the complex-valued function $\varphi(w)$, which is what would be interpreted as the matter field in a physical theory. As an Ad -invariance condition, a section satisfies $(-1) \circ s \circ (-1) = s$:

$$\begin{aligned} (-1) \circ s \circ (-1)(w) &= (-1) \circ s \left(\frac{e^{i2\psi}}{w} \right), \\ &= (-1) \cdot \left(\frac{e^{i2\psi}}{w}, \varphi \left(\frac{e^{i2\psi}}{w} \right) \right), \\ &= \left(w, \left(\frac{|w|}{w} e^{i\psi} \right)^q \varphi \left(\frac{e^{i2\psi}}{w} \right) \right), \\ &\stackrel{!}{=} (w, \varphi(w)). \end{aligned}$$

Whereas as an equivariance condition, a section satisfies $(-1) \circ s = s \circ (-1)$:

$$\begin{aligned} s \circ (-1)(w) &= s \left(\frac{e^{i2\psi}}{w} \right) = \left(\frac{e^{i2\psi}}{w}, \varphi \left(\frac{e^{i2\psi}}{w} \right) \right), \\ (-1) \circ (s)(w) &= (-1) \cdot (w, \varphi(w)) = \left(\frac{e^{i2\psi}}{w}, \left(\frac{w}{|w|} e^{-i\psi} \right)^q \varphi(w) \right), \\ \implies \left(\frac{e^{i2\psi}}{w}, \varphi \left(\frac{e^{i2\psi}}{w} \right) \right) &= \left(\frac{e^{i2\psi}}{w}, \left(\frac{w}{|w|} e^{-i\psi} \right)^q \varphi(w) \right). \end{aligned}$$

An equivariant section of the equivariant Hopf associated line bundle is therefore defined wholly by any complex-valued function φ satisfying

$$\varphi \left(\frac{e^{i2\psi}}{w} \right) = \left(\frac{w}{|w|} e^{-i\psi} \right)^q \varphi(w). \quad (6.28)$$

6.3.3 Equivariant Section Analysis

Equivariant sections of the associated line bundle reduced down to (6.28), which we now analyse. Writing this in terms of polar coordinates $w = r \exp(i\theta)$, (6.28) now reads

$$\varphi \left(\frac{1}{r}, 2\psi - \theta \right) = e^{iq(\theta - \psi)} \varphi(r, \theta). \quad (6.29)$$

We are going to consider the sections of one particular $\mathbb{Z}_2$ -action, the sections of the $\mathbb{Z}_2^0$ -action,

$$\varphi\left(\frac{1}{r}, -\theta\right) = e^{iq\theta}\varphi(r, \theta).$$

Important places to consider are the fixed points of the $\mathbb{Z}_2^\psi$ -actions. These were originally at the intersection of the equator and the $Re(z)$ -axis, however on the complex plane (r, θ) , these are the points $(1, 0)$, and $(1, \pi)$. We consider these separately. At $(1, 0)$, a section must satisfy that

$$\varphi(1, 0) = \varphi(1, 0),$$

which is clearly always satisfied. At the other point, however, we have

$$\varphi(1, -\pi) = e^{iq\pi}\varphi(1, \pi).$$

The θ -coordinate is 2π -periodic, therefore there is no difference between π and $-\pi$, however the phase factor is important to consider.

If q is odd, then $\varphi(1, -\pi) = -\varphi(1, \pi)$ and it must be that there is a Dirichlet boundary condition,

$$\varphi(1, \pi) = 0. \tag{6.30}$$

If q is even, then consider, by writing $1/r = h(r)$,

$$\begin{aligned} \left.\frac{\partial}{\partial r}\varphi(r, \pi)\right|_{r=1} &= \left.\frac{\partial}{\partial r}\varphi(h(r), -\pi)\right|_{r=1}, \\ &= \left.\frac{dh}{dr}\right|_{r=1} \left.\frac{\partial}{\partial h}\varphi(h, -\pi)\right|_{h=1}, \\ &= -\left.\frac{1}{r^2}\right|_{r=1} \left.\frac{\partial}{\partial h}\varphi(h, -\pi)\right|_{h=1}, \\ &= -\left.\frac{\partial}{\partial h}\varphi(h, -\pi)\right|_{h=1}. \end{aligned}$$

By changing the variable from h to r on the right hand side, we recover the left hand side but with differing sign, therefore we see the Neumann boundary condition,

$$\left.\frac{\partial}{\partial r}\varphi(r, \pi)\right|_{r=1} = 0. \tag{6.31}$$

Despite being less trivial, the case $\psi = 0$ still has some characteristics shared with the $\mathbb{Z}_2$ -equivariant $U(1)$ -bundle over $\mathbb{R}^n$.

By the $\mathbb{Z}_2^\psi$ -actions, points at $1/r$ and r are related. Furthermore, $\varphi(0, -\theta)$ must be independent of θ as all values of θ represent the same point. Therefore, by extension, $\lim_{r \rightarrow \infty} \varphi(r, \theta)$ must also be independent of θ , however by taking the limit of the equivariance relation (6.29) as r tends to infinity we have

$$\varphi(0, -\theta) = e^{iq\theta} \lim_{r \rightarrow \infty} \varphi(r, \theta).$$

For $\varphi(0, -\theta)$ to be θ -independent, the phase factor must be cancelled out, therefore $\lim_{r \rightarrow \infty} \varphi(r, \theta) = 0$ and as a consequence,

$$\varphi(0, -\theta) = 0. \tag{6.32}$$

A simple way of finding such a function is by utilising a trick in geometry. We shall define a non-vanishing function in some region $r > 1$, which will, by the equivariance condition, define the function in $r < 1$. We will consider functions that have *compact support* and present definitions [9] in a less general form but tailored to what we require.

Definition 6.22. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ and let f be a function $f : \mathbb{K}^m \rightarrow \mathbb{K}^n$. The support of f is

$$\text{supp}(f) = \overline{\{x \in \mathbb{K} : f(x) \neq 0\}}. \tag{6.33}$$

The overline of the set denotes the set's closure, the set and its boundary. If $\text{supp}(f)$ is a compact set, then f has compact support.

Remark 6.23. Any $[a, b]^n$ on $\mathbb{R}^n$ is compact by the Heine-Borel theorem so any non-zero function on a closed region will have compact support. $\triangle$

We are interested in defining a function in $r > 1$, which will, by the equivariance condition, define the function in $r < 1$. Consider the following example for the construction of such a function.

Example 6.24. We define a non-zero function on the interval $3 < r < 5$, so the support of this function will be the annulus with boundaries at $r = 3$ and $r = 5$. A non-vanishing function on this domain is

$$f(r) = \exp\left(\frac{-1}{1 - (r - 4)^2}\right).$$

This defines a bump in the annulus with maximum at $r = 4$, which is zero at the boundary.

We can extend this to define the section in $r > 1$ as follows.

$$\varphi(r, \theta) = \begin{cases} \exp\left(\frac{-1}{1-(r-4)^2}\right) \Omega(\theta), & \text{for } r \in [3, 5], \\ 0 & \text{otherwise,} \end{cases} \quad (6.34)$$

with $\Omega(\theta)$ some smooth, nowhere-vanishing function of θ . By the equivariance of the section, one can extend this, defining a function $r < 1$ as well. In $0 < r < 1$,

$$\varphi(r, \theta) = \begin{cases} e^{-iq\theta} \exp\left(\frac{-1}{1-(\frac{1}{r}-4)^2}\right) \Omega(-\theta), & \text{for } r \in [\frac{1}{5}, \frac{1}{3}], \\ 0 & \text{otherwise.} \end{cases} \quad (6.35)$$

This satisfies all the boundary conditions of vanishing at the origin, at the $\mathbb{Z}_2^0$ -action boundary points and at infinity. We have in this way found a function φ satisfying the necessary conditions and thus found an equivariant section of the equivariant Hopf associated line bundles. That is

$$\varphi(r, \theta) = \begin{cases} \exp\left(\frac{-1}{1-(r-4)^2}\right) \Omega(\theta), & \text{for } r \in [3, 5], \\ e^{-iq\theta} \exp\left(\frac{-1}{1-(\frac{1}{r}-4)^2}\right) \Omega(-\theta), & \text{for } r \in [\frac{1}{5}, \frac{1}{3}], \\ 0 & \text{otherwise.} \end{cases} \quad (6.36)$$

▽

Remark 6.25. We considered $r \rightarrow 0$ and $r \rightarrow \infty$ since, despite not being in the domain, in the unpunctured 2-sphere these points do exist and by having a function that smoothly approaches these two points, one repairs these punctures. △

Any function with compact support can be extended to define an equivariant section of the equivariant Hopf associated line bundles by the construction in Ex. 6.24. Physically these functions φ are scalar fields of electric charge q as usual for a theory with $U(1)$ as the gauge group.

7 Conclusions

This work heuristically introduced differential geometry through both theory and examples, providing the essential tools for the study of principal bundles and gauge theory. We then provided a detailed overview and exemplification of principal bundle theory and gauge theory, using [8] and [21] as bases and references. The Hopf fibration was thereafter introduced as an interesting example and shown to be a principal bundle, satisfying the conditions in Chapter 3. Presheaves and sheaves were then introduced to study Čech cohomology. Within the Čech cohomological framework, we were able to show that magnetic charges lie within $\check{H}^2(\mathcal{U}, \mathbb{Z})$. As a corollary of this general result we were able to provide a direct proof of the quantisation of magnetic charges, which is of importance because it outlines and emphasises the role that gauge theories play geometrically. The geometric extra data restricts the possibilities of magnetic charge, quantising that which on the outset would not appear to have reason to be quantised. We then introduced the theory of equivariant principal bundles and were successfully able to classify the $\mathbb{Z}_2$ -equivariant principal $U(1)$ -bundles over $\mathbb{R}^n$. We studied the $\mathbb{Z}_2$ -equivariant associated complex line $U(1)$ -bundles over $\mathbb{R}^n$ and analysed properties of the sections, finding that they exhibit either Dirichlet or Neumann boundary conditions, depending on whether the bundle has an odd or even charge. We introduced equivariant Hopf bundles, deriving a family of equivariant $\mathbb{Z}_2$ -actions and formed equivariant Hopf associated line bundles. We were then able to investigate sections of these $\mathbb{Z}_2$ -equivariant Hopf associated line bundles both algebraically and geometrically and provide explicit examples in both descriptions. Through the use of bundle theory, using a local trivialisation, we were able to reduce the problem to one that admits explicit analysis. Within our analysis, we found Dirichlet and Neumann boundary conditions at the fixed points of a chosen $\mathbb{Z}_2$ -action and derived explicitly an equivariant section of the chosen equivariant Hopf associated line bundle.

We were unable to study singular cohomology during this dissertation. Subsequent work would involve getting to terms with singular cohomology to investigate how, in the proof of equivariant magnetic charge, the gauge theoretic quantisation over the integers forms a lattice inside the expected reals. Other future work would include explicitly classifying the $\mathbb{Z}_2$ -equivariant principal $U(1)$ -bundles over $\mathbb{S}^2$. This would provide a classification

which the equivariant Hopf bundles fit and would be in keeping with Proposition 3.33, whose geometry involves principal $U(1)$ -bundles over $\mathbb{S}^2$. We would then move to study equivariant bundles further, looking at different $\mathbb{Z}_2$ -actions on the Hopf fibration and other non-trivial principal bundles.

A Calculation of Pullback Curvature on $\mathbb{C}$

Using coordinates

$$z_1 = z(1 + |z|^2)^{-\frac{1}{2}}, \quad z_2 = (1 + |z|^2)^{-\frac{1}{2}},$$

one calculates

$$\begin{aligned} s^*dz_1 &= (1 + |z|^2)^{-\frac{1}{2}}dz - \frac{1}{2}(1 + |z|^2)^{-\frac{3}{2}}|z|^2dz - \frac{1}{2}(1 + |z|^2)^{-\frac{3}{2}}z^2d\bar{z}, \\ s^*d\bar{z}_1dz_1 &= (1 + |z|^2)^{-1}d\bar{z}dz - \frac{1}{2}|z|^2(1 + |z|^2)^{-2}d\bar{z}dz - \frac{1}{2}|z|^2(1 + |z|^2)^{-2}d\bar{z}dz \\ &\quad + \frac{1}{4}|z|^4(1 + |z|^2)^{-3}d\bar{z}dz - \frac{1}{4}|z|^4(1 + |z|^2)^{-3}d\bar{z}dz, \\ &= (1 + |z|^2)^{-1}d\bar{z}dz - |z|^2(1 + |z|^2)^{-2}d\bar{z}dz, \\ s^*dz_2 &= \frac{1}{2}(1 + |z|^2)^{-\frac{3}{2}}z d\bar{z} - \frac{1}{2}(1 + |z|^2)^{-\frac{3}{2}}\bar{z}dz, \\ s^*d\bar{z}_2dz_2 &= -\frac{1}{4}|z|^2(1 + |z|^2)^{-3}d\bar{z}dz + \frac{1}{4}|z|^2(1 + |z|^2)^{-3}d\bar{z}dz, \\ &= 0. \end{aligned}$$

Combining these results,

$$s^*\Omega = s^*d\bar{z}_1dz_1 + s^*d\bar{z}_2dz_2 = \frac{1}{(1 + |z|^2)^2}(1 + |z|^2 - |z|^2)d\bar{z}dz = \frac{d\bar{z}dz}{(1 + |z|^2)^2} = -\frac{dzd\bar{z}}{(1 + |z|^2)^2}.$$

Giving the pullback curvature form.

B Algebra of Polynomial Functions on $\mathbb{S}^2$

Since $\mathcal{B}$ is the set of $U(1)$ -invariant elements of $\mathcal{A}$ and is generated by $\langle z_1 \bar{z}_1, z_2 \bar{z}_2, z_1 \bar{z}_2, z_2 \bar{z}_1 \rangle$.

We consider now the set $\{x, z, \bar{z}\}$ defined as,

$$x = z_1 \bar{z}_1 - z_2 \bar{z}_2,$$

$$z = 2z_1 \bar{z}_2,$$

plus the sphere relation $x^2 + z\bar{z} = 1$. Note that $(x, z, \bar{z})$ are exactly the coordinates on $\mathbb{S}^2$ in the geometric approach.

One can see that $z_1 \bar{z}_2$ and $z_2 \bar{z}_1$ are defined immediately as

$$z_1 \bar{z}_2 = \frac{z}{2}, \quad z_2 \bar{z}_1 = \frac{\bar{z}}{2}.$$

If one then considers x to isolate $z_1 \bar{z}_1$ as

$$z_1 \bar{z}_1 = x + z_2 \bar{z}_2,$$

then by using the equivalence class relations on this algebra, one has

$$z_1 \bar{z}_1 = x + z_2 \bar{z}_2, \quad \equiv x + 1 - z_1 \bar{z}_1,$$

resulting in

$$z_1 \bar{z}_1 \equiv \frac{1+x}{2}.$$

Similarly, one can start with

$$z_2 \bar{z}_2 = z_1 \bar{z}_1 - x,$$

then by the sphere relation,

$$z_2 \bar{z}_2 \equiv 1 - z_1 \bar{z}_1 - x, \quad \implies z_2 \bar{z}_2 \equiv \frac{1-x}{2}.$$

Since we have defined all of the generators in terms of x , z , and $\bar{z}$, these three variables themselves form generators. Furthermore, one may note that

$$x^2 + z\bar{z} = z_1 \bar{z}_1 + z_2 \bar{z}_2, \tag{B.1}$$

meaning that one can define the sphere relation entirely in terms of x , z , and $\bar{z}$, so one may define $\mathcal{B}$ as

$$\mathcal{B} = \text{Polynomial}(x, z, \bar{z}) / \langle x^2 + z\bar{z} - 1 \rangle. \tag{B.2}$$

Note that x , z , and $\bar{z}$ are the standard coordinates on $\mathbb{S}^2$ and $x^2 + z\bar{z} = 1$ defines $\mathbb{S}^2$, because of this and the similar form to $\mathcal{O}(\mathbb{S}^3)$ we can associate $\mathcal{B}$ with the algebra of polynomial functions on $\mathbb{S}^2$,

$$\mathcal{B} \cong \mathcal{O}(\mathbb{S}^2). \tag{B.3}$$

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